Linear Algebra and Its Applications

Chapter 4

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

Contents

4.1 Vector spaces and subspaces 13

Practice problem 4.1.P1: An affine line fails scalar closure 13

Practice problem 4.1.P2: Every generator belongs to its span 13

Exercise 4.1.1: The first quadrant 13

Exercise 4.1.2: The first and third quadrants together 13

Exercise 4.1.3: The closed unit disk 14

Exercise 4.1.4: A translated line is not closed under addition 14

Exercise 4.1.5: Multiples of a single polynomial 14

Exercise 4.1.6: A fixed quadratic coefficient 14

Exercise 4.1.7: Integer coefficients and real scalars 15

Exercise 4.1.8: Polynomials vanishing at zero 15

Exercise 4.1.9: A one-parameter line in three dimensions 15

Exercise 4.1.10: A line with a zero middle coordinate . . . 15
Exercise 4.1.11: Separate two free parameters . . . 15
Exercise 4.1.12: Separate two parameters in four dimensions . . . 16
Exercise 4.1.13: A set of generators versus its span . . . 16
Exercise 4.1.14: Another vector in the same plane . . . 16
Exercise 4.1.15: A coordinate forced to be negative one . . . 16
Exercise 4.1.16: A coordinate forced to be one . . . 17
Exercise 4.1.17: Three parameters, each with its own coefficient vector . . . 17
Exercise 4.1.18: A permanently zero coordinate is harmless . . . 17
Exercise 4.1.19: The space of oscillating functions . . . 17
Exercise 4.1.20: Continuous functions and an endpoint condition . . . 17
Exercise 4.1.21: Upper triangular matrices form a subspace . . . 18
Exercise 4.1.22: A homogeneous matrix equation . . . 18
Exercise 4.1.23: Five statements about vectors and subspaces . . . 18
Exercise 4.1.24: Five more statements, including the closure quantifiers . . . 18
Exercise 4.1.25: Uniqueness of the zero vector . . . 19
Exercise 4.1.26: Uniqueness of the additive inverse . . . 19
Exercise 4.1.27: Multiplication by the scalar zero . . . 19
Exercise 4.1.28: A scalar times the zero vector . . . 19
Exercise 4.1.29: The scalar negative one gives the additive inverse . . . 20
Exercise 4.1.30: A nonzero scalar cannot annihilate a nonzero vector . . . 20
Exercise 4.1.31: The span is the smallest containing subspace . . . 20
Exercise 4.1.32: Intersection works; union can fail . . . 20
Exercise 4.1.33: The sum of two subspaces . . . 20
Exercise 4.1.34: Spanning the sum by combining the generators . . . 21
Exercise 4.1.35: A checked representation in four dimensions . . . 21
Exercise 4.1.36: Test membership in a column span . . . 21
Exercise 4.1.37: Recognizing two multiple-angle functions . . . 22
Exercise 4.1.38: Recognizing three polynomials in sine . . . 22

4.2 Null spaces, column spaces, and linear transformations . . . 24

Practice problem 4.2.P1: Two descriptions of one plane . . . 24
Practice problem 4.2.P2: Add two attainable outputs . . . 24
Exercise 4.2.1: Test a proposed null-space vector directly . . . 24
Exercise 4.2.2: Another direct multiplication . . . 24

Exercise 4.2.3: Describe a null space using two free variables . . . 25
Exercise 4.2.4: A forced zero coordinate and two free coordinates . . . 25
Exercise 4.2.5: Back substitution in a five-column matrix . . . 25
Exercise 4.2.6: Three free variables . . . 25
Exercise 4.2.7: An affine plane misses zero . . . 26
Exercise 4.2.8: Move all variable terms to one side . . . 26
Exercise 4.2.9: Two homogeneous constraints define a null space . . . 26
Exercise 4.2.10: Another homogeneous description . . . 26
Exercise 4.2.11: A hidden constant obstructs the zero vector . . . 26
Exercise 4.2.12: Another parametrization that never produces zero . . . 27
Exercise 4.2.13: A span in disguise . . . 27
Exercise 4.2.14: A two-generator subspace . . . 27
Exercise 4.2.15: Make the coefficient vectors into columns . . . 27
Exercise 4.2.16: Construct a matrix from three parameters . . . 27
Exercise 4.2.17: Count inputs and outputs . . . 28
Exercise 4.2.18: A rectangular matrix with three inputs . . . 28
Exercise 4.2.19: Five inputs and two outputs . . . 28
Exercise 4.2.20: A row matrix . . . 28
Exercise 4.2.21: Exhibit one vector in each space . . . 28
Exercise 4.2.22: A null relation among three columns . . . 29
Exercise 4.2.23: A vector can belong to both spaces . . . 29
Exercise 4.2.24: Two independent membership checks in four dimensions . . . 29
Exercise 4.2.25: Six statements about null spaces and column spaces . . . 29
Exercise 4.2.26: Six related statements about kernels and ranges . . . 30
Exercise 4.2.27: Scale a known homogeneous solution . . . 30
Exercise 4.2.28: Scale an attainable right-hand side . . . 30
Exercise 4.2.29: Prove the column-space subspace theorem . . . 30
Exercise 4.2.30: The range of a linear transformation is a subspace . . . 31
Exercise 4.2.31: Evaluate a quadratic at two inputs . . . 31
Exercise 4.2.32: Repeat the same evaluation twice . . . 31
Exercise 4.2.33: Add a matrix to its transpose . . . 32
Exercise 4.2.34: The antiderivative with a fixed initial value . . . 32
Exercise 4.2.35: The image of a subspace . . . 32
Exercise 4.2.36: The inverse image of a subspace . . . 33

Exercise 4.2.37: An output that is not a zero-output input . . . 33
Exercise 4.2.38: Another vector lying in both spaces . . . 33
Exercise 4.2.39: Recover column relations and the entire null space . . . 33
Exercise 4.2.40: Find the line where two planes meet . . . 34
4.3 Linearly independent sets; bases . . . 35
Practice problem 4.3.P1: A basis must have the right ambient space . . . 35
Practice problem 4.3.P2: Remove redundant spanning vectors . . . 35
Practice problem 4.3.P3: Spanning every vector of a line is not enough . . . 36
Exercise 4.3.1: Identify a basis using original pivot columns . . . 36
Exercise 4.3.2: Identify a basis using original pivot columns . . . 36
Exercise 4.3.3: Identify a basis using original pivot columns . . . 37
Exercise 4.3.4: Identify a basis using original pivot columns . . . 37
Exercise 4.3.5: Identify a basis using original pivot columns . . . 38
Exercise 4.3.6: Identify a basis using original pivot columns . . . 38
Exercise 4.3.7: Identify a basis using original pivot columns . . . 39
Exercise 4.3.8: Identify a basis using original pivot columns . . . 39
Exercise 4.3.9: A null-space basis from one free variable . . . 40
Exercise 4.3.10: A null-space basis in five coordinates . . . 40
Exercise 4.3.11: A basis for a plane . . . 41
Exercise 4.3.12: A basis for a line . . . 41
Exercise 4.3.13: Null space and original column space . . . 41
Exercise 4.3.14: Two different spaces from one reduction . . . 42
Exercise 4.3.15: Identify a basis using original pivot columns . . . 43
Exercise 4.3.16: Identify a basis using original pivot columns . . . 43
Exercise 4.3.17: Identify a basis using original pivot columns . . . 44
Exercise 4.3.18: Identify a basis using original pivot columns . . . 44
Exercise 4.3.19: Delete a generator using a known relation . . . 45
Exercise 4.3.20: Another known dependence relation . . . 45
Exercise 4.3.21: Basis definitions and row operations . . . 45
Exercise 4.3.22: Minimal spanning and maximal independence . . . 46
Exercise 4.3.23: Four spanning vectors in four dimensions . . . 46
Exercise 4.3.24: An independent square family . . . 46
Exercise 4.3.25: A proposed basis outside the subspace . . . 46
Exercise 4.3.26: A trigonometric spanning set . . . 47

Exercise 4.3.27: The mass–spring vibration space . . . 47
Exercise 4.3.28: A critically damped circuit space . . . 47
Exercise 4.3.29: Too few vectors to span . . . 47
Exercise 4.3.30: Too many vectors to be independent . . . 47
Exercise 4.3.31: Dependence survives a linear map . . . 47
Exercise 4.3.32: Injectivity reflects dependence . . . 48
Exercise 4.3.33: Independence of two quadratic polynomials . . . 48
Exercise 4.3.34: A polynomial dependence by inspection . . . 48
Exercise 4.3.35: Construct a four-vector family with a two-vector basis . . . 48
Exercise 4.3.36: Bases of two spaces and their sum . . . 49
Exercise 4.3.37: Four independent functions by exact evaluations . . . 50
Exercise 4.3.38: Seven cosine powers are independent . . . 51
4.4 Coordinate systems . . . 52
Practice problem 4.4.P1: Both directions of a basis-coordinate matrix . . . 52
Practice problem 4.4.P2: Coordinates of a quadratic polynomial . . . 52
Exercise 4.4.1: Coordinates and basis representations . . . 53
Exercise 4.4.2: Coordinates and basis representations . . . 54
Exercise 4.4.3: Coordinates and basis representations . . . 54
Exercise 4.4.4: Coordinates and basis representations . . . 54
Exercise 4.4.5: Find weights by reducing a basis system . . . 55
Exercise 4.4.6: Find weights by reducing a basis system . . . 55
Exercise 4.4.7: Find weights by reducing a basis system . . . 56
Exercise 4.4.8: Find weights by reducing a basis system . . . 57
Exercise 4.4.9: Coordinates and basis representations . . . 58
Exercise 4.4.10: Coordinates and basis representations . . . 59
Exercise 4.4.11: Coordinates and basis representations . . . 59
Exercise 4.4.12: Coordinates and basis representations . . . 59
Exercise 4.4.13: Coordinates and basis representations . . . 60
Exercise 4.4.14: Coordinates and basis representations . . . 61
Exercise 4.4.15: Coordinates and basis representations . . . 62
Exercise 4.4.16: Coordinates and basis representations . . . 62
Exercise 4.4.17: Coordinates and basis representations . . . 62
Exercise 4.4.18: Coordinates and basis representations . . . 63
Exercise 4.4.19: Coordinates and basis representations . . . 63

Exercise 4.4.20: Coordinates and basis representations . . . 63
Exercise 4.4.21: Coordinates and basis representations . . . 63
Exercise 4.4.22: Coordinates and basis representations . . . 64
Exercise 4.4.23: Coordinates and basis representations . . . 64
Exercise 4.4.24: Coordinates and basis representations . . . 64
Exercise 4.4.25: Coordinates and basis representations . . . 64
Exercise 4.4.26: Coordinates and basis representations . . . 64
Exercise 4.4.27: Test polynomial independence using all four coefficient positions . . . 64
Exercise 4.4.28: Test polynomial independence using all four coefficient positions . . . 65
Exercise 4.4.29: Test polynomial independence using all four coefficient positions . . . 65
Exercise 4.4.30: Test polynomial independence using all four coefficient positions . . . 66
Exercise 4.4.31: Coordinates and basis representations . . . 66
Exercise 4.4.32: Coordinates and basis representations . . . 66
Exercise 4.4.33: Test polynomial independence using all four coefficient positions . . . 67
Exercise 4.4.34: Test polynomial independence using all four coefficient positions . . . 67
Exercise 4.4.35: Coordinates and basis representations . . . 68
Exercise 4.4.36: Coordinates and basis representations . . . 69
Exercise 4.4.37: Coordinates and basis representations . . . 71
Exercise 4.4.38: Coordinates and basis representations . . . 72

4.5 The dimension of a vector space . . . 73

Practice problem 4.5.P1: Dependence alone does not bound the list size . . . 73
Practice problem 4.5.P2: A larger set than a spanning set is dependent . . . 73
Exercise 4.5.1: Find a dimension from a checked basis . . . 73
Exercise 4.5.2: Find a dimension from a checked basis . . . 74
Exercise 4.5.3: Find a dimension from a checked basis . . . 74
Exercise 4.5.4: Find a dimension from a checked basis . . . 75
Exercise 4.5.5: Find a dimension from a checked basis . . . 75
Exercise 4.5.6: Find a dimension from a checked basis . . . 76
Exercise 4.5.7: A homogeneous system defining the zero space . . . 77
Exercise 4.5.8: One constraint in four coordinates . . . 77
Exercise 4.5.9: An equal-coordinate plane . . . 77
Exercise 4.5.10: Three generators of one line . . . 77
Exercise 4.5.11: Find a dimension from a checked basis . . . 77
Exercise 4.5.12: Find a dimension from a checked basis . . . 78

Exercise 4.5.13: Nullity and column dimension from pivots . . . 79
Exercise 4.5.14: Seven columns with four pivots . . . 79
Exercise 4.5.15: A nonconsecutive pivot pattern . . . 79
Exercise 4.5.16: An invertible two-by-two matrix . . . 79
Exercise 4.5.17: Three nonzero triangular pivots . . . 80
Exercise 4.5.18: Two pivots and a zero row . . . 80
Exercise 4.5.19: Dimension definitions and missing hypotheses . . . 80
Exercise 4.5.20: Subspace identity and spanning-set cautions . . . 81
Exercise 4.5.21: Hermite polynomial coefficients . . . 81
Exercise 4.5.22: Laguerre polynomial coefficients . . . 81
Exercise 4.5.23: Hermite coordinates of a cubic . . . 82
Exercise 4.5.24: Laguerre coordinates of a quadratic . . . 82
Exercise 4.5.25: A short set cannot span . . . 82
Exercise 4.5.26: Equal finite dimensions force equality . . . 82
Exercise 4.5.27: Why all polynomials have no finite basis . . . 83
Exercise 4.5.28: Continuous functions contain arbitrarily large independent families . . . 83
Exercise 4.5.29: Compare spanning and independent family sizes . . . 83
Exercise 4.5.30: Counterexamples to incorrect dimension converses . . . 83
Exercise 4.5.31: A linear image cannot gain dimension . . . 83
Exercise 4.5.32: An injective image preserves dimension . . . 84
Exercise 4.5.33: Extend an independent family to a basis . . . 84
Exercise 4.5.34: Cosine multiples give another basis . . . 85

4.6 Rank . . . 86

Practice problem 4.6.P1: Rank and nullity from the given echelon form . . . 86
Practice problem 4.6.P2: Choose the correct rows and columns . . . 86
Practice problem 4.6.P3: The next step toward a null-space basis . . . 86
Practice problem 4.6.P4: Transpose does not change rank . . . 86
Exercise 4.6.1: Three bases from the supplied echelon form . . . 86
Exercise 4.6.2: Three bases from the supplied echelon form . . . 87
Exercise 4.6.3: Three bases from the supplied echelon form . . . 88
Exercise 4.6.4: Three bases from the supplied echelon form . . . 88
Exercise 4.6.5: A rank-three matrix with seven columns . . . 89
Exercise 4.6.6: A rank-two matrix with five columns . . . 89
Exercise 4.6.7: Full row rank does not identify a null space with its coordinate space . . . 89

Exercise 4.6.8: Rank four with six output coordinates . . . 90
Exercise 4.6.9: Recover rank from a known nullity . . . 90
Exercise 4.6.10: Seven columns and five free coordinates . . . 90
Exercise 4.6.11: From nullity to row-space dimension . . . 90
Exercise 4.6.12: A second row-space dimension . . . 90
Exercise 4.6.13: The largest possible rank . . . 90
Exercise 4.6.14: The largest row-space dimension . . . 90
Exercise 4.6.15: The smallest nullity for a wide matrix . . . 90
Exercise 4.6.16: The smallest nullity for a tall matrix . . . 91
Exercise 4.6.17: Five rank statements . . . 91
Exercise 4.6.18: Five statements about row operations . . . 91
Exercise 4.6.19: A one-dimensional kernel of a five-row system . . . 91
Exercise 4.6.20: Two free variables in an eight-variable system . . . 91
Exercise 4.6.21: Onto with just one free direction . . . 92
Exercise 4.6.22: Too many variables for a one-dimensional kernel . . . 92
Exercise 4.6.23: Replace redundant equations . . . 92
Exercise 4.6.24: A unique solution for one right-hand side versus all . . . 92
Exercise 4.6.25: One missing pivot row leaves some constants unattainable . . . 92
Exercise 4.6.26: Full rank for a tall matrix . . . 92
Exercise 4.6.27: Locate the four fundamental spaces . . . 92
Exercise 4.6.28: Two rank–nullity identities . . . 93
Exercise 4.6.29: Onto is equivalent to a trivial transpose kernel . . . 93
Exercise 4.6.30: The augmented-rank consistency criterion . . . 93
Exercise 4.6.31: An outer product has at most one independent column . . . 93
Exercise 4.6.32: Factor a rank-one matrix . . . 93
Exercise 4.6.33: Every nonzero column generates a rank-one column space . . . 94
Exercise 4.6.34: A rank-r matrix is a sum of r rank-one matrices . . . 94
Exercise 4.6.35: Four fundamental bases and two invertible assembled matrices . . . 94
Exercise 4.6.36: Four fundamental bases and two invertible assembled matrices . . . 95
Exercise 4.6.37: Reconstruct the original matrix from pivot columns . . . 97
Exercise 4.6.38: Three generated rank factorizations and a general proof . . . 97
4.7 Change of basis . . . 99
Practice problem 4.7.P1: Read the coordinate labels on the columns . . . 99
Practice problem 4.7.P2: Reverse the change from the example . . . 99

Exercise 4.7.1: Construct columns directly from the given equations . . . 99
Exercise 4.7.2: A singular matrix reveals a printed inconsistency . . . 99
Exercise 4.7.3: The output coordinate system matches the columns . . . 100
Exercise 4.7.4: A three-vector version of the same rule . . . 100
Exercise 4.7.5: Translate three linear-combination equations . . . 100
Exercise 4.7.6: Change from F to D . . . 100
Exercise 4.7.7: Compute both directions in standard coordinates . . . 101
Exercise 4.7.8: Compute both directions in standard coordinates . . . 101
Exercise 4.7.9: Compute both directions in standard coordinates . . . 101
Exercise 4.7.10: Compute both directions in standard coordinates . . . 102
Exercise 4.7.11: Two statements about the direction of a change . . . 102
Exercise 4.7.12: Independence and an algorithm's direction . . . 102
Exercise 4.7.13: Change from a polynomial basis to coefficients . . . 102
Exercise 4.7.14: Represent t^2 using three different polynomials . . . 103
Exercise 4.7.15: Justify the construction of the change matrix . . . 103
Exercise 4.7.16: Justify uniqueness of the change matrix . . . 103
Exercise 4.7.17: Change between powers of cosine and multiple angles . . . 103
Exercise 4.7.18: Integrate after changing the function basis . . . 104
Exercise 4.7.19: Find bases when a change matrix is prescribed . . . 105
Exercise 4.7.20: Compose changes in the same order as their actions . . . 105
4.8 Applications to difference equations . . . 106
Practice problem 4.8.P1: A three-signal basis . . . 106
Exercise 4.8.1: Substitute two exponential signals . . . 106
Exercise 4.8.2: Two roots with opposite signs . . . 106
Exercise 4.8.3: Show the first two verified solutions span . . . 106
Exercise 4.8.4: A basis for the second equation . . . 106
Exercise 4.8.5: A repeated root produces a factor of k . . . 107
Exercise 4.8.6: A cosine-sine pair . . . 107
Exercise 4.8.7: Test the three supplied solutions by initial values . . . 107
Exercise 4.8.8: Test the three supplied solutions by initial values . . . 107
Exercise 4.8.9: Test the three supplied solutions by initial values . . . 107
Exercise 4.8.10: Test the three supplied solutions by initial values . . . 108
Exercise 4.8.11: Too few solutions for a third-order equation . . . 108
Exercise 4.8.12: Too few solutions for a fourth-order equation . . . 108

Exercise 4.8.13: Two distinct fractional roots . . . 108
Exercise 4.8.14: Two positive integer roots . . . 108
Exercise 4.8.15: A leading coefficient different from one . . . 108
Exercise 4.8.16: A familiar difference equation . . . 109
Exercise 4.8.17: Income approaches its equilibrium . . . 109
Exercise 4.8.18: A second stable income model . . . 109
Exercise 4.8.19: Two real roots for the bending-moment recurrence . . . 109
Exercise 4.8.20: Impose the two boundary values . . . 110
Exercise 4.8.21: Compute and plot a moving average . . . 110
Exercise 4.8.22: A filter separates two frequencies . . . 110
Exercise 4.8.23: A loan paid off by a final smaller payment . . . 111
Exercise 4.8.24: Monthly deposits and compound interest . . . 112
Exercise 4.8.25: A quadratic particular solution . . . 113
Exercise 4.8.26: A linear particular solution . . . 113
Exercise 4.8.27: Another linear particular solution . . . 113
Exercise 4.8.28: An affine particular solution with roots of magnitude five . . . 113
Exercise 4.8.29: A fourth-order equation as a first-order vector recurrence . . . 113
Exercise 4.8.30: A third-order companion system . . . 114
Exercise 4.8.31: Remove an irrelevant common shift . . . 114
Exercise 4.8.32: The order depends on the last nonzero coefficient . . . 114
Exercise 4.8.33: Independence does not require an invertible Casorati sample for arbitrary signals . . . 114
Exercise 4.8.34: Sampling can destroy independence . . . 114
Exercise 4.8.35: A linear difference operator on signals . . . 115
Exercise 4.8.36: Add a homogeneous solution to a particular solution . . . 115
Exercise 4.8.37: One-sided shifts have only a one-sided inverse . . . 115
4.9 Applications to Markov chains . . . 116
Practice problem 4.9.P1: A city resident's next two locations . . . 116
Practice problem 4.9.P2: Test the steady-state equation rather than its appearance . . . 116
Practice problem 4.9.P3: A long-run migration percentage . . . 116
Exercise 4.9.1: Radio listeners at two station breaks . . . 116
Exercise 4.9.2: Three equally treated food choices . . . 117
Exercise 4.9.3: Healthy and ill students . . . 117
Exercise 4.9.4: Good, indifferent, and bad weather . . . 117

Exercise 4.9.5: Normalize a two-state balance equation . . . 118
Exercise 4.9.6: A second two-state equilibrium . . . 118
Exercise 4.9.7: Three-state balance with two useful equations . . . 118
Exercise 4.9.8: A steady state despite zero transition entries . . . 118
Exercise 4.9.9: One zero can disappear under squaring . . . 118
Exercise 4.9.10: An absorbing state prevents regularity . . . 119
Exercise 4.9.11: Long-run radio preferences . . . 119
Exercise 4.9.12: No long-run preference between symmetric foods . . . 119
Exercise 4.9.13: The long-run illness rate . . . 119
Exercise 4.9.14: Long-run good weather . . . 119
Exercise 4.9.15: A migration percentage in the long run . . . 119
Exercise 4.9.16: Allocate a car fleet across three locations . . . 120
Exercise 4.9.17: A stochastic matrix has eigenvalue one . . . 120
Exercise 4.9.18: An explicit steady state for every two-state matrix . . . 120
Exercise 4.9.19: A row of ones records total probability . . . 120
Exercise 4.9.20: The square is stochastic . . . 121
Exercise 4.9.21: Observe convergence of all columns of powers . . . 121
Exercise 4.9.22: A reproducible comparison of two numerical methods . . . 122

4.S Chapter 4 supplementary exercises . . . 124

Exercise 4.S.1: Twenty statements tying the chapter together . . . 124
Exercise 4.S.2: A parametrization has a redundant generator . . . 125
Exercise 4.S.3: Describe a plane by a homogeneous equation . . . 125
Exercise 4.S.4: A scalar multiplier cannot depend on the input variable . . . 125
Exercise 4.S.5: Delete redundant polynomials from the given list . . . 125
Exercise 4.S.6: A basis of a known two-dimensional span . . . 126
Exercise 4.S.7: The homogeneous information that certifies every right-hand side . . . 126
Exercise 4.S.8: Equal finite dimensions force equality of nested spaces . . . 126
Exercise 4.S.9: Rank and nullity of injective and surjective maps . . . 126
Exercise 4.S.10: A maximal independent set must span . . . 126
Exercise 4.S.11: A minimal finite spanning set is independent . . . 127
Exercise 4.S.12: Rank cannot increase beyond either factor's rank . . . 127
Exercise 4.S.13: An invertible left factor preserves rank . . . 127
Exercise 4.S.14: An invertible right factor preserves rank . . . 127
Exercise 4.S.15: A zero product limits two ranks . . . 127

Exercise 4.S.16: Rank is subadditive 127
Exercise 4.S.17: Find an invertible square submatrix of full rank 128
Exercise 4.S.18: Expand four controlled time steps 128
Exercise 4.S.19: Check the rank of the controllability matrix 128
Exercise 4.S.20: Check the rank of the controllability matrix 129
Exercise 4.S.21: Check the rank of the controllability matrix 129
Exercise 4.S.22: Check the rank of the controllability matrix 129

4.1 Vector spaces and subspaces

A vector need not be an arrow: a polynomial, a matrix, or a function can be a vector when addition and scalar multiplication satisfy the vector-space axioms. A subset H of a known vector space is a subspace when (i) it contains the zero vector, (ii) $u + v \in H$ for *every* $u, v \in H$, and (iii) $cu \in H$ for *every* $u \in H$ and real scalar c. One counterexample disproves closure; one successful example does not prove it. A span is automatically a subspace.

Practice problem 4.1.P1: An affine line fails scalar closure

What we must show. For $H = \{(3s, 2 + 5s) : s \in \mathbb{R}\}$, find a vector whose scalar multiple leaves H. Take $s = 0$. Then $u = (0, 2) \in H$. Its multiple $2u = (0, 4)$ is not in H: the equation $3s = 0$ would require $s = 0$, but then the second coordinate would be 2, not 4. Thus scalar multiplication is not closed on H.

Conclusion. H is not a vector space under the usual operations.

Practice problem 4.1.P2: Every generator belongs to its span

For $W = \text{Span}\{v_1, \ldots, v_p\}$ and a fixed k between 1 and p, write

$$v_k = 0v_1 + \cdots + 0v_{k-1} + 1v_k + 0v_{k+1} + \cdots + 0v_p.$$

This is a permitted linear combination of the generators, hence $v_k \in W$. In particular, choose $k = 1$ for the first part of the question. A span contains all its generators, even if some are zero or repeated.

Exercise 4.1.1: The first quadrant

(a) Addition preserves both inequalities.

If $u = (u_1, u_2)$ and $v = (v_1, v_2)$ have nonnegative coordinates, then $u_1 + v_1 \geq 0$ and $u_2 + v_2 \geq 0$. Therefore $u + v$ is still in the first quadrant.

(b) Negative scalars reveal the obstruction.

The vector $u = (1, 0)$ belongs to the set, but $(-1)u = (-1, 0)$ does not.

Conclusion. Addition is closed, but scalar multiplication is not; the first quadrant is not a vector space.

Exercise 4.1.2: The first and third quadrants together

(a) Any scalar multiple stays in the set.

Here $W = \{(x, y) : xy \geq 0\}$. If $u = (x, y) \in W$, then the product of the coordinates of cu is $(cx)(cy) = c^2xy \geq 0$, since $c^2 \geq 0$. This proves closure under multiplication by every real scalar, including zero.

(b) Addition can leave the set.

Choose $u = (2, 1)$ and $v = (-1, -2)$. Both coordinate products equal $2 > 0$, so both vectors are in W. Their sum is $(1, -1)$, whose coordinate product is $-1 < 0$. Thus $u + v \notin W$.

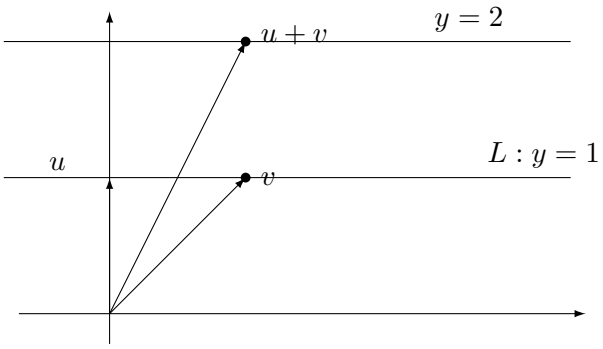
Conclusion. W passes the scalar test but fails the addition test.

Exercise 4.1.3: The closed unit disk

The vector $u = (1, 0)$ is in the disk because $1^2 + 0^2 = 1$. But $2u = (2, 0)$ is outside it, since $2^2 + 0^2 = 4 > 1$. Thus the disk is not closed under scalar multiplication and is not a subspace. It is not enough that the disk contains zero: closure must hold for arbitrarily large as well as small scalars.

Exercise 4.1.4: A translated line is not closed under addition

What we must show. Supply a geometric counterexample, rather than just assert that a line misses zero. Consider the line $L = \{(x, 1) : x \in \mathbb{R}\}$. Take $u = (0, 1)$ and $v = (1, 1)$. Both endpoints lie on L, but $u + v = (1, 2)$ lies on the parallel line $y = 2$. The parallelogram construction for adding u and v therefore ends off L.



More generally write any line not through zero as $p + \text{Span}\{d\}$, where $p \notin \text{Span}\{d\}$. Then $p \in L$ but $2p \notin L$, for otherwise $2p = p + td$ would imply $p = td$, a contradiction. Adding p to itself already fails.

Exercise 4.1.5: Multiples of a single polynomial

The set $\{at^2 : a \in \mathbb{R}\}$ equals $\text{Span}\{t^2\}$, a subspace of P_2 (and of P_n for every $n \geq 2$). Explicitly its zero is obtained with $a = 0$, $at^2 + bt^2 = (a + b)t^2$ stays in the set, and $c(at^2) = (ca)t^2$ stays in the set.

Conclusion. It is a subspace.

Exercise 4.1.6: A fixed quadratic coefficient

The set consists of polynomials $a + t^2$, whose coefficient of t^2 is always 1. It does not contain the zero polynomial: no choice of a makes $a + t^2$ zero for every t. Equivalently multiplying t^2 by zero produces a polynomial outside the set. Thus this subset of P_2 is not a subspace.

Exercise 4.1.7: Integer coefficients and real scalars

The constant polynomial $p(t) = 1$ has integer coefficients and degree at most 3. The real scalar $1/2$ is permitted in the ambient vector space, but $(1/2)p(t) = 1/2$ does not have integer coefficients. Scalar closure fails.

Conclusion. This set is not a real subspace of P_3. The choice of scalar field matters.

Exercise 4.1.8: Polynomials vanishing at zero

Let $H = \{p \in P_n : p(0) = 0\}$. The zero polynomial is in H. If $p, q \in H$, then $(p+q)(0) = p(0)+q(0) = 0$. For real c, $(cp)(0) = cp(0) = 0$. These equations prove both closure properties; sums and scalar multiples still have degree at most n.

Conclusion. H is a subspace. For $n \geq 1$ it is $\text{Span}\{t, t^2, \ldots, t^n\}$; for $n = 0$ it is just the zero polynomial.

Exercise 4.1.9: A one-parameter line in three dimensions

Factor out the arbitrary real parameter:

$$(-2t, 5t, 3t) = t(-2, 5, 3).$$

Thus choose $v = (-2, 5, 3)$, and $H = \text{Span}\{v\}$. This is equality in both directions: every displayed vector is a multiple of v, and every real multiple of v is obtained by choosing that number as t. A span is a subspace, so H is a subspace of $\mathbb{R}^3$.

Exercise 4.1.10: A line with a zero middle coordinate

Here $(3t, 0, -7t) = t(3, 0, -7)$. As t runs through all real numbers, these are exactly all linear combinations of the single vector $(3, 0, -7)$. Consequently $H = \text{Span}\{(3, 0, -7)\}$ is a subspace of $\mathbb{R}^3$. The zero middle coordinate does not make it a different kind of vector space.

Exercise 4.1.11: Separate two free parameters

For arbitrary $b, c \in \mathbb{R}$,

$$\begin{pmatrix} 2b + 3c \\ -b \\ 2c \end{pmatrix} = b \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} + c \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}.$$

Take $u = (2, -1, 0)$ and $v = (3, 0, 2)$. Every original vector is in their span, and an arbitrary linear combination $bu + cv$ has exactly the required form. Therefore $W = \text{Span}\{u, v\}$, and spans are subspaces of the ambient vector space.

Exercise 4.1.12: Separate two parameters in four dimensions

Collect the coefficients of s and t in each coordinate:

$$\begin{pmatrix} 2s+4t \\ 2s \\ 2s-3t \\ 5t \end{pmatrix} = s\begin{pmatrix} 2 \\ 2 \\ 2 \\ 0 \end{pmatrix} + t\begin{pmatrix} 4 \\ 0 \\ -3 \\ 5 \end{pmatrix}.$$

Since s, t are unrestricted, W is precisely the span of the two displayed constant vectors. The span theorem proves that W is a subspace of $\mathbb{R}^4$.

Exercise 4.1.13: A set of generators versus its span

What we must show. Distinguish membership in a finite set from membership in all its linear combinations. The given vectors are $v_1 = (1, 0, -1)$, $v_2 = (2, 1, 3)$, $v_3 = (4, 2, 6)$, and $w = (3, 1, 2)$.

(a) Inspect the three individual vectors.

The vector w equals none of v_1, v_2, v_3, so $w \notin \{v_1, v_2, v_3\}$. Those three vectors are distinct, so that set has exactly three elements.

(b) Count the span.

Since $v_1 \neq 0$, the multiples tv_1 give distinct vectors for distinct real t. Thus the span contains infinitely many vectors.

(c) Find an actual combination.

Adding coordinates gives $v_1 + v_2 = (1+2, 0+1, -1+3) = (3, 1, 2) = w$. Hence w belongs to the span even though it is not one of its three generators.

Exercise 4.1.14: Another vector in the same plane

Use the generators from the preceding exercise and $w = (1, 3, 14)$. Because $v_3 = 2v_2$, combinations of the first two vectors suffice. Set $av_1 + bv_2 = w$. Coordinate comparison gives

$$a + 2b = 1, \qquad b = 3, \qquad -a + 3b = 14.$$

The middle equation gives $b = 3$, the first gives $a = -5$, and the last checks: $-(-5) + 3(3) = 14$. Therefore $w = -5v_1 + 3v_2$ lies in the required span.

Exercise 4.1.15: A coordinate forced to be negative one

Every vector in the set has second coordinate -1. The zero vector has second coordinate zero and is therefore absent. A vector space must contain zero, so no spanning set for this W exists: W is not a vector space.

Exercise 4.1.16: A coordinate forced to be one

Every vector has first coordinate 1, regardless of the values of a, b. Thus zero is absent. This violates the first subspace condition and shows that W is not a vector space under the inherited operations.

Exercise 4.1.17: Three parameters, each with its own coefficient vector

Expanding coordinate by coordinate gives

$$\begin{pmatrix} 2a-b \\ 3b-c \\ 3c-a \\ 3b \end{pmatrix} = a\begin{pmatrix} 2 \\ 0 \\ -1 \\ 0 \end{pmatrix} + b\begin{pmatrix} -1 \\ 3 \\ 0 \\ 3 \end{pmatrix} + c\begin{pmatrix} 0 \\ -1 \\ 3 \\ 0 \end{pmatrix}.$$

With all three parameters arbitrary, W equals the span of these three vectors, so it is a subspace of $\mathbb{R}^4$. These vectors form a valid spanning set; one need not prove independence merely to supply a spanning set.

Exercise 4.1.18: A permanently zero coordinate is harmless

The required decomposition is

$$\begin{pmatrix} 4a+3b \\ 0 \\ a+3b+c \\ 3b-2c \end{pmatrix} = a\begin{pmatrix} 4 \\ 0 \\ 1 \\ 0 \end{pmatrix} + b\begin{pmatrix} 3 \\ 0 \\ 3 \\ 3 \end{pmatrix} + c\begin{pmatrix} 0 \\ 0 \\ 1 \\ -2 \end{pmatrix}.$$

Every such combination has the specified form and vice versa. Thus the three coefficient vectors span W, which is a subspace. A coordinate fixed at zero is compatible with subspaces; a coordinate fixed at nonzero 1 was not.

Exercise 4.1.19: The space of oscillating functions

Fix ω and let $f(t) = \cos(\omega t)$ and $g(t) = \sin(\omega t)$. The set of functions $c_1 f + c_2 g$, with arbitrary real c_1, c_2, is exactly $\mathrm{Span}\{f, g\}$ inside the vector space of all real-valued functions on $\mathbb{R}$. Hence it is a subspace. Explicitly, adding two functions adds their two coefficients, and scalar multiplication multiplies both coefficients; neither operation changes the fixed frequency ω. Choosing both coefficients zero gives the zero function. The span argument remains valid if $\omega = 0$.

Exercise 4.1.20: Continuous functions and an endpoint condition

(a) State the three required facts.

The function identically zero on $[a, b]$ is continuous. The sum of any two continuous functions on $[a, b]$ is continuous, and every real scalar multiple of a continuous function there is continuous. These are the subspace tests, so $C[a, b]$ is a subspace of all real-valued functions on that interval.

(b) Preserve equality of the endpoint values.

Let $H = \{f \in C[a,b] : f(a) = f(b)\}$. The zero function is in H. If $f, g \in H$, then

$$(f+g)(a) = f(a) + g(a) = f(b) + g(b) = (f+g)(b).$$

For every real c, $(cf)(a) = cf(a) = cf(b) = (cf)(b)$. Both new functions are continuous by part (a), so both belong to H.

Conclusion. H is a subspace of $C[a,b]$.

Exercise 4.1.21: Upper triangular matrices form a subspace

Take $H = \{\left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right) : a, b, d \in \mathbb{R}\}$. Zero is obtained with $a = b = d = 0$. Adding two such matrices still gives a zero entry in position $(2,1)$, and multiplying by a scalar preserves that zero entry. Therefore H is a subspace of $M_{2\times 2}$. One explicit spanning set is

$$\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}.$$

Exercise 4.1.22: A homogeneous matrix equation

Let F be the fixed 3×2 matrix, and let $H = \{A \in M_{2\times 4} : FA = 0\}$. The zero 2×4 matrix is in H, because multiplying it by F gives the zero 3×4 matrix. For $A, B \in H$, distributivity gives $F(A+B) = FA + FB = 0$. For real c, $F(cA) = c(FA) = 0$. Thus both $A + B$ and cA belong to H.

Conclusion. H is a subspace of $M_{2\times 4}$. The matrix sizes ensure each product is defined.

Exercise 4.1.23: Five statements about vectors and subspaces

(a) **False.** A zero function must vanish at *every* input. For example $f(t) = t$ has $f(0) = 0$ but is not the zero function.

(b) **False.** Vectors can be functions, matrices, or polynomials, not only arrows in three-dimensional space. An element of P_2, such as t^2, is a vector in that vector space.

(c) **False.** Containing zero is necessary but insufficient. The unit disk contains zero but fails scalar closure, as Exercise 4.1.3 shows.

(d) **True.** A subspace inherits the vector-space operations and all axioms; the subspace conditions ensure the operations stay inside it.

(e) **False.** The chapter introduction describes the major shuttle control systems in terms of *digital*, not analog, signals.

Exercise 4.1.24: Five more statements, including the closure quantifiers

(a) **True.** A vector means an element of a vector space; its physical appearance is irrelevant.

(b) **True.** The identity $(-1)u = -u$ follows from distributivity and uniqueness of the additive inverse; Exercise 4.1.29 supplies the proof.

(c) **True.** Every vector space is a subspace of itself, using its own zero vector and the two operations.

(d) **False.** As defined here, elements of $\mathbb{R}^2$ are ordered pairs, not ordered triples in $\mathbb{R}^3$. The plane $\{(x, y, 0) : x, y \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^3$ isomorphic to $\mathbb{R}^2$, but it is a different set.

(e) **False as written.** The stated conditions merely say that $u, v, u + v$ are in H and that cu is in H for particular choices. Closure requires these conclusions for *all* $u, v \in H$ and *all* real c. For instance $H = \{(0, 0), (1, 0)\}$ satisfies the listed conditions with $u = v = 0$ and $c = 0$ but is not closed under multiplication by 2. If the omitted universal quantifiers are supplied, the resulting statement is exactly the correct subspace test.

Exercise 4.1.25: Uniqueness of the zero vector

The missing axiom is **4**, the additive-identity axiom (with the commutative form justified by Axiom 2, as the exercise specifies). If w is another additive identity, then using it as an identity gives $0 + w = 0$, whereas using the original zero as an identity gives $0 + w = w$. The same expression equals both 0 and w, hence $w = 0$.

Exercise 4.1.26: Uniqueness of the additive inverse

The blanks are **(a) 3, (b) 5, (c) 4**. To see each step, suppose $u + w = 0$ and add $-u$ to both sides:

$$(-u) + (u + w) = (-u) + 0.$$

Associativity (Axiom 3) changes the left side to $((-u) + u) + w$. The inverse axiom (Axiom 5, with commutativity) changes $(-u)+u$ to zero. The identity axiom (Axiom 4) then changes $0+w = (-u)+0$ to $w = -u$.

Conclusion. There can be only one additive inverse of a given vector.

Exercise 4.1.27: Multiplication by the scalar zero

The blanks are **(a) 8, (b) 3, (c) 5, (d) 4**. Distributivity over scalar addition (Axiom 8) gives $0u = (0 + 0)u = 0u + 0u$. Add the additive inverse $-(0u)$ to both sides. Associativity (3) groups the last two terms on the right as $0u + (-(0u))$; the inverse axiom (5) changes each such pair to zero. The result is $0 = 0u + 0$, which the identity axiom (4) reduces to $0 = 0u$. Here the first zero in $0u$ is a scalar; the zero on the right is a vector.

Exercise 4.1.28: A scalar times the zero vector

The blanks are **(a) 4, (b) 7, (c) 3, (d) 5, (e) 4**. First $0 = 0 + 0$ by the identity axiom (4). Thus $c0 = c(0 + 0) = c0 + c0$ by distributivity over vector addition (7). Add the inverse of $c0$ to both sides. Associativity (3) allows the right-hand inverse pair to be grouped; Axiom 5 reduces the inverse pairs to zero; and Axiom 4 removes the remaining added zero. Hence $0 = c0$. This holds for every scalar c, including negative and zero scalars.

Exercise 4.1.29: The scalar negative one gives the additive inverse

Axiom 10 says $u = 1u$. Axiom 8 then gives

$$u + (-1)u = 1u + (-1)u = (1 + (-1))u = 0u = 0,$$

where the last equality is Exercise 4.1.27. Therefore $(-1)u$ is an additive inverse of u. By uniqueness of that inverse (Exercise 4.1.26), $(-1)u = -u$.

Exercise 4.1.30: A nonzero scalar cannot annihilate a nonzero vector

Suppose $cu = 0$ and $c \neq 0$. Then $1/c$ is an allowed scalar. Multiply both sides by it and use Axioms 9 and 10 and Exercise 4.1.28:

$$u = 1u = ((1/c)c)u = (1/c)(cu) = (1/c)0 = 0.$$

The nonzero hypothesis is needed to form $1/c$; when $c = 0$, the equation $cu = 0$ holds for every u and does not force $u = 0$.

Exercise 4.1.31: The span is the smallest containing subspace

Suppose the subspace H contains u and v. An arbitrary member of $\text{Span}\{u, v\}$ has form $au + bv$ for real a, b. Scalar closure puts au, bv in H, and additive closure puts $au+bv$ in H. Thus $\text{Span}\{u, v\} \subseteq H$. Since the span itself is a subspace that contains u, v, it is the smallest such subspace, in the sense of inclusion.

Exercise 4.1.32: Intersection works; union can fail

Check the intersection's three properties.

Both H and K contain zero, so their intersection does too. If $u, v \in H \cap K$, then they belong to both subspaces; hence $u + v$ belongs to both, so $u + v \in H \cap K$. Likewise any scalar multiple cu belongs to both, hence to their intersection. Thus $H \cap K$ is a subspace.

Give a concrete counterexample for a union.

Let $H = \{(x, 0) : x \in \mathbb{R}\}$ and $K = \{(0, y) : y \in \mathbb{R}\}$, the two coordinate axes. Their union contains $(1, 0)$ and $(0, 1)$, but not their sum $(1, 1)$. Thus a union of subspaces need not be a subspace.

Exercise 4.1.33: The sum of two subspaces

(a) Verify closure, keeping each summand in its own subspace.

Zero is $0 + 0$ with the first zero in H and the second in K. For $x = h_1 + k_1$ and $y = h_2 + k_2$ in $H + K$, associativity and commutativity give

$$x + y = (h_1 + h_2) + (k_1 + k_2) \in H + K,$$

since $h_1 + h_2 \in H$ and $k_1 + k_2 \in K$. For a real scalar c, $cx = ch_1 + ck_1 \in H + K$ by both scalar closures. Therefore $H + K$ is a subspace of V.

(b) Put the missing summand equal to zero.

If $h \in H$, then $h = h + 0$ with $0 \in K$, so $H \subseteq H + K$. If $k \in K$, write $k = 0 + k$ to obtain $K \subseteq H + K$. With their inherited operations, both are subspaces of $H + K$.

Exercise 4.1.34: Spanning the sum by combining the generators

Let $H = \text{Span}\{u_1, \ldots, u_p\}$ and $K = \text{Span}\{v_1, \ldots, v_q\}$.

Every element of the sum belongs to the combined span.

If $x = h + k \in H + K$, write $h = \sum_i a_i u_i$ and $k = \sum_j b_j v_j$. Then $x = \sum_i a_i u_i + \sum_j b_j v_j$, a combination of the combined list.

Every element of the combined span belongs to the sum.

Conversely split any combination of that combined list into its u-part $h \in H$ and its v-part $k \in K$. It then equals $h + k \in H + K$. These two inclusions prove equality of the sets.

Exercise 4.1.35: A checked representation in four dimensions

What we must show. Express $w = (9, -4, -4, 7)$ as a combination of the given v_1, v_2, v_3. Set $av_1 + bv_2 + cv_3 = w$. The augmented system and its reduced form are

$$\left[\begin{array}{rrr|r} 8 & -4 & -7 & 9 \\ -4 & 3 & 6 & -4 \\ -3 & -2 & -5 & -4 \\ 9 & -8 & -18 & 7 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

The three coefficient values are $1, -2, 1$. Rather than trust the reduction alone, check them in the original vectors:

$$\begin{pmatrix} 8 \\ -4 \\ -3 \\ 9 \end{pmatrix} - 2\begin{pmatrix} -4 \\ 3 \\ -2 \\ -8 \end{pmatrix} + \begin{pmatrix} -7 \\ 6 \\ -5 \\ -18 \end{pmatrix} = \begin{pmatrix} 8 + 8 - 7 \\ -4 - 6 + 6 \\ -3 + 4 - 5 \\ 9 + 16 - 18 \end{pmatrix} = \begin{pmatrix} 9 \\ -4 \\ -4 \\ 7 \end{pmatrix}.$$

Conclusion. $w = v_1 - 2v_2 + v_3$ belongs to the stated span.

Exercise 4.1.36: Test membership in a column span

The columns of A span precisely the vectors Ax. Therefore solve $Ax = y$:

$$\left[\begin{array}{rrr|r} 3 & -5 & -9 & -4 \\ 8 & 7 & -6 & -8 \\ -5 & -8 & 3 & 6 \\ 2 & -2 & -9 & -5 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1/5 \\ 0 & 1 & 0 & -2/5 \\ 0 & 0 & 1 & 3/5 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

There is no inconsistent row. The candidate is $x = (-1/5, -2/5, 3/5)$. Checking the original multiplication gives

$$Ax = \tfrac{1}{5}(-3 + 10 - 27,\ -8 - 14 - 18,\ 5 + 16 + 9,\ -2 + 4 - 27) = (-4, -8, 6, -5) = y.$$

Conclusion. y is in the subspace spanned by the columns of A.

Exercise 4.1.37: Recognizing two multiple-angle functions

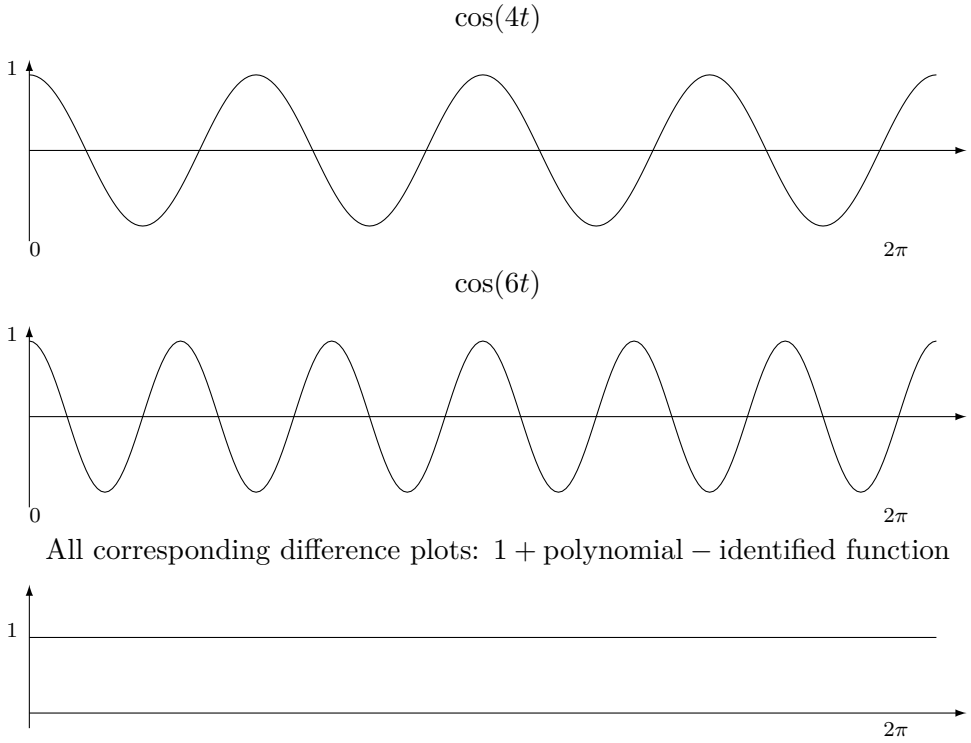
What we must show. Identify the two given polynomials in $\cos t$, then verify them algebraically and graphically. Write $c = \cos t$. The double-angle formula gives $\cos 2t = 2c^2 - 1$. Apply it a second time:

$$\cos 4t = 2(2c^2 - 1)^2 - 1 = 8c^4 - 8c^2 + 1 = f(t).$$

For the other polynomial use $\cos 3u = 4\cos^3 u - 3\cos u$ with $u = 2t$:

$$\cos 6t = 4(2c^2 - 1)^3 - 3(2c^2 - 1) = 32c^6 - 48c^4 + 18c^2 - 1 = g(t).$$

Thus the first graph has four full cosine oscillations on $[0, 2\pi]$ and the second has six. The requested difference graphs are exactly $1 + f(t) - \cos 4t = 1$ and $1 + g(t) - \cos 6t = 1$. A numerical graph supports the recognition, while the displayed identities prove it for every t.



Exercise 4.1.38: Recognizing three polynomials in sine

Put $s = \sin t$. The imaginary part of $(\cos t + i \sin t)^3 = \cos 3t + i \sin 3t$ gives

$$\sin 3t = 3s\cos^2 t - s^3 = 3s(1 - s^2) - s^3 = 3s - 4s^3 = f(t).$$

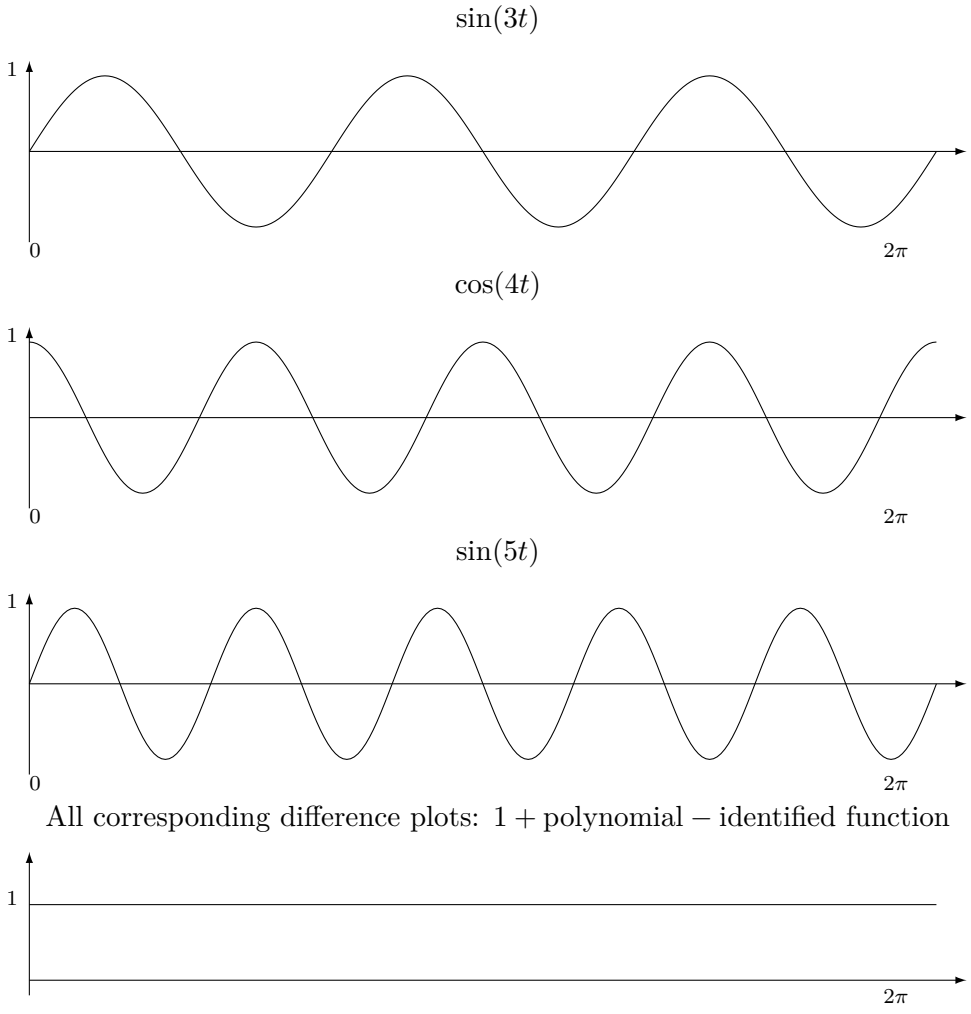
The double-angle identity $\cos 2t = 1 - 2s^2$ gives

$$\cos 4t = 2(1 - 2s^2)^2 - 1 = 1 - 8s^2 + 8s^4 = g(t).$$

Likewise the imaginary part of the fifth power gives

$$\begin{aligned}\sin 5t &= 5\cos^4 t\, s - 10\cos^2 t\, s^3 + s^5 \\ &= 5(1 - s^2)^2 s - 10(1 - s^2)s^3 + s^5 = 5s - 20s^3 + 16s^5 = h(t).\end{aligned}$$

(The power identities used here also follow by repeated angle-addition formulas.) The plots have three sine cycles, four cosine cycles, and five sine cycles, respectively, on $[0, 2\pi]$. Subtracting each identified function from one plus its original polynomial gives the constant 1, as requested.



4.2 Null spaces, column spaces, and linear transformations

For an $m \times n$ matrix A, the null space $\operatorname{Nul} A$ contains the *inputs* $x \in \mathbb{R}^n$ satisfying $Ax = 0$. The column space $\operatorname{Col} A$ contains the *outputs* $Ax \in \mathbb{R}^m$. Do not confuse these two roles. To test $w \in \operatorname{Nul} A$, multiply Aw and compare with zero. To test $w \in \operatorname{Col} A$, solve $Ax = w$ and ask whether a solution exists. A homogeneous linear equation defines a subspace; a nonhomogeneous equation usually describes a translated subspace and does not even contain zero. For a linear map $T : V \to W$, the corresponding terms are kernel $\{v : T(v) = 0\}$ and range $\{T(v) : v \in V\}$.

Practice problem 4.2.P1: Two descriptions of one plane

For $W = \{(a, b, c) : a - 3b - c = 0\}$, first observe that $W = \operatorname{Nul}[\,1 \;\; -3 \;\; -1\,]$. The null space theorem therefore makes it a subspace of $\mathbb{R}^3$. Alternatively solve $a = 3b + c$. Then

$$(a, b, c) = b(3, 1, 0) + c(1, 0, 1).$$

As b, c vary freely, this proves $W = \operatorname{Span}\{(3, 1, 0), (1, 0, 1)\}$, a subspace by the span theorem. The first description is implicit (an equation); the second is explicit (generators and arbitrary coefficients).

Practice problem 4.2.P2: Add two attainable outputs

Suppose $Ax = v$ and $Ay = w$ have solutions x, y. By distributivity, $A(x + y) = Ax + Ay = v + w$, so the equation with right-hand side $v + w$ is also consistent. Equivalently $v, w \in \operatorname{Col} A$, and the column space is closed under addition. We need not find either of the original solutions.

Exercise 4.2.1: Test a proposed null-space vector directly

What we must show. Decide whether $w = (1, 3, -4)$ is in $\operatorname{Nul} A$ for the displayed matrix. Multiply using the original rows, retaining the signs:

$$\begin{pmatrix} 3 & -5 & -3 \\ 6 & -2 & 0 \\ -8 & 4 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix} = \begin{pmatrix} 3 - 15 + 12 \\ 6 - 6 + 0 \\ -8 + 12 - 4 \end{pmatrix} = 0.$$

Every coordinate is zero, so $w \in \operatorname{Nul} A$. A vector belongs to the null space precisely when this multiplication gives zero, not merely when its coordinates resemble a row or column of A.

Exercise 4.2.2: Another direct multiplication

With $w = (1, -1, 1)$, compute

$$\begin{pmatrix} 2 & 6 & 4 \\ -3 & 2 & 5 \\ -5 & -4 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 - 6 + 4 \\ -3 - 2 + 5 \\ -5 + 4 + 1 \end{pmatrix} = 0.$$

Thus $w \in \operatorname{Nul} A$. All three equations must hold; checking only the first row would not have been enough.

Exercise 4.2.3: Describe a null space using two free variables

The homogeneous system for $A = \left(\begin{smallmatrix} 1 & 2 & 4 & 0 \\ 0 & 1 & 3 & -2 \end{smallmatrix}\right)$ is

$$x_1 + 2x_2 + 4x_3 = 0, \qquad x_2 + 3x_3 - 2x_4 = 0.$$

The pivot variables are x_1, x_2; choose $x_3 = s, x_4 = t$ freely. The second equation gives $x_2 = -3s + 2t$, and substitution in the first gives $x_1 = -2(-3s + 2t) - 4s = 2s - 4t$. Therefore

$$x = s \begin{pmatrix} 2 \\ -3 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -4 \\ 2 \\ 0 \\ 1 \end{pmatrix}.$$

Both parameters may be any real number, so the two displayed vectors span $\operatorname{Nul} A$; this formula describes every solution and only solutions.

Exercise 4.2.4: A forced zero coordinate and two free coordinates

For $A = \left(\begin{smallmatrix} 1 & -3 & 2 & 0 \\ 0 & 0 & 3 & 0 \end{smallmatrix}\right)$, the second equation is $3x_3 = 0$, so $x_3 = 0$. The first becomes $x_1 = 3x_2$. Letting $x_2 = s$ and $x_4 = t$ gives

$$\operatorname{Nul} A = \left\{ s \begin{pmatrix} 3 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} : s, t \in \mathbb{R} \right\}.$$

In particular the fourth column being zero leaves the fourth input free; it does not require the fourth input to be zero.

Exercise 4.2.5: Back substitution in a five-column matrix

The nonzero rows give

$$x_1 - 4x_2 + 2x_4 = 0, \qquad x_3 - 5x_4 = 0, \qquad 2x_5 = 0.$$

The last equation gives $x_5 = 0$. Set the free variables $x_2 = s, x_4 = t$. Then $x_3 = 5t$ and $x_1 = 4s - 2t$. Thus

$$\operatorname{Nul} A = \operatorname{Span} \left\{ \begin{pmatrix} 4 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 0 \\ 5 \\ 1 \\ 0 \end{pmatrix} \right\}.$$

Each generating vector is obtained by giving one free variable value 1 and the other value 0; their arbitrary combinations preserve both free variables.

Exercise 4.2.6: Three free variables

The matrix gives two effective equations:

$$x_1 + 3x_2 - 4x_3 - 3x_4 + x_5 = 0, \qquad x_2 - 3x_3 + x_4 = 0.$$

The zero third row adds no condition. Set $x_3 = r, x_4 = s, x_5 = t$. Then $x_2 = 3r - s$ and $x_1 = -3(3r - s) + 4r + 3s - t = -5r + 6s - t$. Consequently

$$\operatorname{Nul} A = \operatorname{Span}\left\{\begin{pmatrix}-5\\3\\1\\0\\0\end{pmatrix}, \begin{pmatrix}6\\-1\\0\\1\\0\end{pmatrix}, \begin{pmatrix}-1\\0\\0\\0\\1\end{pmatrix}\right\}.$$

Exercise 4.2.7: An affine plane misses zero

The equation is $a+b+c=2$. At $(a,b,c)=(0,0,0)$ it would read $0=2$, which is false. Hence the zero vector is not in W; W is not a vector space under the usual operations. The fact that the equation is linear in the variables does not make its *nonhomogeneous* solution set a subspace.

Exercise 4.2.8: Move all variable terms to one side

The relation $3r - 2 = 3s + t$ becomes $3r - 3s - t = 2$. Zero does not satisfy this relation, because its left side is zero. Therefore the given set is not a vector space.

Exercise 4.2.9: Two homogeneous constraints define a null space

Rewrite $p - 3q = 4s$ and $2p = s + 5r$ as

$$p - 3q + 0r - 4s = 0, \qquad 2p + 0q - 5r - s = 0.$$

Thus W is the null space of $\left(\begin{smallmatrix}1 & -3 & 0 & -4\\ 2 & 0 & -5 & -1\end{smallmatrix}\right)$, and the null space theorem makes it a subspace of $\mathbb{R}^4$. Both right-hand sides after rearrangement are zero; this is the essential property.

Exercise 4.2.10: Another homogeneous description

Move all terms to the left: $3a + b - c + 0d = 0$ and $a + b + 2c - 2d = 0$. Then

$$W = \operatorname{Nul}\begin{pmatrix}3 & 1 & -1 & 0\\ 1 & 1 & 2 & -2\end{pmatrix}.$$

The null space of a matrix is a subspace of its input space, here $\mathbb{R}^4$. This proves the requested vector-space property without finding parameters.

Exercise 4.2.11: A hidden constant obstructs the zero vector

For $(s - 2t, 3 + 3s, 3s + t, 2s)$ to be zero, its fourth coordinate forces $s = 0$. But then its second coordinate is 3, not zero. No choice of t can fix that contradiction. Thus zero is absent, and W is not a vector space. Merely setting all parameters zero would give only one candidate; the argument above checks that *no* parameters produce zero.

Exercise 4.2.12: Another parametrization that never produces zero

If $(3p-5q, 4q, p, q+1) = 0$, the third coordinate forces $p = 0$ and the second forces $q = 0$. The fourth would then be 1. This contradiction shows that zero cannot occur, so W is not a vector space.

Exercise 4.2.13: A span in disguise

For arbitrary real c, d,

$$(c-6d, d, c) = c(1,0,1) + d(-6,1,0).$$

Both directions of this parametrization hold, so $W = \text{Span}\{(1,0,1), (-6,1,0)\}$. The span theorem makes it a subspace of $\mathbb{R}^3$.

Exercise 4.2.14: A two-generator subspace

Collecting the coefficients of s, t gives

$$(-s+3t, s-2t, 5s-t) = s(-1,1,5) + t(3,-2,-1).$$

Since s, t are unrestricted real numbers, W is exactly the span of these two vectors, hence is a subspace of $\mathbb{R}^3$.

Exercise 4.2.15: Make the coefficient vectors into columns

What we must show. Find a matrix whose column space is the given parametrized set. Separate the coefficients of r, s, t:

$$\begin{pmatrix} 2s+t \\ r-s+2t \\ 3r+s \\ 2r-s-t \end{pmatrix} = r\begin{pmatrix} 0 \\ 1 \\ 3 \\ 2 \end{pmatrix} + s\begin{pmatrix} 2 \\ -1 \\ 1 \\ -1 \end{pmatrix} + t\begin{pmatrix} 1 \\ 2 \\ 0 \\ -1 \end{pmatrix}.$$

Therefore one suitable matrix is

$$A = \begin{pmatrix} 0 & 2 & 1 \\ 1 & -1 & 2 \\ 3 & 1 & 0 \\ 2 & -1 & -1 \end{pmatrix}.$$

By definition its column space contains exactly the linear combinations just listed. Other matrices can have the same column space; uniqueness is not requested.

Exercise 4.2.16: Construct a matrix from three parameters

Here the required equality is

$$\begin{pmatrix} b-c \\ 2b+3d \\ b+3c-3d \\ c+d \end{pmatrix} = b\begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix} + c\begin{pmatrix} -1 \\ 0 \\ 3 \\ 1 \end{pmatrix} + d\begin{pmatrix} 0 \\ 3 \\ -3 \\ 1 \end{pmatrix}.$$

Thus take $A = \begin{pmatrix} 1 & -1 & 0 \\ 2 & 0 & 3 \\ 1 & 3 & -3 \\ 0 & 1 & 1 \end{pmatrix}$. For $x = (b, c, d)$, the product Ax gives the original form, and every such form is obtained from some x. This proves equality with $\operatorname{Col} A$.

Exercise 4.2.17: Count inputs and outputs

The displayed matrix has four rows and two columns, so its mapping is $A : \mathbb{R}^2 \to \mathbb{R}^4$.

Conclusion. (a) $\operatorname{Nul} A$ is a subspace of $\mathbb{R}^2$, so $k = 2$. (b) $\operatorname{Col} A$ is a subspace of $\mathbb{R}^4$, so $k = 4$.

These are ambient-space dimensions, not necessarily the dimensions of the two subspaces.

Exercise 4.2.18: A rectangular matrix with three inputs

The matrix has four rows and three columns. A vector x in Ax must therefore have three coordinates, whereas the output has four.

Conclusion. (a) $k = 3$ for the ambient space of $\operatorname{Nul} A$. (b) $k = 4$ for the ambient space of $\operatorname{Col} A$.

Exercise 4.2.19: Five inputs and two outputs

The matrix has size 2×5, so Ax is defined for $x \in \mathbb{R}^5$ and belongs to $\mathbb{R}^2$.

Conclusion. (a) $\operatorname{Nul} A \subseteq \mathbb{R}^5$, so $k = 5$. (b) $\operatorname{Col} A \subseteq \mathbb{R}^2$, so $k = 2$.

Exercise 4.2.20: A row matrix

Here $A = [\,1 \;\; -3 \;\; 2 \;\; 0 \;\; -5\,]$ has one row and five columns. It maps a five-coordinate input to one real number.

Conclusion. (a) $k = 5$ for $\operatorname{Nul} A$. (b) $k = 1$ for $\operatorname{Col} A$.

Exercise 4.2.21: Exhibit one vector in each space

Using the matrix in Exercise 4.2.17, its second column equals $-2/3$ times its first. Therefore $2a_1 + 3a_2 = 0$, giving the nonzero null vector $(2, 3)$. A direct check is

$$\begin{pmatrix} 6 & -4 \\ -3 & 2 \\ -9 & 6 \\ 9 & -6 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 12 - 12 \\ -6 + 6 \\ -18 + 18 \\ 18 - 18 \end{pmatrix} = 0.$$

The first column $(6, -3, -9, 9)$ is a nonzero member of $\operatorname{Col} A$. It equals $A(1, 0)$, so its membership follows directly from the definition.

Exercise 4.2.22: A null relation among three columns

For the matrix in Exercise 4.2.18, choose $x = (-1, -1, 1)$. Multiplication verifies

$$Ax = \begin{pmatrix} -5+2+3 \\ 1+0-1 \\ 0+2-2 \\ 5-7+2 \end{pmatrix} = 0.$$

Thus this is a nonzero vector of $\operatorname{Nul} A$. The first column $(5, -1, 0, -5)$ is a nonzero vector in $\operatorname{Col} A$, because it is the output $A(1, 0, 0)$.

Exercise 4.2.23: A vector can belong to both spaces

For $A = \left(\begin{smallmatrix} -2 & 4 \\ -1 & 2 \end{smallmatrix}\right)$ and $w = (2, 1)$, the negative of the first column is w. Hence $A(-1, 0) = w$, proving $w \in \operatorname{Col} A$. For null-space membership test the different multiplication:

$$Aw = (-4+4, -2+2) = (0, 0).$$

Thus $w \in \operatorname{Nul} A$ as well. These two statements are compatible; a square matrix's input and output spaces have the same ambient space, though they serve different roles.

Exercise 4.2.24: Two independent membership checks in four dimensions

The matrix and vector are

$$A = \begin{pmatrix} 10 & -8 & -2 & -2 \\ 0 & 2 & 2 & -2 \\ 1 & -1 & 6 & 0 \\ 1 & 1 & 0 & -2 \end{pmatrix}, \qquad w = \begin{pmatrix} 2 \\ 2 \\ 0 \\ 2 \end{pmatrix}.$$

Column-space membership.

The first two columns add to w, so $A(1, 1, 0, 0) = w$. This is a concrete solution of $Ax = w$, proving $w \in \operatorname{Col} A$.

Null-space membership.

Use w itself as the input:

$$Aw = (20 - 16 - 4,\ 4 - 4,\ 2 - 2,\ 2 + 2 - 4) = (0, 0, 0, 0).$$

Conclusion. w belongs to both $\operatorname{Col} A$ and $\operatorname{Nul} A$.

Exercise 4.2.25: Six statements about null spaces and column spaces

(a) **True.** $\operatorname{Nul} A = \{x : Ax = 0\}$ is the definition.

(b) **False.** An $m \times n$ matrix has n input coordinates, so its null space is in $\mathbb{R}^n$, not generally $\mathbb{R}^m$.

(c) **True.** The possible outputs of $x \mapsto Ax$ are precisely all combinations of the columns, hence form $\operatorname{Col} A$.

(d) **False.** One consistent equation only says that this particular b is in the column space. For instance $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $b = (1, 0)$ give a consistent equation but $\operatorname{Col} A$ is just the horizontal axis.

(e) **True.** The kernel contains zero and is closed under addition and scalar multiplication because the map is linear.

(f) **True.** $\operatorname{Col} A = \{Ax : x \in \mathbb{R}^n\}$; this is the output description of the span of the columns.

Exercise 4.2.26: Six related statements about kernels and ranges

(a) **True.** A null space is a subspace of the input space, hence a vector space.

(b) **True.** Columns of an $m \times n$ matrix have m coordinates; their combinations belong to $\mathbb{R}^m$.

(c) **False.** Solutions of $Ax = b$ are inputs x, not outputs. The column space instead contains those right-hand sides b for which a solution exists.

(d) **True.** The kernel of $x \mapsto Ax$ is, by definition, the set of inputs taken to zero, exactly $\operatorname{Nul} A$.

(e) **True.** The range is a subspace of the target vector space: add two outputs or multiply an output by a scalar and linearity produces another output.

(f) **True.** A homogeneous linear differential equation is $L(f) = 0$ for a linear differential operator L. Its solutions are the kernel of L on the vector space of functions with the required derivatives.

Exercise 4.2.27: Scale a known homogeneous solution

The given solution vector is $u = (3, 2, -1)$. The proposed vector is $10u$. The solution set of the homogeneous system is a null space, which is a vector space and is therefore closed under scalar multiplication. Consequently $10u = (30, 20, -10)$ is another solution. This uses no substitution or row operations, as the exercise requests.

Exercise 4.2.28: Scale an attainable right-hand side

Both systems have the same coefficient matrix A. Their right-hand sides are $b = (0, 1, 9)$ and $5b = (0, 5, 45)$, respectively. Since the first is consistent, there is an input u with $Au = b$. Linearity gives $A(5u) = 5Au = 5b$, so the second system is consistent too. Equivalently $b \in \operatorname{Col} A$ and column spaces are closed under scalar multiplication.

Exercise 4.2.29: Prove the column-space subspace theorem

A general member of $\operatorname{Col} A$ has form Ax, where $x \in \mathbb{R}^n$.

(a) Zero is an output.

Take $x = 0$. Then $Ax = A0 = 0$, so the zero vector of $\mathbb{R}^m$ lies in $\operatorname{Col} A$.

(b) Two outputs can be added.

For inputs $x, w \in \mathbb{R}^n$, distributivity gives $Ax + Aw = A(x + w)$. Since $x + w$ is another allowed input, this is an element of $\operatorname{Col} A$.

(c) An output can be multiplied by any scalar.

For real c, the identity $c(Ax) = A(cx)$ expresses it as an output as well.

Conclusion. The three conditions prove that $\operatorname{Col} A$ is a subspace of $\mathbb{R}^m$.

Exercise 4.2.30: The range of a linear transformation is a subspace

Let $R = \{T(x) : x \in V\} \subseteq W$. First $T(0) = 0$, so $0 \in R$. For arbitrary $y = T(x)$ and $z = T(v)$ in R, linearity gives

$$y + z = T(x) + T(v) = T(x + v) \in R.$$

For any real scalar c, $cy = cT(x) = T(cx) \in R$. The inputs $x + v, cx$ are in V because V is a vector space. Thus R passes all subspace tests inside W. One need not assume that T is onto W.

Exercise 4.2.31: Evaluate a quadratic at two inputs

The map is $T(p) = (p(0), p(1))$ from P_2 into $\mathbb{R}^2$.

(a) Check the two linearity identities at both coordinates.

For polynomials p, q and scalar c,

$$T(p + q) = (p(0) + q(0), p(1) + q(1)) = T(p) + T(q), \qquad T(cp) = cT(p).$$

Thus T is linear.

(b) Solve the kernel equations and then describe the range.

Write $p(t) = a + bt + ct^2$. The equations $p(0) = p(1) = 0$ give $a = 0$ and $b + c = 0$, hence $p(t) = c(t^2 - t)$. The kernel is therefore $\operatorname{Span}\{t^2 - t\}$. For any desired output (r, s), the polynomial $p(t) = r + (s - r)t$ satisfies $p(0) = r$ and $p(1) = s$. Therefore the range is all of $\mathbb{R}^2$.

Conclusion. One kernel generator is $t^2 - t$, and T is onto $\mathbb{R}^2$.

Exercise 4.2.32: Repeat the same evaluation twice

Here $T(p) = (p(0), p(0))$. With $p(t) = a + bt + ct^2$, the output is (a, a). It is zero exactly when $a = 0$, leaving b, c arbitrary. Thus

$$\ker T = \operatorname{Span}\{t, t^2\}, \qquad \operatorname{range} T = \{(a, a) : a \in \mathbb{R}\} = \operatorname{Span}\{(1, 1)\}.$$

The output has two coordinates, but those coordinates must be equal; therefore the map is not onto the whole target $\mathbb{R}^2$.

Exercise 4.2.33: Add a matrix to its transpose

Define $T(A) = A + A^T$ on $M_{2\times 2}$.

(a) Verify linearity using transpose rules.

For matrices A, C and real scalar r,

$$T(A+C) = A + C + (A+C)^T = (A + A^T) + (C + C^T), \qquad T(rA) = r(A + A^T).$$

These are exactly the two linearity identities.

(b) Produce a preimage of any symmetric matrix.

If $B^T = B$, choose $A = B/2$. Then $T(A) = B/2 + B^T/2 = B$.

(c) Establish equality of the range with the symmetric matrices.

Every output satisfies $(A + A^T)^T = A^T + A = A + A^T$, so every output is symmetric. Conversely part (b) realizes every symmetric matrix as an output. Both inclusions prove the claimed equality of sets.

(d) Solve the zero-output condition entry by entry.

For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$,

$$T(A) = \begin{pmatrix} 2a & b+c \\ b+c & 2d \end{pmatrix}.$$

This is zero exactly when $a = d = 0$ and $c = -b$. Therefore

$$\ker T = \left\{ b \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} : b \in \mathbb{R} \right\}.$$

Exercise 4.2.34: The antiderivative with a fixed initial value

For continuous f on $[0, 1]$, the specified antiderivative is uniquely

$$T(f)(x) = \int_0^x f(t)\, dt.$$

Existence follows from the Fundamental Theorem of Calculus. For uniqueness, two such antiderivatives differ by a constant, and their value zero at the left endpoint forces that constant to be zero. Linearity of the integral gives $T(f+g)(x) = T(f)(x) + T(g)(x)$ and $T(cf)(x) = cT(f)(x)$ for every x. Thus these are equalities of functions, and T is a linear transformation. If $T(f)$ is the zero function, differentiating it gives $f(x) = 0$ on $(0, 1)$, and continuity gives zero at the endpoints too. Conversely the zero function has zero antiderivative with the prescribed initial value.

Conclusion. The kernel consists only of the zero function.

Exercise 4.2.35: The image of a subspace

Let U be a subspace of V. Since $0 \in U$ and $T(0) = 0$, zero is in $T(U)$. For $y = T(u)$ and $z = T(v)$ with $u, v \in U$, linearity gives $y + z = T(u+v) \in T(U)$ because $u + v \in U$. Likewise $cy = T(cu) \in T(U)$ since $cu \in U$ for every real c. These three tests prove that $T(U)$ is a subspace of W. The conclusion concerns the image of the whole subspace, not just a list of selected outputs.

Exercise 4.2.36: The inverse image of a subspace

Let $U = \{x \in V : T(x) \in Z\}$, where Z is a subspace of W. Since $T(0) = 0 \in Z$, zero belongs to U. If $x, y \in U$, then $T(x), T(y) \in Z$, and $T(x + y) = T(x) + T(y) \in Z$ by additive closure of Z. Thus $x + y \in U$. For real c, $T(cx) = cT(x) \in Z$, so $cx \in U$. Therefore U is a subspace of V. This argument uses the preimage as a set; it does not require T to have an inverse function.

Exercise 4.2.37: An output that is not a zero-output input

The displayed data are

$$A = \begin{pmatrix} 7 & 6 & -4 & 1 \\ -5 & -1 & 0 & -2 \\ 9 & -11 & 7 & -3 \\ 19 & -9 & 7 & 1 \end{pmatrix}, \qquad w = (1, 1, -1, -3).$$

To test null-space membership, multiply:

$$Aw = (7 + 6 + 4 - 3,\ -5 - 1 + 6,\ 9 - 11 - 7 + 9,\ 19 - 9 - 7 - 3) = (14, 0, 0, 0) \neq 0.$$

Therefore $w \notin \operatorname{Nul} A$. For the column space, the second column plus the third minus the fourth is

$$(6, -1, -11, -9) + (-4, 0, 7, 7) - (1, -2, -3, 1) = (1, 1, -1, -3) = w.$$

Thus $A(0, 1, 1, -1) = w$, which proves $w \in \operatorname{Col} A$.

Conclusion. w is in the column space only.

Exercise 4.2.38: Another vector lying in both spaces

Here

$$A = \begin{pmatrix} -8 & 5 & -2 & 0 \\ -5 & 2 & 1 & -2 \\ 10 & -8 & 6 & -3 \\ 3 & -2 & 1 & 0 \end{pmatrix}, \qquad w = (1, 2, 1, 0).$$

First $Aw = (-8 + 10 - 2,\ -5 + 4 + 1,\ 10 - 16 + 6,\ 3 - 4 + 1) = 0$, so $w \in \operatorname{Nul} A$. For the distinct column-space test use $x = (-2, -3, 0, 1)$:

$$Ax = (16 - 15,\ 10 - 6 - 2,\ -20 + 24 - 3,\ -6 + 6) = (1, 2, 1, 0) = w.$$

Conclusion. w belongs to both spaces, verified by two different products.

Exercise 4.2.39: Recover column relations and the entire null space

The given matrix is

$$A = \begin{pmatrix} 5 & 1 & 2 & 2 & 0 \\ 3 & 3 & 2 & -1 & -12 \\ 8 & 4 & 4 & -5 & 12 \\ 2 & 1 & 1 & 0 & -2 \end{pmatrix}, \qquad B = [\, a_1\ a_2\ a_4 \,].$$

(a) Exhibit the two column combinations.

Coordinate-wise verification gives

$$a_3 = \tfrac{1}{3}a_1 + \tfrac{1}{3}a_2, \qquad a_5 = \tfrac{10}{3}a_1 - \tfrac{26}{3}a_2 - 4a_4.$$

Both are combinations of columns of B, so $a_3, a_5 \in \operatorname{Col} B$.

(b) Solve all homogeneous equations.

Exact row reduction gives

$$A \sim \begin{pmatrix} 1 & 0 & 1/3 & 0 & 10/3 \\ 0 & 1 & 1/3 & 0 & -26/3 \\ 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Take the free variables $x_3 = s, x_5 = t$. Then $x_1 = -s/3 - 10t/3$, $x_2 = -s/3 + 26t/3$, and $x_4 = 4t$. Hence

$$\operatorname{Nul} A = \operatorname{Span}\left\{ \begin{pmatrix} -1/3 \\ -1/3 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -10/3 \\ 26/3 \\ 0 \\ 4 \\ 1 \end{pmatrix} \right\}.$$

These formulas also check the two column relations in part (a): moving all terms to one side gives the corresponding zero combinations.

(c) Explain both failures of invertibility separately.

The first null-space generator is nonzero, so T is not one-to-one: it maps that generator and zero to the same output. There are only three pivots among four rows, so the columns do not span $\mathbb{R}^4$; hence T is not onto $\mathbb{R}^4$. Its having five inputs does not by itself make it onto.

Exercise 4.2.40: Find the line where two planes meet

What we must show. Construct a nonzero vector in both spans and show that its span is the entire intersection. Let $w = c_1v_1 + c_2v_2 = c_3v_3 + c_4v_4$. Move the last two terms left and solve the homogeneous system with columns $v_1, v_2, -v_3, -v_4$:

$$\begin{pmatrix} 5 & 1 & -2 & 0 \\ 3 & 3 & 1 & 12 \\ 8 & 4 & -5 & 28 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & -10/3 \\ 0 & 1 & 0 & 26/3 \\ 0 & 0 & 1 & -4 \end{pmatrix}.$$

There is one free variable, $c_4 = t$, and $c_1 = 10t/3$, $c_2 = -26t/3$, $c_3 = 4t$. Taking $t = 1$ gives

$$w = \tfrac{10}{3}(5, 3, 8) - \tfrac{26}{3}(1, 3, 4) = (8, -16, -8).$$

Independently the other plane gives $4(2, -1, 5) + (0, -12, -28) = (8, -16, -8)$, confirming both memberships. Every possible set of coefficients in the homogeneous system is a multiple of this one, so every common vector is a multiple of w.

Conclusion. $H \cap K = \operatorname{Span}\{(8, -16, -8)\} = \operatorname{Span}\{(1, -2, -1)\}$.

4.3 Linearly independent sets; bases

A basis for H must consist of vectors *in* H, span exactly H, and be linearly independent. Spanning a larger space is not enough. Independence means that the only zero combination has every coefficient zero. Removing a generator already in the span of the others does not change the span.

For a matrix, row operations preserve the solutions of its homogeneous system and all linear relations between columns. Consequently the pivot indices in RREF identify which *original* columns form a basis for the column space. Reduced columns themselves need not belong to the original column space. In the numerical work below, columns are denoted v_j in their printed order; a displayed RREF and the explicit column relations provide checkable elimination and spanning certificates.

Practice problem 4.3.P1: A basis must have the right ambient space

The two vectors are not scalar multiples: the first would require multiplier -2, but then its second entry would be 4, not 7. They are independent, but two vectors cannot span $\mathbb{R}^3$ because a 3×2 coefficient matrix cannot have a pivot in every row. Nor can they be a basis of $\mathbb{R}^2$: they have three coordinates and are not elements of that space.

Conclusion. They are a basis for their two-dimensional span in $\mathbb{R}^3$, not for $\mathbb{R}^3$ or $\mathbb{R}^2$.

Practice problem 4.3.P2: Remove redundant spanning vectors

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 6 & 2 & -4 \\ -3 & 2 & -2 & -8 \\ 4 & -1 & 3 & 9 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & \frac{4}{5} & 2 \\ 0 & 1 & \frac{1}{5} & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \\ -1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = (\frac{4}{5})v_1 + (\frac{1}{5})v_2$$
$$v_4 = (2)v_1 + (-1)v_2$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 2.

Practice problem 4.3.P3: Spanning every vector of a line is not enough

Neither $e_1 = (1,0,0)^T$ nor $e_2 = (0,1,0)^T$ lies in $H = \{(s,s,0)^T : s \in \mathbb{R}\}$, since their first two coordinates differ. Their span is the entire xy-plane, which is larger than H. Thus they cannot be a basis of H, despite the valid expression $(s,s,0)^T = se_1 + se_2$. A basis of H is instead $\{(1,1,0)^T\}$: it is nonzero and its multiples are exactly H.

Conclusion. No. Membership in H and equality of the span with H are missing.

Exercise 4.3.1: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 3.

Conclusion. The set is a basis for $\mathbb{R}^3$: it is independent and spans.

Exercise 4.3.2: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_2 = 0$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 2.

Conclusion. The set is not a basis for $\mathbb{R}^3$. It is dependent and does not span $\mathbb{R}^3$.

Exercise 4.3.3: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 0 & 1 & -1 \\ -3 & -4 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ -4 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = (1)v_1 + (-1)v_2$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 2.

Conclusion. The set is not a basis for $\mathbb{R}^3$. It is dependent and does not span $\mathbb{R}^3$.

Exercise 4.3.4: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 2 & 2 & -8 \\ -1 & -3 & 5 \\ 1 & 2 & 4 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \\ 2 \end{bmatrix}, \begin{bmatrix} -8 \\ 5 \\ 4 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 3.

Conclusion. The set is a basis for $\mathbb{R}^3$: it is independent and spans.

Exercise 4.3.5: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 3 & -3 & 0 & 0 \\ -3 & 7 & 0 & -3 \\ 0 & 0 & 0 & 5 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 4. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 3 \\ -3 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 7 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -3 \\ 5 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = 0$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 3.

Conclusion. The set is not a basis for $\mathbb{R}^3$. It is dependent and spans $\mathbb{R}^3$.

Exercise 4.3.6: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & -4 \\ 2 & 3 \\ -4 & 6 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ -4 \end{bmatrix}, \begin{bmatrix} -4 \\ 3 \\ 6 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 2.

Conclusion. The set is not a basis for $\mathbb{R}^3$. It is independent and does not span $\mathbb{R}^3$.

Exercise 4.3.7: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} -2 & 6 \\ 3 & -1 \\ 0 & 5 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} -2 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 \\ -1 \\ 5 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 2.

Conclusion. The set is not a basis for $\mathbb{R}^3$. It is independent and does not span $\mathbb{R}^3$.

Exercise 4.3.8: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 0 & 2 & 0 \\ -2 & 3 & -1 & 0 \\ 3 & -1 & 5 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 4. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This

proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = (2)v_1 + (1)v_2$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 3.

Conclusion. The set is not a basis for $\mathbb{R}^3$. It is dependent and spans $\mathbb{R}^3$.

Exercise 4.3.9: A null-space basis from one free variable

What we must show. Solve the homogeneous system and turn its free-variable parameters into basis vectors.

$$A = \begin{bmatrix} 1 & 0 & -2 & -2 \\ 0 & 1 & 1 & 4 \\ 3 & -1 & -7 & 3 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The free variable columns are 3. The pivot equations are

$$\begin{aligned} x_1 &= 2x_3 \\ x_2 &= -x_3 \\ x_4 &= 0 \end{aligned}$$

Set one free variable to 1 and the other free variables to 0, in turn. This gives

$$\mathcal{B} = \left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

Every homogeneous solution is a linear combination of these vectors, with its free-variable values as coefficients. Conversely substituting each displayed vector into Ax gives zero. They are independent because each has a different free coordinate equal to 1, with the other free coordinates zero. Thus they satisfy both parts of the basis definition.

Exercise 4.3.10: A null-space basis in five coordinates

What we must show. Solve the homogeneous system and turn its free-variable parameters into basis vectors.

$$A = \begin{bmatrix} 1 & 1 & -2 & 1 & 5 \\ 0 & 1 & 0 & -1 & -2 \\ 0 & 0 & -8 & 0 & 16 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 2 & 3 \\ 0 & 1 & 0 & -1 & -2 \\ 0 & 0 & 1 & 0 & -2 \end{bmatrix}.$$

The free variable columns are 4, 5. The pivot equations are

$$\begin{aligned} x_1 &= -2x_4 - 3x_5 \\ x_2 &= x_4 + 2x_5 \\ x_3 &= 2x_5 \end{aligned}$$

Set one free variable to 1 and the other free variables to 0, in turn. This gives

$$\mathcal{B} = \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 2 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Every homogeneous solution is a linear combination of these vectors, with its free-variable values as coefficients. Conversely substituting each displayed vector into Ax gives zero. They are independent because each has a different free coordinate equal to 1, with the other free coordinates zero. Thus they satisfy both parts of the basis definition.

Exercise 4.3.11: A basis for a plane

Solve the single homogeneous equation for x: $x = 3y - 2z$. Then

$$(x, y, z)^T = y(3, 1, 0)^T + z(-2, 0, 1)^T.$$

Every vector in the plane has this form. The displayed vectors lie in the plane, and a zero combination has second coordinate $y = 0$ and third coordinate $z = 0$. Thus they are independent and span precisely the plane.

Conclusion. $\{(3, 1, 0)^T, (-2, 0, 1)^T\}$ is a basis.

Exercise 4.3.12: A basis for a line

The line equation $y = -3x$ means that every vector is $(x, -3x)^T = x(1, -3)^T$. The vector $(1, -3)^T$ is nonzero, so the one-element set containing it is independent. Its multiples are exactly the line, so it spans the required space.

Conclusion. $\{(1, -3)^T\}$ is a basis.

Exercise 4.3.13: Null space and original column space

Null space: use the row-equivalent matrix.

What we must show. Solve the homogeneous system and turn its free-variable parameters into basis vectors.

$$A = \begin{bmatrix} -2 & 4 & -2 & -4 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 6 & 5 \\ 0 & 1 & \frac{5}{2} & \frac{3}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The free variable columns are 3, 4. The pivot equations are

$$x_1 = -6x_3 - 5x_4$$
$$x_2 = -\frac{5x_3}{2} - \frac{3x_4}{2}$$

Set one free variable to 1 and the other free variables to 0, in turn. This gives

$$\mathcal{B} = \left\{ \begin{bmatrix} -6 \\ -\frac{5}{2} \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ -\frac{3}{2} \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Every homogeneous solution is a linear combination of these vectors, with its free-variable values as coefficients. Conversely substituting each displayed vector into Ax gives zero. They are independent because each has a different free coordinate equal to 1, with the other free coordinates zero. Thus they satisfy both parts of the basis definition.

Column space: return to the original matrix.

The pivot indices are 1 and 2, so a column-space basis is $\{(-2, 2, -3)^T, (4, -6, 8)^T\}$. Do not take columns of the echelon matrix instead: row operations need not preserve the actual column space.

Exercise 4.3.14: Two different spaces from one reduction

Null space: solve the homogeneous equations.

What we must show. Solve the homogeneous system and turn its free-variable parameters into basis vectors.

$$A = \begin{bmatrix} 1 & 2 & 3 & -4 & 8 \\ 1 & 2 & 0 & 2 & 8 \\ 2 & 4 & -3 & 10 & 9 \\ 3 & 6 & 0 & 6 & 9 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 0 & 2 & 0 \\ 0 & 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The free variable columns are 2, 4. The pivot equations are

$$\begin{aligned} x_1 &= -2x_2 - 2x_4 \\ x_3 &= 2x_4 \\ x_5 &= 0 \end{aligned}$$

Set one free variable to 1 and the other free variables to 0, in turn. This gives

$$\mathcal{B} = \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

Every homogeneous solution is a linear combination of these vectors, with its free-variable values as coefficients. Conversely substituting each displayed vector into Ax gives zero. They are independent because each has a different free coordinate equal to 1, with the other free coordinates zero. Thus they satisfy both parts of the basis definition.

Column space: retain the original pivot columns.

The pivot indices are 1,3,5. Thus a basis for $\operatorname{Col} A$ is

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ -3 \\ 0 \end{bmatrix}, \begin{bmatrix} 8 \\ 8 \\ 9 \\ 9 \end{bmatrix} \right\}.$$

Exercise 4.3.15: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 0 & 2 & 2 & 3 \\ 0 & 1 & -2 & -1 & -1 \\ -2 & 2 & -8 & 10 & -6 \\ 3 & 3 & 0 & 3 & 9 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 2 & 0 & 0 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 4, 5. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 10 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ -6 \\ 9 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = (2)v_1 + (-2)v_2$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 4.

Exercise 4.3.16: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & -2 & 3 & 5 & 2 \\ 0 & 0 & -1 & -3 & -1 \\ 0 & 0 & 1 & 3 & 1 \\ 1 & 2 & -1 & -4 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & -\frac{5}{2} & 0 \\ 0 & 1 & 0 & \frac{3}{4} & \frac{1}{2} \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ 1 \\ -1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This

proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_4 = (-\frac{5}{2})v_1 + (\frac{3}{4})v_2 + (3)v_3$$
$$v_5 = (\frac{1}{2})v_2 + (1)v_3$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 3.

Exercise 4.3.17: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 2 & 4 & -2 & 8 & -8 \\ 0 & 0 & -4 & 4 & 4 \\ -4 & 2 & 0 & 8 & 0 \\ -6 & -4 & 1 & -3 & 0 \\ 0 & 4 & -7 & 15 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 3, 5. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 2 \\ 0 \\ -4 \\ -6 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 2 \\ -4 \\ 4 \end{bmatrix}, \begin{bmatrix} -2 \\ -4 \\ 0 \\ 1 \\ -7 \end{bmatrix}, \begin{bmatrix} -8 \\ 4 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_4 = (-1)v_1 + (2)v_2 + (-1)v_3$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 4.

Exercise 4.3.18: Identify a basis using original pivot columns

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} -3 & 3 & 0 & 6 & -6 \\ 2 & 0 & 2 & -2 & 3 \\ 6 & -9 & -4 & -14 & 0 \\ 0 & 0 & 0 & 0 & -1 \\ -7 & 6 & -1 & 13 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 3, 5. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} -3 \\ 2 \\ 6 \\ 0 \\ -7 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ -9 \\ 0 \\ 6 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ -4 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} -6 \\ 3 \\ 0 \\ -1 \\ 0 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_4 = (2)v_2 + (-1)v_3$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 4.

Exercise 4.3.19: Delete a generator using a known relation

The supplied relation gives $v_3 = (4/3)v_1 + (5/3)v_2$. Therefore every combination of all three vectors is a combination of v_1, v_2, so those two still span H. They are not scalar multiples: if $v_2 = cv_1$, the first coordinate gives $c = 1/4$, but the second would be $-3/4$ rather than 9. Thus they are independent.

Conclusion. $\{(4, -3, 7)^T, (1, 9, -2)^T\}$ is a basis for H.

Exercise 4.3.20: Another known dependence relation

Rearrange $2v_1 - v_2 - v_3 = 0$ as $v_3 = 2v_1 - v_2$. The third generator is redundant, so $H = \text{Span}\{v_1, v_2\}$. If $v_2 = cv_1$, the first coordinate gives $c = 4/3$, while the second coordinate would require $3 = (4/3)4 = 16/3$, a contradiction. Hence the two retained vectors are independent.

Conclusion. $\{(3, 4, -2, -5)^T, (4, 3, 2, 4)^T\}$ is a basis for H.

Exercise 4.3.21: Basis definitions and row operations

(a) False. A singleton $\{v\}$ is independent whenever $v \neq 0$, because $cv = 0$ then forces $c = 0$; only the zero singleton is dependent.

(b) False. Spanning is only one requirement. The set $\{e_1, 2e_1\}$ spans a line but is dependent and hence is not a basis.

(c) True. An invertible square matrix has a pivot in every row and column, so its columns both span $\mathbb{R}^n$ and are independent.

(d) False. One can append redundant vectors to a spanning set. A basis is a *minimal* spanning set, equivalently a maximal independent set, not a largest spanning set.

(e) False. If A is replaced by an elementary row-equivalent matrix EA, the relations $Ac = 0$ and $EAc = 0$ are equivalent because E is invertible. Thus precisely the same coefficient relations remain.

Exercise 4.3.22: Minimal spanning and maximal independence

(a) False. $\{e_1\}$ is independent in $\mathbb{R}^2$ but does not span it.

(b) True. Repeatedly delete a vector expressible in terms of the others. Each deletion preserves the span, and finiteness ensures the process stops at an independent spanning subset, hence a basis.

(c) True, with "as large as possible" meaning that no further vector of the space can be added while preserving independence. If an independent set did not span, a vector outside its span could be adjoined without creating dependence. Thus a maximal independent set spans and is a basis.

(d) False. The usual free-variable construction always gives independent vectors: in their free-coordinate positions they have distinct unit entries.

(e) False. Use pivot columns of A, not of its row-reduced matrix. For example $A = (1, 1)^T$ reduces to $B = (1, 0)^T$; their column spaces are different lines.

Exercise 4.3.23: Four spanning vectors in four dimensions

Put the four spanning vectors into a 4×4 matrix A. Because they span $\mathbb{R}^4$, its echelon form has a pivot in every row. There are four rows, hence four pivots; all four columns are pivot columns. Therefore $Ac = 0$ has no free variable and only the zero solution. The columns are independent as well as spanning.

Conclusion. The four vectors form a basis for $\mathbb{R}^4$.

Exercise 4.3.24: An independent square family

The n independent columns of their $n \times n$ matrix give a pivot in every column. Hence there are n pivots, one in each row too. A pivot in every row means the columns span $\mathbb{R}^n$. They were already independent, so they satisfy both basis requirements.

Conclusion. $\{v_1, \ldots, v_n\}$ is a basis for $\mathbb{R}^n$.

Exercise 4.3.25: A proposed basis outside the subspace

A basis of H must be a subset of H. Here $v_1 = (1, 0, 1)^T$ and $v_3 = (0, 1, 0)^T$ have unequal second and third coordinates, so neither belongs to H. The supplied expansion does indeed express every $(s, t, t)^T$ using the three vectors, but it does not show that their span is contained in H. Their span is larger: the three vectors are actually independent and span $\mathbb{R}^3$. A correct basis of H is $\{(1, 0, 0)^T, (0, 1, 1)^T\}$, since $(s, t, t)^T = s(1, 0, 0)^T + t(0, 1, 1)^T$ uniquely.

Conclusion. No. Unique representation of the vectors of H does not repair the failure of the proposed basis vectors to lie in H.

Exercise 4.3.26: A trigonometric spanning set

Use the identity $\sin 2t = 2\sin t\cos t$. It shows the third function is half the second, so $\sin t$ and $\sin 2t$ span the same subspace as all three. If $a\sin t + b\sin 2t = 0$ for every t, evaluate at $t = \pi/2$ to obtain $a = 0$, then at $t = \pi/4$ to obtain $b = 0$. Thus the two retained functions are independent.

Conclusion. $\{\sin t, \sin 2t\}$ is a basis of the specified function space.

Exercise 4.3.27: The mass–spring vibration space

Exercise 4.1.19 gives all displacements as $y(t) = c_1\cos(\omega t) + c_2\sin(\omega t)$, with fixed physical frequency $\omega > 0$. This already proves spanning by the two displayed functions. If $a\cos(\omega t) + b\sin(\omega t) = 0$ identically, $t = 0$ gives $a = 0$, and $t = \pi/(2\omega)$ gives $b = 0$.

Conclusion. $\{\cos(\omega t), \sin(\omega t)\}$ is a basis for the vibration space. If a nonphysical zero frequency were allowed, the sine function would vanish and the space would instead have basis $\{1\}$.

Exercise 4.3.28: A critically damped circuit space

The given description $v(t) = e^{-bt}(c_1 + c_2 t)$ means $H = \text{Span}\{e^{-bt}, te^{-bt}\}$. To check independence, suppose $ae^{-bt} + cte^{-bt} = 0$ for every t. Since $e^{-bt} > 0$, division gives $a + ct = 0$ identically. At $t = 0$ we get $a = 0$; at $t = 1$ we then get $c = 0$.

Conclusion. $\{e^{-bt}, te^{-bt}\}$ is a basis for the critically damped circuit's solution space H.

Exercise 4.3.29: Too few vectors to span

Place the $k < n$ vectors into an $n \times k$ matrix. At most k pivots are possible, so at least one of the n rows has no pivot. By the spanning criterion of Section 1.4, the columns cannot span $\mathbb{R}^n$. A basis must span the whole space.

Conclusion. A set with fewer than n vectors cannot be a basis of $\mathbb{R}^n$.

Exercise 4.3.30: Too many vectors to be independent

There are $k > n$ columns but only n rows. The homogeneous system with these columns therefore has at least one free variable, so it has a nonzero coefficient solution. This is a nontrivial dependence relation among the k vectors. A basis must be independent.

Conclusion. A set with more than n vectors cannot be a basis of $\mathbb{R}^n$.

Exercise 4.3.31: Dependence survives a linear map

Dependence provides scalars $c_1, \ldots, c_p$, not all zero, such that $\sum_j c_j v_j = 0$. Apply the linear map T:

$$\sum_j c_j T(v_j) = T\left(\sum_j c_j v_j\right) = T(0) = 0.$$

The coefficients have not changed, so they are still not all zero. This is a nontrivial dependence relation for the image vectors. Taking the contrapositive, independent images force the original vectors to be independent.

Conclusion. Linear maps preserve dependence, although they can turn an independent family into a dependent one.

Exercise 4.3.32: Injectivity reflects dependence

A dependence relation among the images has the form $\sum_j c_j T(v_j) = 0$ with not all c_j zero. By linearity,

$$T\left(\sum_j c_j v_j\right) = 0 = T(0).$$

Injectivity says equal images have equal inputs, so $\sum_j c_j v_j = 0$. The same nonzero coefficient list proves original dependence. The contrapositive states that an injective linear map carries an independent family to an independent image family.

Conclusion. Injectivity prevents a nonzero input combination from being collapsed to zero.

Exercise 4.3.33: Independence of two quadratic polynomials

Suppose $a(1+t^2) + b(1-t^2) = 0$ as a polynomial identity. Its constant coefficient is $a+b$, and its t^2 coefficient is $a-b$. Both must be zero, so addition and subtraction give $a = b = 0$. The polynomials belong to $\mathcal{P}_3$ because their degrees are at most three, even though their actual degree is only two.

Conclusion. Yes, the two polynomials are linearly independent in $\mathcal{P}_3$; they do not form a basis of that larger space.

Exercise 4.3.34: A polynomial dependence by inspection

The identity $(1+t) + (1-t) - 2 = 0$ gives $p_1 + p_2 - p_3 = 0$. Thus p_3 is redundant. To check the remaining pair, a zero combination $a(1+t) + b(1-t) = 0$ has constant coefficient $a+b = 0$ and linear coefficient $a-b = 0$, so $a = b = 0$.

Conclusion. $\{1+t, 1-t\}$ is a basis of the specified span, and the displayed relation proves the original three-function set dependent.

Exercise 4.3.35: Construct a four-vector family with a two-vector basis

Set $v_1 = u_1$, $v_2 = u_1 + u_2$, $v_3 = u_2$, and $v_4 = u_1 - u_2$. Every v_j lies in $\text{Span}\{u_1, u_2\}$, and $v_1 = u_1, v_3 = u_2$ already generate that space. Since a subset of an independent set is independent, $\{u_1, u_2\}$ is independent.

Conclusion. These four choices give $\{v_1, v_3\} = \{u_1, u_2\}$ as a basis of their common span. The additional u_3, u_4 need not be used.

Exercise 4.3.36: Bases of two spaces and their sum

Find a basis for H.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 2 & 4 \\ 0 & -1 & 1 \\ -1 & 1 & -4 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = (3)v_1 + (-1)v_2$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 2.

Find a basis for K.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} -2 & 2 & -1 \\ -2 & 3 & 4 \\ -1 & 2 & 6 \\ 3 & -6 & -2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} -2 \\ -2 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 2 \\ -6 \end{bmatrix}, \begin{bmatrix} -1 \\ 4 \\ 6 \\ -2 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 3.

Find a basis for their sum.

The sum $H + K$ consists exactly of combinations of all six generators. Use the concatenated matrix, numbering its columns $1, \ldots, 6$ in the order $u_1, u_2, u_3, v_1, v_2, v_3$.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 0 & 3 & -2 & 2 & -1 \\ 2 & 2 & 4 & -2 & 3 & 4 \\ 0 & -1 & 1 & -1 & 2 & 6 \\ -1 & 1 & -4 & 3 & -6 & -2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 3 & -2 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 5, 6. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 2 \\ -6 \end{bmatrix}, \begin{bmatrix} -1 \\ 4 \\ 6 \\ -2 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = (3)v_1 + (-1)v_2$$
$$v_4 = (-2)v_1 + (1)v_2$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 4.

Exercise 4.3.37: Four independent functions by exact evaluations

Assume $c_1 t + c_2 \sin t + c_3 \cos 2t + c_4 \sin t \cos t = 0$ for every real t. Evaluating at the exact inputs $0, \pi, \pi/2, \pi/4$ gives, successively,

$$\begin{aligned} c_3 &= 0, \\ \pi c_1 + c_3 &= 0, \\ (\pi/2)c_1 + c_2 - c_3 &= 0, \\ (\pi/4)c_1 + (\sqrt{2}/2)c_2 + (1/2)c_4 &= 0. \end{aligned}$$

The first equation gives $c_3 = 0$, the second $c_1 = 0$, the third $c_2 = 0$, and the last $c_4 = 0$. All divisions are by nonzero constants. Thus the only identical zero combination is trivial.

Conclusion. The four functions are linearly independent. Exact evaluation points avoid confusing small numerical roundoff with an actual zero.

Exercise 4.3.38: Seven cosine powers are independent

Suppose $c_0 + c_1 \cos t + \cdots + c_6 \cos^6 t = 0$ for every real t. Choose seven exact inputs $t_j = \arccos r_j$, where

$$(r_1, \ldots, r_7) = (-1, -2/3, -1/3, 0, 1/3, 2/3, 1).$$

Each is allowed since $r_j \in [-1, 1]$. Their evaluation equations are $\sum_{k=0}^{6} c_k r_j^k = 0$ for $j = 1, \ldots, 7$. This is the 7×7 system whose jth row is $[1, r_j, r_j^2, \ldots, r_j^6]$. Its RREF is I_7; equivalently, the polynomial $q(r) = \sum_{k=0}^{6} c_k r^k$ has seven distinct roots but degree at most six, so it must be identically zero. All c_k therefore vanish.

Conclusion. $\{1, \cos t, \ldots, \cos^6 t\}$ is an independent family. The seven evaluations are a finite exact certificate, as requested.

4.4 Coordinate systems

An ordered basis $\mathcal{B} = (b_1, \ldots, b_n)$ assigns to each vector x its unique list of weights $[x]_{\mathcal{B}} = (c_1, \ldots, c_n)^T$ in the expression $x = \sum_j c_j b_j$. Changing the basis order changes the meaning of this list.

For a basis in $\mathbb{R}^n$, put $P_{\mathcal{B}} = [b_1 \ \cdots \ b_n]$. The two directions are

$$x = P_{\mathcal{B}}[x]_{\mathcal{B}}, \qquad [x]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}x.$$

The first assembles a vector from weights; the second finds its weights. For polynomials use the standard coefficient order $1, t, t^2, \ldots$. A missing power has coefficient zero and must still occupy its position.

Practice problem 4.4.P1: Both directions of a basis-coordinate matrix

(a) Check independence and spanning.

The matrix

$$P_{\mathcal{B}} = \begin{bmatrix} 1 & -3 & 3 \\ 0 & 4 & -6 \\ 0 & 0 & 3 \end{bmatrix}$$

is triangular with three nonzero pivots. It is invertible, so its columns are independent and span $\mathbb{R}^3$; they are a basis.

(b)–(c) State the matrix and direction correctly.

The displayed matrix is the change from $\mathcal{B}$ coordinates to standard coordinates, and the equation is $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$.

(d) Solve for the weights.

For $x = (-8, 2, 3)^T$, the last equation is $3c_3 = 3$, so $c_3 = 1$. The middle equation $4c_2 - 6c_3 = 2$ gives $c_2 = 2$. The first $c_1 - 3c_2 + 3c_3 = -8$ gives $c_1 = -5$.

Conclusion. $[x]_{\mathcal{B}} = (-5, 2, 1)^T$, and multiplication by the matrix gives back $(-8, 2, 3)^T$.

Practice problem 4.4.P2: Coordinates of a quadratic polynomial

Compare coefficients in the fixed standard order $1, t, t^2$.

Write $p = c_1 b_1 + c_2 b_2 + c_3 b_3$. Equality of polynomials means equality of the constant, linear, and quadratic coefficients, so the coefficient matrix has the coefficient vectors of b_1, b_2, b_3 as columns.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & -1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (1)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ 0 & -1 & 1 & -3 \\ 0 & 1 & 1 & -1 \end{array}\right].$$

Use $R_2 \leftarrow (-1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ 0 & 1 & -1 & 3 \\ 0 & 1 & 1 & -1 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 2 & -4 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{2})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -2 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-1)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 6 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = 5 \\ c_2 = 1 \\ c_3 = -2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} 5 \\ 1 \\ -2 \end{bmatrix}$.

Exercise 4.4.1: Coordinates and basis representations

What we must show. Assemble the vector from its given ordered basis weights. The entries of $[x]_{\mathcal{B}}$ multiply the corresponding basis vectors; they are not generally the standard coordinates of x.

$$x = \begin{bmatrix} 3 & -4 \\ -5 & 6 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ -7 \end{bmatrix}.$$

Coordinate by coordinate the calculation is

$$x = \begin{bmatrix} (3)(5) + (-4)(3) \\ (-5)(5) + (6)(3) \end{bmatrix} = \begin{bmatrix} 3 \\ -7 \end{bmatrix}.$$

Conclusion. The standard-coordinate vector is the final column displayed above.

Exercise 4.4.2: Coordinates and basis representations

What we must show. Assemble the vector from its given ordered basis weights. The entries of $[x]_{\mathcal{B}}$ multiply the corresponding basis vectors; they are not generally the standard coordinates of x.

$$x = \begin{bmatrix} 3 & -4 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 5 \end{bmatrix} = \begin{bmatrix} -26 \\ 1 \end{bmatrix}.$$

Coordinate by coordinate the calculation is

$$x = \begin{bmatrix} (3)(-2) + (-4)(5) \\ (2)(-2) + (1)(5) \end{bmatrix} = \begin{bmatrix} -26 \\ 1 \end{bmatrix}.$$

Conclusion. The standard-coordinate vector is the final column displayed above.

Exercise 4.4.3: Coordinates and basis representations

What we must show. Assemble the vector from its given ordered basis weights. The entries of $[x]_{\mathcal{B}}$ multiply the corresponding basis vectors; they are not generally the standard coordinates of x.

$$x = \begin{bmatrix} 1 & 5 & 4 \\ -2 & 0 & -3 \\ 3 & -2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} -7 \\ 4 \\ 3 \end{bmatrix}.$$

Coordinate by coordinate the calculation is

$$x = \begin{bmatrix} (1)(1) + (5)(0) + (4)(-2) \\ (-2)(1) + (0)(0) + (-3)(-2) \\ (3)(1) + (-2)(0) + (0)(-2) \end{bmatrix} = \begin{bmatrix} -7 \\ 4 \\ 3 \end{bmatrix}.$$

Conclusion. The standard-coordinate vector is the final column displayed above.

Exercise 4.4.4: Coordinates and basis representations

What we must show. Assemble the vector from its given ordered basis weights. The entries of $[x]_{\mathcal{B}}$ multiply the corresponding basis vectors; they are not generally the standard coordinates of x.

$$x = \begin{bmatrix} -2 & 3 & 4 \\ 2 & 0 & -1 \\ 0 & 2 & 3 \end{bmatrix} \begin{bmatrix} -3 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 8 \\ -5 \\ 1 \end{bmatrix}.$$

Coordinate by coordinate the calculation is

$$x = \begin{bmatrix} (-2)(-3) + (3)(2) + (4)(-1) \\ (2)(-3) + (0)(2) + (-1)(-1) \\ (0)(-3) + (2)(2) + (3)(-1) \end{bmatrix} = \begin{bmatrix} 8 \\ -5 \\ 1 \end{bmatrix}.$$

Conclusion. The standard-coordinate vector is the final column displayed above.

Exercise 4.4.5: Find weights by reducing a basis system

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{rr|r} 1 & 3 & -1 \\ -2 & -5 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-2)R_1$ to obtain

$$\left[\begin{array}{rr|r} 1 & 3 & -1 \\ 0 & 1 & -1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (3)R_2$ to obtain

$$\left[\begin{array}{rr|r} 1 & 0 & 2 \\ 0 & 1 & -1 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = 2 \\ c_2 = -1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$.

Exercise 4.4.6: Find weights by reducing a basis system

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{rr|r} 1 & 2 & -1 \\ -4 & -3 & -6 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-4)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & 5 & -10 \end{array}\right].$$

Use $R_2 \leftarrow (\frac{1}{5})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & 1 & -2 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & -2 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = 3 \\ c_2 = -2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$.

Exercise 4.4.7: Find weights by reducing a basis system

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & -3 & 2 & 8 \\ -1 & 4 & -2 & -9 \\ -3 & 9 & 4 & 6 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-1)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -3 & 2 & 8 \\ 0 & 1 & 0 & -1 \\ -3 & 9 & 4 & 6 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-3)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -3 & 2 & 8 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 10 & 30 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{10})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -3 & 2 & 8 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -3 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-3)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = -1 \\ c_2 = -1 \\ c_3 = 3 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} -1 \\ -1 \\ 3 \end{bmatrix}$.

Exercise 4.4.8: Find weights by reducing a basis system

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 1 & 0 & -1 & 0 \\ 3 & 8 & 3 & -2 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (1)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -2 & -2 & 0 \\ 3 & 8 & 3 & -2 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (3)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -2 & -2 & 0 \\ 0 & 2 & 0 & -2 \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{1}{2})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & -2 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (2)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -2 & -2 \end{array}\right].$$

Use $R_3 \leftarrow (-\frac{1}{2})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (1)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (1)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = 1 \\ c_2 = -1 \\ c_3 = 1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$.

Exercise 4.4.9: Coordinates and basis representations

The map from basis coordinates to standard coordinates places each ordered basis vector in the corresponding matrix column:

$$P_{\mathcal{B}} = \begin{bmatrix} 1 & 2 \\ -3 & -5 \end{bmatrix}.$$

Indeed, multiplying by $(c_1, \ldots, c_n)^T$ gives $c_1 b_1 + \cdots + c_n b_n = x$, so this matrix has exactly the required direction.

Exercise 4.4.10: Coordinates and basis representations

The map from basis coordinates to standard coordinates places each ordered basis vector in the corresponding matrix column:

$$P_{\mathcal{B}} = \begin{bmatrix} 3 & 2 & 1 \\ 0 & 2 & -2 \\ 6 & -4 & 3 \end{bmatrix}.$$

Indeed, multiplying by $(c_1, \ldots, c_n)^T$ gives $c_1 b_1 + \cdots + c_n b_n = x$, so this matrix has exactly the required direction.

Exercise 4.4.11: Coordinates and basis representations

Invert the ordered basis matrix.

For a 2×2 matrix, the inverse is the adjugate divided by the nonzero determinant. Here

$$P_{\mathcal{B}} = \begin{bmatrix} 1 & -3 \\ -2 & 5 \end{bmatrix}, \qquad \det P_{\mathcal{B}} = -1, \qquad P_{\mathcal{B}}^{-1} = \begin{bmatrix} -5 & -3 \\ -2 & -1 \end{bmatrix}.$$

Apply the inverse to the actual standard vector.

$$[x]_{\mathcal{B}} = \begin{bmatrix} -5 & -3 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ -5 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}.$$

Multiplying the answer by $P_{\mathcal{B}}$ returns x, so the direction of the change has been checked.

Exercise 4.4.12: Coordinates and basis representations

Invert the ordered basis matrix.

For a 2×2 matrix, the inverse is the adjugate divided by the nonzero determinant. Here

$$P_{\mathcal{B}} = \begin{bmatrix} 1 & 2 \\ -1 & -1 \end{bmatrix}, \qquad \det P_{\mathcal{B}} = 1, \qquad P_{\mathcal{B}}^{-1} = \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix}.$$

Apply the inverse to the actual standard vector.

$$[x]_{\mathcal{B}} = \begin{bmatrix} -1 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -8 \\ 5 \end{bmatrix}.$$

Multiplying the answer by $P_{\mathcal{B}}$ returns x, so the direction of the change has been checked.

Exercise 4.4.13: Coordinates and basis representations

Compare coefficients in the fixed standard order $1, t, t^2$.

Write $p = c_1 b_1 + c_2 b_2 + c_3 b_3$. Equality of polynomials means equality of the constant, linear, and quadratic coefficients, so the coefficient matrix has the coefficient vectors of b_1, b_2, b_3 as columns.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 4 \\ 1 & 1 & 1 & 7 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (1)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & 0 & 6 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & -2 & 2 \end{array}\right].$$

Use $R_3 \leftarrow (-\frac{1}{2})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (1)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = 2 \\ c_2 = 6 \\ c_3 = -1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 6 \\ -1 \end{bmatrix}$.

Exercise 4.4.14: Coordinates and basis representations

Compare coefficients in the fixed standard order $1, t, t^2$.

Write $p = c_1 b_1 + c_2 b_2 + c_3 b_3$. Equality of polynomials means equality of the constant, linear, and quadratic coefficients, so the coefficient matrix has the coefficient vectors of b_1, b_2, b_3 as columns.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 3 \\ -1 & -1 & 1 & -6 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-1)R_1$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 3 \\ 0 & -1 & 3 & -5 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-1)R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 2 & -2 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{1}{2})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (2)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-1)R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = 3 \\ c_2 = 2 \\ c_3 = -1 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}$.

Exercise 4.4.15: Coordinates and basis representations

(a) True. A basis with n ordered members produces exactly n real weights, hence a coordinate vector in $\mathbb{R}^n$, whatever type of object x itself may be.

(b) False. For a basis of $\mathbb{R}^n$, $P_{\mathcal{B}}$ maps weights to standard vectors: $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$. The coordinate map uses $P_{\mathcal{B}}^{-1}$ instead.

(c) False. The four independent polynomials $1, t, t^2, t^3$ would have independent images under an isomorphism, but four vectors cannot be independent in $\mathbb{R}^3$. In fact $\mathcal{P}_3$ corresponds to $\mathbb{R}^4$.

Exercise 4.4.16: Coordinates and basis representations

(a) True. For the standard basis, $x = x_1e_1 + \cdots + x_ne_n$, so its basis weights are exactly its usual coordinates.

(b) False. The coordinate mapping goes from x to $[x]_{\mathcal{B}}$. The displayed reverse direction is its inverse, assembling a vector from its weights.

(c) True. Every two-dimensional plane through the origin has a two-vector basis. Its coordinate map is a linear bijection onto $\mathbb{R}^2$. An affine plane not through zero would not be a vector space under the usual operations.

Exercise 4.4.17: Coordinates and basis representations

One representation is $5v_1 - 2v_2 + 0v_3 = (1, 1)^T$. There is also the nontrivial relation $5v_1 - v_2 + v_3 = 0$, since

$$5(1, -3)^T - (2, -8)^T + (-3, 7)^T = (0, 0)^T.$$

Add that zero relation to the first representation. It gives $10v_1 - 3v_2 + v_3 = (1, 1)^T$. The two weight triples $(5, -2, 0)$ and $(10, -3, 1)$ differ, and direct substitution verifies both.

Conclusion. The failure of independence permits distinct representations of the same vector.

Exercise 4.4.18: Coordinates and basis representations

The jth basis vector has the representation

$$b_j = 0b_1 + \cdots + 0b_{j-1} + 1b_j + 0b_{j+1} + \cdots + 0b_n.$$

A basis makes representations unique, so these are exactly its coordinates. The coefficient column is e_j, with a 1 in place j and zero elsewhere.

Conclusion. $[b_j]_{\mathcal{B}} = e_j$ for each j, the columns of I_n.

Exercise 4.4.19: Coordinates and basis representations

The assumed existence of a representation for every vector says Span $S = V$. To establish independence, compare two representations of zero. One uses all coefficients zero. If a nontrivial zero combination existed, it would give a different representation of zero, contradicting uniqueness. Thus the only zero combination is trivial.

Conclusion. S is independent and spans V, hence is a basis.

Exercise 4.4.20: Coordinates and basis representations

Because the four generators span, write $w = \sum_{j=1}^{4} k_j v_j$. Their dependence supplies $a_1, \ldots, a_4$, not all zero, with $\sum_j a_j v_j = 0$. For every real t,

$$w = \sum_{j=1}^{4} (k_j + t a_j) v_j.$$

At least one a_j is nonzero, so changing t changes that coefficient. The resulting coefficient lists are distinct, although the resulting vector remains w.

Conclusion. Every $w \in V$ has infinitely many representations in the dependent spanning family, in particular more than one.

Exercise 4.4.21: Coordinates and basis representations

The basis matrix and its determinant are

$$P_{\mathcal{B}} = \begin{bmatrix} 1 & -2 \\ -4 & 9 \end{bmatrix}, \qquad \det P_{\mathcal{B}} = 9 - 8 = 1.$$

Thus the matrix implementing the coordinate map is its inverse:

$$A = P_{\mathcal{B}}^{-1} = \begin{bmatrix} 9 & 2 \\ 4 & 1 \end{bmatrix}.$$

Indeed $AP_{\mathcal{B}} = I_2$, so $Ax = AP_{\mathcal{B}}[x]_{\mathcal{B}} = [x]_{\mathcal{B}}$ for every x.

Conclusion. $A = \begin{bmatrix} 9 & 2 \\ 4 & 1 \end{bmatrix}$.

Exercise 4.4.22: Coordinates and basis representations

Form $P_{\mathcal{B}} = [b_1 \ \cdots \ b_n]$. Its columns are a basis, so it is invertible. Since $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$, multiply by the inverse to obtain

$$[x]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}x.$$

Conclusion. The implementing matrix is $A = [b_1 \ \cdots \ b_n]^{-1}$, not the basis matrix itself.

Exercise 4.4.23: Coordinates and basis representations

Suppose $[u]_{\mathcal{B}} = [w]_{\mathcal{B}} = (c_1, \ldots, c_n)^T$. By the definition of coordinates, $u = \sum_j c_j b_j$ and $w = \sum_j c_j b_j$. The right sides are identical, so $u = w$.

Conclusion. Equal coordinate outputs imply equal vector inputs; the coordinate map is one-to-one.

Exercise 4.4.24: Coordinates and basis representations

Take any $y = (y_1, \ldots, y_n)^T \in \mathbb{R}^n$ and define $u = y_1 b_1 + \cdots + y_n b_n \in V$. This is a legitimate vector because V is closed under sums and scalar multiplication. The defining representation has weight list y, and its uniqueness gives $[u]_{\mathcal{B}} = y$.

Conclusion. Every vector of $\mathbb{R}^n$ is reached, so the coordinate map is onto.

Exercise 4.4.25: Coordinates and basis representations

Let $C(x) = [x]_{\mathcal{B}}$. It is linear and one-to-one. For any scalars c_j,

$$\sum_j c_j u_j = 0 \quad \Longleftrightarrow \quad C\left(\sum_j c_j u_j\right) = 0 \quad \Longleftrightarrow \quad \sum_j c_j C(u_j) = 0.$$

The forward implication in the first equivalence uses $C(0) = 0$; the reverse uses injectivity. The second equivalence uses linearity. Thus the two homogeneous equations have exactly the same coefficient solutions. One family has only the trivial zero combination if and only if the other does.

Conclusion. The original vectors are independent exactly when their coordinate vectors are independent.

Exercise 4.4.26: Coordinates and basis representations

If $w = \sum_j c_j u_j$, linearity gives $[w]_{\mathcal{B}} = \sum_j c_j [u_j]_{\mathcal{B}}$. Conversely suppose the latter relation holds. Linearity rewrites it as $C(w) = C(\sum_j c_j u_j)$. Injectivity of C then implies $w = \sum_j c_j u_j$. The coefficients themselves are the same in both representations.

Conclusion. Span membership is preserved in both directions by coordinates.

Exercise 4.4.27: Test polynomial independence using all four coefficient positions

Move the problem to coefficient coordinates.

Use the ordered basis $(1, t, t^2, t^3)$ for $\mathcal{P}_3$, so each polynomial gives one coefficient column. Expanding powers first where necessary gives

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 0 & -3 & 2 \\ 2 & 0 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

There is a pivot in every column, so the coefficient columns are independent. The coordinate map is injective and linear, so the polynomials are independent as well.

Conclusion. The set is linearly independent.

Exercise 4.4.28: Test polynomial independence using all four coefficient positions

Move the problem to coefficient coordinates.

Use the ordered basis $(1, t, t^2, t^3)$ for $\mathcal{P}_3$, so each polynomial gives one coefficient column. Expanding powers first where necessary gives

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ -2 & 0 & -2 \\ -1 & 2 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

There is a pivot in every column, so the coefficient columns are independent. The coordinate map is injective and linear, so the polynomials are independent as well.

Conclusion. The set is linearly independent.

Exercise 4.4.29: Test polynomial independence using all four coefficient positions

Move the problem to coefficient coordinates.

Use the ordered basis $(1, t, t^2, t^3)$ for $\mathcal{P}_3$, so each polynomial gives one coefficient column. Expanding powers first where necessary gives

$$A = \begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & -3 \\ 1 & -2 & 3 \\ 0 & 1 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

A nonzero zero-combination coefficient vector is $\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$. It gives the polynomial relation

$$(-1)p_1 + (1)p_2 + (1)p_3 = 0.$$

Thus the polynomials are dependent.

Conclusion. The set is linearly dependent.

Exercise 4.4.30: Test polynomial independence using all four coefficient positions

Move the problem to coefficient coordinates.

Use the ordered basis $(1, t, t^2, t^3)$ for $\mathcal{P}_3$, so each polynomial gives one coefficient column. Expanding powers first where necessary gives

$$A = \begin{bmatrix} 8 & 9 & 1 \\ -12 & -6 & 6 \\ 6 & 1 & -5 \\ -1 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

A nonzero zero-combination coefficient vector is $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$. It gives the polynomial relation

$$(1)p_1 + (-1)p_2 + (1)p_3 = 0.$$

Thus the polynomials are dependent.

Conclusion. The set is linearly dependent.

Exercise 4.4.31: Coordinates and basis representations

Use the coefficient order $(1, t, t^2)$. Spanning $\mathcal{P}_2$ is equivalent to having a pivot in every row of the coefficient matrix.

(a) Reduce the coefficient matrix.

$$\begin{bmatrix} 1 & -3 & -4 & 1 \\ -3 & 5 & 5 & 0 \\ 5 & -7 & -6 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & \frac{5}{4} & -\frac{5}{4} \\ 0 & 1 & \frac{7}{4} & -\frac{3}{4} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Only 2 rows contain pivots, so the set does not span $\mathcal{P}_2$.

(b) Reduce the coefficient matrix.

$$\begin{bmatrix} 0 & 1 & -3 & 2 \\ 5 & -8 & 4 & -3 \\ 1 & -2 & 2 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & -4 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

All three rows contain pivots, so the set spans $\mathcal{P}_2$.

Exercise 4.4.32: Coordinates and basis representations

(a) Check the three coefficient columns.

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & -3 & -3 \end{bmatrix}, \qquad \det A = -1 \neq 0.$$

Therefore these columns form a basis of $\mathbb{R}^3$, so the polynomials form a basis of $\mathcal{P}_2$.

(b) Assemble the specified polynomial.

$$q = -p_1 + p_2 + 2p_3 = -(1 + t^2) + (t - 3t^2) + 2(1 + t - 3t^2) = 1 + 3t - 10t^2.$$

Conclusion. $q(t) = 1 + 3t - 10t^2$.

Exercise 4.4.33: Test polynomial independence using all four coefficient positions

Move the problem to coefficient coordinates.

Use the ordered basis $(1, t, t^2, t^3)$ for $\mathcal{P}_3$, so each polynomial gives one coefficient column. Expanding powers first where necessary gives

$$A = \begin{bmatrix} 3 & 5 & 0 & 1 \\ 7 & 1 & 1 & 16 \\ 0 & 0 & -2 & -6 \\ 0 & -2 & 0 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

A nonzero zero-combination coefficient vector is $\begin{bmatrix} -2 \\ 1 \\ -3 \\ 1 \end{bmatrix}$. It gives the polynomial relation

$$(-2)p_1 + (1)p_2 + (-3)p_3 + (1)p_4 = 0.$$

Thus the polynomials are dependent.

Conclusion. The polynomials do not form a basis of $\mathcal{P}_3$, since they are dependent.

Exercise 4.4.34: Test polynomial independence using all four coefficient positions

Move the problem to coefficient coordinates.

Use the ordered basis $(1, t, t^2, t^3)$ for $\mathcal{P}_3$, so each polynomial gives one coefficient column. Expanding powers first where necessary gives

$$A = \begin{bmatrix} 5 & 9 & 6 & 0 \\ -3 & 1 & -2 & 0 \\ 4 & 8 & 5 & 0 \\ 2 & -6 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & \frac{3}{4} & 0 \\ 0 & 1 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

A nonzero zero-combination coefficient vector is $\begin{bmatrix} -\frac{3}{4} \\ -\frac{1}{4} \\ 1 \\ 0 \end{bmatrix}$. It gives the polynomial relation

$$(-\frac{3}{4})p_1 + (-\frac{1}{4})p_2 + (1)p_3 = 0.$$

Thus the polynomials are dependent.

Conclusion. The polynomials do not form a basis of $\mathcal{P}_3$, since they are dependent.

Exercise 4.4.35: Coordinates and basis representations

Check that the ordered generators are independent.

The coefficient matrix has a pivot in every column, as the reduction below verifies. Hence the generators form a basis of their stated span H.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{cc|c} 11 & 14 & 19 \\ -5 & -8 & -13 \\ 10 & 13 & 18 \\ 7 & 10 & 15 \end{array}\right].$$

Use $R_1 \leftarrow (\frac{1}{11})R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{14}{11} & \frac{19}{11} \\ -5 & -8 & -13 \\ 10 & 13 & 18 \\ 7 & 10 & 15 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-5)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{14}{11} & \frac{19}{11} \\ 0 & -\frac{18}{11} & -\frac{48}{11} \\ 10 & 13 & 18 \\ 7 & 10 & 15 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (10)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{14}{11} & \frac{19}{11} \\ 0 & -\frac{18}{11} & -\frac{48}{11} \\ 0 & \frac{3}{11} & \frac{8}{11} \\ 7 & 10 & 15 \end{array}\right].$$

Use $R_4 \leftarrow R_4 - (7)R_1$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{14}{11} & \frac{19}{11} \\ 0 & -\frac{18}{11} & -\frac{48}{11} \\ 0 & \frac{3}{11} & \frac{8}{11} \\ 0 & \frac{12}{11} & \frac{32}{11} \end{array}\right].$$

Use $R_2 \leftarrow (-\frac{11}{18})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{14}{11} & \frac{19}{11} \\ 0 & 1 & \frac{8}{3} \\ 0 & \frac{3}{11} & \frac{8}{11} \\ 0 & \frac{12}{11} & \frac{32}{11} \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (\frac{3}{11})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{14}{11} & \frac{19}{11} \\ 0 & 1 & \frac{8}{3} \\ 0 & 0 & 0 \\ 0 & \frac{12}{11} & \frac{32}{11} \end{array}\right].$$

Use $R_4 \leftarrow R_4 - (\frac{12}{11})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & \frac{14}{11} & \frac{19}{11} \\ 0 & 1 & \frac{8}{3} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (\frac{14}{11})R_2$ to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & -\frac{5}{3} \\ 0 & 1 & \frac{8}{3} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = -\frac{5}{3} \\ c_2 = \frac{8}{3} \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} -\frac{5}{3} \\ \frac{8}{3} \end{bmatrix}$.

Verify membership directly.

The computed weights satisfy $\begin{bmatrix} 11 & 14 \\ -5 & -8 \\ 10 & 13 \\ 7 & 10 \end{bmatrix} \begin{bmatrix} -\frac{5}{3} \\ \frac{8}{3} \end{bmatrix} = \begin{bmatrix} 19 \\ -13 \\ 18 \\ 15 \end{bmatrix}$, so $x \in H$ and the listed coordinates are the unique basis weights.

Exercise 4.4.36: Coordinates and basis representations

Check that the ordered generators are independent.

The coefficient matrix has a pivot in every column, as the reduction below verifies. Hence the generators form a basis of their stated span H.

What we must show. Analyze the system encoded by the displayed augmented matrix. The last column contains the constants; all preceding columns contain variable coefficients.

Plan. Move from left to right to make pivots and clear the entries below them. Then clear entries above the pivots. Each operation is reversible, so the equations keep exactly the same solutions.

$$\left[\begin{array}{ccc|c} -6 & 8 & -9 & 4 \\ 4 & -3 & 5 & 7 \\ -9 & 7 & -8 & -8 \\ 4 & -3 & 3 & 3 \end{array}\right].$$

Use $R_1 \leftarrow (-\frac{1}{6})R_1$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 4 & -3 & 5 & 7 \\ -9 & 7 & -8 & -8 \\ 4 & -3 & 3 & 3 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (4)R_1$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & \frac{7}{3} & -1 & \frac{29}{3} \\ -9 & 7 & -8 & -8 \\ 4 & -3 & 3 & 3 \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-9)R_1$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & \frac{7}{3} & -1 & \frac{29}{3} \\ 0 & -5 & \frac{11}{2} & -14 \\ 4 & -3 & 3 & 3 \end{array}\right].$$

Use $R_4 \leftarrow R_4 - (4)R_1$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & \frac{7}{3} & -1 & \frac{29}{3} \\ 0 & -5 & \frac{11}{2} & -14 \\ 0 & \frac{7}{3} & -3 & \frac{17}{3} \end{array}\right].$$

Use $R_2 \leftarrow (\frac{3}{7})R_2$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{7} & \frac{29}{7} \\ 0 & -5 & \frac{11}{2} & -14 \\ 0 & \frac{7}{3} & -3 & \frac{17}{3} \end{array}\right].$$

Use $R_3 \leftarrow R_3 - (-5)R_2$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{7} & \frac{29}{7} \\ 0 & 0 & \frac{47}{14} & \frac{47}{7} \\ 0 & \frac{7}{3} & -3 & \frac{17}{3} \end{array}\right].$$

Use $R_4 \leftarrow R_4 - (\frac{7}{3})R_2$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{7} & \frac{29}{7} \\ 0 & 0 & \frac{47}{14} & \frac{47}{7} \\ 0 & 0 & -2 & -4 \end{array}\right].$$

Use $R_3 \leftarrow (\frac{14}{47})R_3$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{7} & \frac{29}{7} \\ 0 & 0 & 1 & 2 \\ 0 & 0 & -2 & -4 \end{array}\right].$$

Use $R_4 \leftarrow R_4 - (-2)R_3$ to obtain

$$\left[\begin{array}{rrr|r} 1 & -\frac{4}{3} & \frac{3}{2} & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{7} & \frac{29}{7} \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (\frac{3}{2})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -\frac{4}{3} & 0 & -\frac{11}{3} \\ 0 & 1 & -\frac{3}{7} & \frac{29}{7} \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Use $R_2 \leftarrow R_2 - (-\frac{3}{7})R_3$ to obtain

$$\left[\begin{array}{ccc|c} 1 & -\frac{4}{3} & 0 & -\frac{11}{3} \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Use $R_1 \leftarrow R_1 - (-\frac{4}{3})R_2$ to obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Read the equations, keeping every free variable.

Every variable column has a pivot. Thus there are no free variables, and the solution is unique.

$$\begin{cases} c_1 = 3 \\ c_2 = 5 \\ c_3 = 2 \end{cases}$$

Substituting these expressions in the original equations makes each left side equal its given constant. This also checks that the parameters are unrestricted.

Conclusion. The displayed values give the unique solution.

Conclusion. Since the unknowns are basis weights, $[x]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix}$.

Verify membership directly.

The computed weights satisfy $\begin{bmatrix} -6 & 8 & -9 \\ 4 & -3 & 5 \\ -9 & 7 & -8 \\ 4 & -3 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \\ -8 \\ 3 \end{bmatrix}$, so $x \in H$ and the listed coordinates are the unique basis weights.

Exercise 4.4.37: Coordinates and basis representations

What we must show. Convert lattice weights into ordinary spatial coordinates. The basis columns are given in angstroms, so use $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$ rather than the inverse direction:

$$x = \begin{bmatrix} \frac{13}{5} & 0 & 0 \\ -\frac{3}{2} & 3 & 0 \\ 0 & 0 & \frac{24}{5} \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{4} \\ \frac{1}{6} \end{bmatrix} = \begin{bmatrix} \frac{13}{10} \\ 0 \\ \frac{4}{5} \end{bmatrix}.$$

Coordinatewise, the first value is $2.6(1/2) = 1.3$, the second is the sum of the contributions from the first two basis columns, and the third is 4.8 times the third lattice coordinate.

Conclusion. The standard-coordinate location is $(\frac{13}{10}, 0, \frac{4}{5})^T$ angstroms.

Exercise 4.4.38: Coordinates and basis representations

What we must show. Convert lattice weights into ordinary spatial coordinates. The basis columns are given in angstroms, so use $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$ rather than the inverse direction:

$$x = \begin{bmatrix} \frac{13}{5} & 0 & 0 \\ -\frac{3}{2} & 3 & 0 \\ 0 & 0 & \frac{24}{5} \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{3} \end{bmatrix} = \begin{bmatrix} \frac{13}{10} \\ \frac{3}{4} \\ \frac{8}{5} \end{bmatrix}.$$

Coordinatewise, the first value is $2.6(1/2) = 1.3$, the second is the sum of the contributions from the first two basis columns, and the third is 4.8 times the third lattice coordinate.

Conclusion. The standard-coordinate location is $(\frac{13}{10}, \frac{3}{4}, \frac{8}{5})^T$ angstroms.

4.5 The dimension of a vector space

The dimension of a finite-dimensional space is the number of vectors in any basis. A spanning list need not have that size: it can contain redundant vectors. An independent list can be too short to span.

To analyze a parametrized subspace, split its vector expression into one coefficient vector per parameter, then remove any dependent generators. For A with n columns and r pivots, $\dim \operatorname{Col} A = r$ and $\dim \operatorname{Nul} A = n - r$: the null-space basis has one vector per free variable. The zero space has the empty basis and dimension zero.

Practice problem 4.5.P1: Dependence alone does not bound the list size

False. The singleton $S = \{0\}$ is dependent because $1 \cdot 0 = 0$ is a nontrivial relation, yet it has only one element. In the nonzero space V with dimension $p \geq 1$, one is not greater than p.

Conclusion. A short set can be dependent; being too large is sufficient for dependence but is not necessary.

Practice problem 4.5.P2: A larger set than a spanning set is dependent

The finite spanning set S contains a basis, obtained by deleting redundant vectors. Thus $\dim V \leq |S|$. If T contains more vectors than S, then $|T| > |S| \geq \dim V$. Any family with more vectors than the dimension is dependent. If T is infinite, take already a finite subset of more than $\dim V$ elements to obtain a dependence relation.

Conclusion. True. The conclusion uses that S spans, not that S itself must be a basis.

Exercise 4.5.1: Find a dimension from a checked basis

Separate the independent parameter contributions.

The given vector expression is a linear combination of the following coefficient columns, in the order of its printed parameters. Their column space is exactly the requested subspace.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & -2 \\ 1 & 1 \\ 0 & 3 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 2.

Conclusion. A basis is the set of original pivot columns displayed above, and the dimension is 2.

Exercise 4.5.2: Find a dimension from a checked basis

Separate the independent parameter contributions.

The given vector expression is a linear combination of the following coefficient columns, in the order of its printed parameters. Their column space is exactly the requested subspace.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 2 & 0 \\ 0 & -4 \\ -2 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ -4 \\ 0 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 2.

Conclusion. A basis is the set of original pivot columns displayed above, and the dimension is 2.

Exercise 4.5.3: Find a dimension from a checked basis

Separate the independent parameter contributions.

The given vector expression is a linear combination of the following coefficient columns, in the order of its printed parameters. Their column space is exactly the requested subspace.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 0 & 0 & 2 \\ 1 & -1 & 0 \\ 0 & 1 & -3 \\ 1 & 2 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -3 \\ 0 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 3.

Conclusion. A basis is the set of original pivot columns displayed above, and the dimension is 3.

Exercise 4.5.4: Find a dimension from a checked basis

Separate the independent parameter contributions.

The given vector expression is a linear combination of the following coefficient columns, in the order of its printed parameters. Their column space is exactly the requested subspace.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 0 \\ 3 & -1 \\ 1 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -1 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 2.

Conclusion. A basis is the set of original pivot columns displayed above, and the dimension is 2.

Exercise 4.5.5: Find a dimension from a checked basis

Separate the independent parameter contributions.

The given vector expression is a linear combination of the following coefficient columns, in the order of its printed parameters. Their column space is exactly the requested subspace.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & -2 & 0 \\ 2 & 0 & 5 \\ 0 & -2 & 2 \\ -3 & 0 & 6 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \\ -3 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 5 \\ 2 \\ 6 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 3.

Conclusion. A basis is the set of original pivot columns displayed above, and the dimension is 3.

Exercise 4.5.6: Find a dimension from a checked basis

Separate the independent parameter contributions.

The given vector expression is a linear combination of the following coefficient columns, in the order of its printed parameters. Their column space is exactly the requested subspace.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 3 & 0 & -1 \\ 0 & -1 & -3 \\ -7 & 6 & 5 \\ -3 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 3 \\ 0 \\ -7 \\ -3 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 6 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ -3 \\ 5 \\ 1 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This

proves independence. There are no remaining generators to express: every original column is included. The column span consequently has dimension 3.

Conclusion. A basis is the set of original pivot columns displayed above, and the dimension is 3.

Exercise 4.5.7: A homogeneous system defining the zero space

The second equation gives $b = 2c$. Substitution into the third, $2b - c = 0$, gives $4c - c = 3c = 0$, so $c = 0$ and then $b = 0$. The first equation now yields $a = 0$. Thus the subspace contains only $(0, 0, 0)$.

Conclusion. (a) The empty set is a basis of the zero subspace. (b) Its dimension is 0. The singleton containing zero is not a basis, because it is dependent.

Exercise 4.5.8: One constraint in four coordinates

Solve for a: $a = 3b - c$, with b, c, d unrestricted. Then

$$(a, b, c, d)^T = b(3, 1, 0, 0)^T + c(-1, 0, 1, 0)^T + d(0, 0, 0, 1)^T.$$

This proves spanning. For a zero combination, coordinates 2,3,4 force the three coefficients to be zero, proving independence.

Conclusion. (a) The three displayed coefficient vectors form a basis. (b) The dimension is 3.

Exercise 4.5.9: An equal-coordinate plane

A vector whose first and third coordinates agree has form $(s, t, s)^T = s(1, 0, 1)^T + t(0, 1, 0)^T$. These two coefficient vectors are independent: a zero combination has first coordinate equal to the first weight and second coordinate equal to the second weight. Both therefore vanish.

Conclusion. The dimension is 2, with basis $\{(1, 0, 1)^T, (0, 1, 0)^T\}$.

Exercise 4.5.10: Three generators of one line

The second and third generators are -2 and -3 times $(1, -5)^T$. Thus all three generate the same line as this one nonzero vector. A nonzero singleton is independent, so $\{(1, -5)^T\}$ is a basis.

Conclusion. $\dim H = 1$, not three: redundant generators do not add dimensions.

Exercise 4.5.11: Find a dimension from a checked basis

Find a basis rather than counting the generators.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & 3 & -2 & 5 \\ 0 & 1 & -1 & 2 \\ 2 & 1 & 1 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 4. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \\ 2 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_3 = (1)v_1 + (-1)v_2$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 3.

Conclusion. A basis is the set of original pivot columns displayed above, and the dimension is 3.

Exercise 4.5.12: Find a dimension from a checked basis

Find a basis rather than counting the generators.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} 1 & -3 & -2 & -3 \\ -2 & -6 & 3 & 5 \\ 0 & 0 & 5 & 5 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 3. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ -6 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 3 \\ 5 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$v_4 = (-1)v_1 + (1)v_3$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 3.

Conclusion. A basis is the set of original pivot columns displayed above, and the dimension is 3.

Exercise 4.5.13: Nullity and column dimension from pivots

Count pivots and variable columns separately.

$$A = \begin{bmatrix} 1 & -6 & 9 & 0 & -2 \\ 0 & 1 & 2 & -4 & 5 \\ 0 & 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 21 & 0 & \frac{164}{5} \\ 0 & 1 & 2 & 0 & \frac{29}{5} \\ 0 & 0 & 0 & 1 & \frac{1}{5} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are 5 variable columns and 3 pivots, in columns 1, 2, 4. The free-variable columns are 3, 5.

Conclusion. $\dim \operatorname{Col} A = 3$ and $\dim \operatorname{Nul} A = 5 - 3 = 2$. The column-space dimension counts pivot columns; the null-space dimension counts free variables.

Exercise 4.5.14: Seven columns with four pivots

Count pivots and variable columns separately.

$$A = \begin{bmatrix} 1 & 2 & -4 & 3 & -2 & 6 & 0 \\ 0 & 0 & 0 & 1 & 0 & -3 & 7 \\ 0 & 0 & 0 & 0 & 1 & 4 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & -4 & 0 & 0 & 23 & 0 \\ 0 & 0 & 0 & 1 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

There are 7 variable columns and 4 pivots, in columns 1, 4, 5, 7. The free-variable columns are 2, 3, 6.

Conclusion. $\dim \operatorname{Col} A = 4$ and $\dim \operatorname{Nul} A = 7 - 4 = 3$. The column-space dimension counts pivot columns; the null-space dimension counts free variables.

Exercise 4.5.15: A nonconsecutive pivot pattern

Count pivots and variable columns separately.

$$A = \begin{bmatrix} 1 & 2 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 0 & 0 & -3 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

There are 5 variable columns and 3 pivots, in columns 1, 3, 4. The free-variable columns are 2, 5.

Conclusion. $\dim \operatorname{Col} A = 3$ and $\dim \operatorname{Nul} A = 5 - 3 = 2$. The column-space dimension counts pivot columns; the null-space dimension counts free variables.

Exercise 4.5.16: An invertible two-by-two matrix

Count pivots and variable columns separately.

$$A = \begin{bmatrix} 3 & 2 \\ -6 & 5 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

There are 2 variable columns and 2 pivots, in columns 1, 2. There are no free-variable columns.

Conclusion. $\dim \operatorname{Col} A = 2$ and $\dim \operatorname{Nul} A = 2 - 2 = 0$. The column-space dimension counts pivot columns; the null-space dimension counts free variables.

Exercise 4.5.17: Three nonzero triangular pivots

Count pivots and variable columns separately.

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

There are 3 variable columns and 3 pivots, in columns 1, 2, 3. There are no free-variable columns.

Conclusion. $\dim \operatorname{Col} A = 3$ and $\dim \operatorname{Nul} A = 3 - 3 = 0$. The column-space dimension counts pivot columns; the null-space dimension counts free variables.

Exercise 4.5.18: Two pivots and a zero row

Count pivots and variable columns separately.

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

There are 3 variable columns and 2 pivots, in columns 1, 2. The free-variable columns are 3.

Conclusion. $\dim \operatorname{Col} A = 2$ and $\dim \operatorname{Nul} A = 3 - 2 = 1$. The column-space dimension counts pivot columns; the null-space dimension counts free variables.

Exercise 4.5.19: Dimension definitions and missing hypotheses

(a) True. The original pivot columns form a basis of the column space, so their number equals its dimension.

(b) False as stated: a plane not through zero is not a subspace at all. For example $z = 1$ is a plane in $\mathbb{R}^3$ but excludes the zero vector. A plane *through the origin* is a two-dimensional subspace.

(c) False. $\mathcal{P}_4$ has basis $1, t, t^2, t^3, t^4$, of size five.

(d) False. An independent set can be shorter than n; for example $\{e_1\}$ is independent in $\mathbb{R}^2$ but does not span it.

(e) True. A spanning family of p vectors gives $\dim V \leq p$. More than p vectors means more than the dimension, forcing dependence.

Exercise 4.5.20: Subspace identity and spanning-set cautions

(a) False. Elements of $\mathbb{R}^2$ have two coordinates and are not literally elements of $\mathbb{R}^3$. The embedded set $\{(x, y, 0)\}$ is a subspace isomorphic to $\mathbb{R}^2$, which is a different assertion.

(b) False. Nullity counts *free* variables. For $A = I_n$ there are n variables but no free variables, so $\dim \operatorname{Nul} A = 0$.

(c) False. A one-dimensional line is spanned by its infinitely many vectors; an infinite spanning set need not be independent or minimal.

(d) False. A spanning set can be redundant; $\{e_1, e_2, e_1 + e_2\}$ spans $\mathbb{R}^2$ but is dependent.

(e) True. A three-vector basis of such a subspace is independent in $\mathbb{R}^3$, so it also spans $\mathbb{R}^3$. Hence the subspace is the whole space.

Exercise 4.5.21: Hermite polynomial coefficients

Record every coefficient, including zeros.

Relative to the standard basis $(1, t, t^2, t^3)$, the polynomial coefficient matrix is

$$A = \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 2 & 0 & -12 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 8 \end{bmatrix}.$$

It is triangular, with nonzero diagonal entries 1, 2, 4, 8. Thus its determinant is $64 \neq 0$, and its four columns are independent and span $\mathbb{R}^4$. The standard coefficient map is a linear bijection between $\mathcal{P}_3$ and $\mathbb{R}^4$, so the same two properties hold for the four polynomials.

Conclusion. The four Hermite polynomials form a basis of $\mathcal{P}_3$.

Exercise 4.5.22: Laguerre polynomial coefficients

Record every coefficient, including zeros.

Relative to the standard basis $(1, t, t^2, t^3)$, the polynomial coefficient matrix is

$$A = \begin{bmatrix} 1 & 1 & 2 & 6 \\ 0 & -1 & -4 & -18 \\ 0 & 0 & 1 & 9 \\ 0 & 0 & 0 & -1 \end{bmatrix}.$$

It is triangular, with nonzero diagonal entries 1, -1, 1, -1. Thus its determinant is $1 \neq 0$, and its four columns are independent and span $\mathbb{R}^4$. The standard coefficient map is a linear bijection between $\mathcal{P}_3$ and $\mathbb{R}^4$, so the same two properties hold for the four polynomials.

Conclusion. The four Laguerre polynomials form a basis of $\mathcal{P}_3$.

Exercise 4.5.23: Hermite coordinates of a cubic

What we must show. Express the given polynomial in the specified ordered basis. The coefficient equation uses the same ordered coefficient columns as in the preceding basis calculation:

$$\begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 2 & 0 & -12 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 8 \end{bmatrix} c = \begin{bmatrix} -1 \\ 0 \\ 8 \\ 8 \end{bmatrix}.$$

Solve from the last coefficient upward. This gives $c_4 = 1$, $c_3 = 2$, $c_2 = 6$, $c_1 = 3$. For a direct check, multiply the coefficient matrix by these weights; the result is exactly the target coefficient column displayed on the right.

Conclusion. $[p]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 6 \\ 2 \\ 1 \end{bmatrix}$.

Exercise 4.5.24: Laguerre coordinates of a quadratic

What we must show. Express the given polynomial in the specified ordered basis. The coefficient equation uses the same ordered coefficient columns as in the preceding basis calculation:

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & -1 & -4 \\ 0 & 0 & 1 \end{bmatrix} c = \begin{bmatrix} 5 \\ 5 \\ -2 \end{bmatrix}.$$

Solve from the last coefficient upward. This gives $c_3 = -2$, $c_2 = 3$, $c_1 = 6$. For a direct check, multiply the coefficient matrix by these weights; the result is exactly the target coefficient column displayed on the right.

Conclusion. $[p]_{\mathcal{B}} = \begin{bmatrix} 6 \\ 3 \\ -2 \end{bmatrix}$.

Exercise 4.5.25: A short set cannot span

Suppose S did span V. A finite spanning set contains a basis, obtained by deleting redundant vectors. That basis would have at most $|S| < n$ members, contradicting the fact that every basis of V has n members. Thus the assumed spanning property is impossible.

Conclusion. Fewer than n vectors cannot span an n-dimensional space.

Exercise 4.5.26: Equal finite dimensions force equality

Choose a basis $h_1, \ldots, h_n$ for H. Because $H \subseteq V$, these are also n independent vectors in the n-dimensional space V. The basis theorem says they span V. But their span is already H, so $V = H$.

Conclusion. A subspace with the full finite ambient dimension is the entire space.

Exercise 4.5.27: Why all polynomials have no finite basis

Take any finite set of polynomials. Choose an integer $d \geq 0$ at least as large as all their degrees. Every linear combination of them still has degree at most d, since addition and scalar multiplication cannot create a nonzero coefficient of t^{d+1}. Thus this set cannot span the polynomial t^{d+1}. No finite spanning set, and hence no finite basis, exists for all polynomials.

Conclusion. The polynomial space $\mathcal{P}$ is infinite-dimensional.

Exercise 4.5.28: Continuous functions contain arbitrarily large independent families

For every integer $m \geq 0$, the $m+1$ functions $1, t, t^2, \ldots, t^m$ are continuous and linearly independent on $\mathbb{R}$: an identical zero polynomial has every coefficient zero. If $C(\mathbb{R})$ had finite dimension d, take $m = d$. It would then contain $d+1$ independent vectors, contradicting the dimension bound.

Conclusion. $C(\mathbb{R})$ is infinite-dimensional because it contains independent families of arbitrarily large finite size.

Exercise 4.5.29: Compare spanning and independent family sizes

(a) True. Deleting redundancies from the p spanning vectors gives a basis with at most p members, so $\dim V \leq p$.

(b) True. An independent family extends to a basis; the basis therefore contains at least its original p vectors, so $\dim V \geq p$.

(c) True. Take any p-vector basis and adjoin the zero vector. Since $V \neq \{0\}$, we have $p \geq 1$, and zero is not a basis member. The resulting $p+1$ distinct vectors still span V.

Exercise 4.5.30: Counterexamples to incorrect dimension converses

(a) False. In $V = \mathbb{R}^{p+1}$, the p distinct vectors $0, e_1, \ldots, e_{p-1}$ are dependent, but $\dim V = p+1 > p$. For $p = 1$ this example is simply the singleton $\{0\}$ in $\mathbb{R}^2$.

(b) True. If instead $\dim V = d \leq p$, take a d-vector basis and, if necessary, add distinct redundant multiples of one nonzero basis vector until there are p distinct members. This p-element set spans, contradicting the assumption. Thus $d > p$.

(c) False in general. For $p = 3$, the two nonzero vectors $e_1, 2e_1$ in $\mathbb{R}^3$ are dependent, despite there being $p - 1 = 2$ of them. The special boundary case $p = 2$ would be true, since a single nonzero vector is independent; it does not make the general statement true.

Exercise 4.5.31: A linear image cannot gain dimension

Let $h_1, \ldots, h_d$ be a basis of H, where $d = \dim H$. For any $h \in H$, write $h = \sum_j c_j h_j$. Linearity gives $T(h) = \sum_j c_j T(h_j)$, so the d vectors $T(h_j)$ span $T(H)$. Removing redundant images gives a basis with at most d vectors. They may all vanish, in which case the image is the zero space of dimension zero.

Conclusion. $\dim T(H) \le d = \dim H$. Images cannot need more independent directions than the inputs supplying them.

Exercise 4.5.32: An injective image preserves dimension

As in Exercise 31, the images of a basis $h_1, \dots, h_d$ span $T(H)$. If $\sum_j c_j T(h_j) = 0$, linearity gives $T(\sum_j c_j h_j) = T(0)$. Injectivity yields $\sum_j c_j h_j = 0$; independence of the basis forces every $c_j = 0$. Thus the images are also independent and form a d-element basis of $T(H)$. Taking $H = V$ and using surjectivity when it is assumed gives $T(V) = W$, so $\dim W = \dim V$.

Conclusion. $\dim T(H) = \dim H$ for an injective T; in particular finite- dimensional isomorphic vector spaces have equal dimensions.

Exercise 4.5.33: Extend an independent family to a basis

(a) Append the standard coordinate directions.

Form $A = [v_1\ v_2\ v_3\ e_1\ \cdots\ e_5]$. The following reduction and pivot-column certificate give a basis; columns 4 through 8 refer to the standard vectors in order.

Put the generators in columns and locate the pivots.

$$A = \begin{bmatrix} -9 & 9 & 6 & 1 & 0 & 0 & 0 & 0 \\ -7 & 4 & 7 & 0 & 1 & 0 & 0 & 0 \\ 8 & 1 & -8 & 0 & 0 & 1 & 0 & 0 \\ -5 & 6 & 5 & 0 & 0 & 0 & 1 & 0 \\ 7 & -7 & -7 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & -\frac{1}{3} & 0 & 0 & 1 & \frac{3}{7} \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & \frac{5}{7} \\ 0 & 0 & 1 & -\frac{1}{3} & 0 & 0 & 0 & -\frac{3}{7} \\ 0 & 0 & 0 & 0 & 1 & 0 & 3 & \frac{22}{7} \\ 0 & 0 & 0 & 0 & 0 & 1 & -9 & -\frac{53}{7} \end{bmatrix}.$$

The pivot columns are 1, 2, 3, 5, 6. Thus a basis for the column span consists of the corresponding *original* columns:

$$\mathcal{B} = \left\{ \begin{bmatrix} -9 \\ -7 \\ 8 \\ -5 \\ 7 \end{bmatrix}, \begin{bmatrix} 9 \\ 4 \\ 1 \\ 6 \\ -7 \end{bmatrix}, \begin{bmatrix} 6 \\ 7 \\ -8 \\ 5 \\ -7 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

Check both properties required of a basis.

If a combination of these pivot columns is zero, applying the row operations produces a combination of distinct standard coordinate columns equal to zero. Each coefficient must therefore vanish. This proves independence. To see explicitly why they span all the original generators, read each nonpivot column from the reduced matrix:

$$\begin{aligned} v_4 &= (-\frac{1}{3})v_1 + (-\frac{1}{3})v_3 \\ v_7 &= (1)v_1 + (1)v_2 + (3)v_5 + (-9)v_6 \\ v_8 &= (\frac{3}{7})v_1 + (\frac{5}{7})v_2 + (-\frac{3}{7})v_3 + (\frac{22}{7})v_5 + (-\frac{53}{7})v_6 \end{aligned}$$

Row operations preserve these linear relations, so the displayed relations hold for the original columns too. The column span consequently has dimension 5.

(b) Explain why the construction works generally.

The original independent columns $v_1, \ldots, v_k$ each add a new pivot as the columns are processed left to right: none lies in the span of its predecessors. Thus all are retained among the pivot columns. Appending every e_j makes the full column space $\mathbb{R}^n$, because those columns alone already span $\mathbb{R}^n$. The pivot columns therefore form a basis of the whole space that contains all the original independent vectors.

Exercise 4.5.34: Cosine multiples give another basis

(a) Use the given identities to read coefficient columns.

In the ordered basis $\mathcal{B} = (1, \cos t, \ldots, \cos^6 t)$, the coordinate columns of $\mathcal{C} = (1, \cos t, \cos 2t, \ldots, \cos 6t)$ are

$$P = \begin{bmatrix} 1 & 0 & -1 & 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & -3 & 0 & 5 & 0 \\ 0 & 0 & 2 & 0 & -8 & 0 & 18 \\ 0 & 0 & 0 & 4 & 0 & -20 & 0 \\ 0 & 0 & 0 & 0 & 8 & 0 & -48 \\ 0 & 0 & 0 & 0 & 0 & 16 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 32 \end{bmatrix}.$$

The jth column is the coordinate vector of the jth function in $\mathcal{C}$. The matrix is triangular with diagonal $1, 1, 2, 4, 8, 16, 32$, whose product is $32768 \neq 0$. Hence its columns are independent. Because coordinates preserve dependence relations, the functions in $\mathcal{C}$ are independent too.

(b) Compare the number of independent functions with the dimension.

By Exercise 4.3.38, $\mathcal{B}$ is a seven-element basis of H, so $\dim H = 7$. The seven independent members of $\mathcal{C}$ lie in H by the supplied identities. An independent family of seven vectors in a seven-dimensional space is a basis.

Conclusion. $\mathcal{C}$ is a basis for H. The displayed columns give all requested $\mathcal{B}$ coordinates.

4.6 Rank

For an $m \times n$ matrix, the rank r is the number of pivots, the dimension of its column space, and the dimension of its row space. The rank–nullity equation is

$$\dim \operatorname{Nul} A = n - r.$$

Notice the number of *columns* on the right. Row operations preserve the row space and null space, but may change the column space. Consequently use pivot columns of the *original* matrix for a column-space basis; use nonzero rows of an echelon form for a row-space basis. A basis for a null space comes from solving the homogeneous equations with free variables.

Practice problem 4.6.P1: Rank and nullity from the given echelon form

The given echelon matrix B has two nonzero rows and two pivot positions. Thus $\operatorname{rank} A = 2$. Since A has five columns, rank–nullity gives $\dim \operatorname{Nul} A = 5 - 2 = 3$.

Practice problem 4.6.P2: Choose the correct rows and columns

The pivot columns of B are columns $1, 2$. Therefore a column-space basis uses the original columns $(2, 1, -7, 4)$ and $(-1, -2, 8, -5)$ of A. A row-space basis uses the two nonzero rows of B:

$$(1, -2, -4, 3, -2), \qquad (0, 3, 9, -12, 12).$$

The former vectors have four entries and live in $\mathbb{R}^4$; the latter have five entries and live in $\mathbb{R}^5$. This is a useful check on their roles.

Practice problem 4.6.P3: The next step toward a null-space basis

Continue reducing B to reduced echelon form: divide its second row by 3, then add twice that new row to its first. The nonzero rows become $(1, 0, 2, -5, 6)$ and $(0, 1, 3, -4, 4)$. From there take x_3, x_4, x_5 free, express x_1, x_2 in terms of them, and separate their coefficient vectors. It is the free variables, not the pivot columns, that directly produce this basis.

Practice problem 4.6.P4: Transpose does not change rank

The rank theorem gives $\operatorname{rank} A^T = \operatorname{rank} A = 2$. Therefore every row echelon form of A^T has exactly two pivot columns.

Exercise 4.6.1: Three bases from the supplied echelon form

The supplied B has 2 pivots, so $\operatorname{rank} A = 2$ and $\dim \operatorname{Nul} A = 4 - 2 = 2$, without row reducing A. Its pivot columns are 1, 2.

Column and row bases use different source matrices.

A matrix whose columns form a basis of $\operatorname{Col} A$ is

$$C = \begin{pmatrix} 1 & -4 \\ -1 & 2 \\ 5 & -6 \end{pmatrix}.$$

These are the pivot columns of A, not B. A matrix whose rows form a basis of $\operatorname{Row} A$ is

$$R = \begin{pmatrix} 1 & 0 & -1 & 5 \\ 0 & -2 & 5 & -6 \end{pmatrix}.$$

Solve the homogeneous equations for the free variables.

Back substitution in $Bx = 0$ gives a basis matrix

$$N = \begin{pmatrix} 2 & -5 \\ 5 & -3 \\ 2 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \operatorname{Nul} A = \{Nz : z \in \mathbb{R}^2\}.$$

The columns of N are the coefficient vectors of the free parameters (some have been multiplied by nonzero scalars to avoid fractions). Each solves $Bx = 0$, and hence $Ax = 0$. They are independent because their free-coordinate entries isolate the parameters; their number equals the number of free variables. Thus they span the entire null space, not only some solutions.

Exercise 4.6.2: Three bases from the supplied echelon form

The supplied B has 3 pivots, so $\operatorname{rank} A = 3$ and $\dim \operatorname{Nul} A = 5 - 3 = 2$, without row reducing A. Its pivot columns are 1, 3, 5.

Column and row bases use different source matrices.

A matrix whose columns form a basis of $\operatorname{Col} A$ is

$$C = \begin{pmatrix} 1 & 4 & 2 \\ 2 & 6 & -3 \\ 3 & 3 & -3 \\ 3 & 0 & 0 \end{pmatrix}.$$

These are the pivot columns of A, not B. A matrix whose rows form a basis of $\operatorname{Row} A$ is

$$R = \begin{pmatrix} 1 & 3 & 4 & -1 & 2 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & -5 \end{pmatrix}.$$

Solve the homogeneous equations for the free variables.

Back substitution in $Bx = 0$ gives a basis matrix

$$N = \begin{pmatrix} -3 & -3 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \operatorname{Nul} A = \{Nz : z \in \mathbb{R}^2\}.$$

The columns of N are the coefficient vectors of the free parameters (some have been multiplied by nonzero scalars to avoid fractions). Each solves $Bx = 0$, and hence $Ax = 0$. They are independent because their free-coordinate entries isolate the parameters; their number equals the number of free variables. Thus they span the entire null space, not only some solutions.

Exercise 4.6.3: Three bases from the supplied echelon form

The supplied B has 3 pivots, so $\operatorname{rank} A = 3$ and $\dim \operatorname{Nul} A = 6 - 3 = 3$, without row reducing A. Its pivot columns are 1, 2, 5.

Column and row bases use different source matrices.

A matrix whose columns form a basis of $\operatorname{Col} A$ is

$$C = \begin{pmatrix} 2 & 6 & 3 \\ -2 & -3 & 0 \\ 4 & 9 & 3 \\ -2 & 3 & 3 \end{pmatrix}.$$

These are the pivot columns of A, not B. A matrix whose rows form a basis of $\operatorname{Row} A$ is

$$R = \begin{pmatrix} 2 & 6 & -6 & 6 & 3 & 6 \\ 0 & 3 & 0 & 3 & 3 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 \end{pmatrix}.$$

Solve the homogeneous equations for the free variables.

Back substitution in $Bx = 0$ gives a basis matrix

$$N = \begin{pmatrix} 3 & 0 & -3 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \operatorname{Nul} A = \{Nz : z \in \mathbb{R}^3\}.$$

The columns of N are the coefficient vectors of the free parameters (some have been multiplied by nonzero scalars to avoid fractions). Each solves $Bx = 0$, and hence $Ax = 0$. They are independent because their free-coordinate entries isolate the parameters; their number equals the number of free variables. Thus they span the entire null space, not only some solutions.

Exercise 4.6.4: Three bases from the supplied echelon form

The supplied B has 5 pivots, so $\operatorname{rank} A = 5$ and $\dim \operatorname{Nul} A = 6 - 5 = 1$, without row reducing A. Its pivot columns are 1, 2, 3, 5, 6.

Column and row bases use different source matrices.

A matrix whose columns form a basis of $\operatorname{Col} A$ is

$$C = \begin{pmatrix} 1 & 1 & -2 & 1 & -2 \\ 1 & 2 & -3 & -2 & -3 \\ 1 & -1 & 0 & 1 & 6 \\ 1 & -2 & 2 & -3 & 0 \\ 1 & -2 & 1 & 2 & -1 \end{pmatrix}.$$

These are the pivot columns of A, not B. A matrix whose rows form a basis of $\operatorname{Row} A$ is

$$R = \begin{pmatrix} 1 & 1 & -2 & 0 & 1 & -2 \\ 0 & 1 & -1 & 0 & -3 & -1 \\ 0 & 0 & 1 & 1 & -13 & -1 \\ 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Solve the homogeneous equations for the free variables.

Back substitution in $Bx = 0$ gives a basis matrix

$$N = \begin{pmatrix} -1 \\ -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \operatorname{Nul} A = \{Nz : z \in \mathbb{R}^1\}.$$

The columns of N are the coefficient vectors of the free parameters (some have been multiplied by nonzero scalars to avoid fractions). Each solves $Bx = 0$, and hence $Ax = 0$. They are independent because their free-coordinate entries isolate the parameters; their number equals the number of free variables. Thus they span the entire null space, not only some solutions.

Exercise 4.6.5: A rank-three matrix with seven columns

Rank–nullity gives $\dim \operatorname{Nul} A = 7 - 3 = 4$. The row-space dimension equals rank, so $\dim \operatorname{Row} A = 3$. Transpose preserves rank, so $\operatorname{rank} A^T = 3$.

Exercise 4.6.6: A rank-two matrix with five columns

There are $5 - 2 = 3$ nonpivot columns, giving $\dim \operatorname{Nul} A = 3$. The rank theorem gives $\dim \operatorname{Row} A = 2$ and $\operatorname{rank} A^T = 2$.

Exercise 4.6.7: Full row rank does not identify a null space with its coordinate space

Four pivot columns in a 4×7 matrix mean rank 4, so its columns span all of $\mathbb{R}^4$: $\operatorname{Col} A = \mathbb{R}^4$. Rank–nullity gives nullity $7 - 4 = 3$. But $\operatorname{Nul} A$ is a three-dimensional subspace of $\mathbb{R}^7$, not the set $\mathbb{R}^3$ of triples. Equality $\operatorname{Nul} A = \mathbb{R}^3$ is therefore false; a choice of basis gives an isomorphism with $\mathbb{R}^3$, not literal equality of the sets.

Exercise 4.6.8: Rank four with six output coordinates

The nullity is $8-4=4$. The column space is a four-dimensional subspace of $\mathbb{R}^6$, because the columns have six entries. Thus it is not literally $\mathbb{R}^4$; dimension does not change the number of entries in its vectors.

Exercise 4.6.9: Recover rank from a known nullity

The matrix has six columns and nullity three, so its rank is $6-3=3$. Thus $\dim \operatorname{Col} A = 3$. But the columns have four entries, so $\operatorname{Col} A$ is a subspace of $\mathbb{R}^4$, not literally $\mathbb{R}^3$.

Exercise 4.6.10: Seven columns and five free coordinates

By rank–nullity, $\operatorname{rank} A = 7-5=2$. Column-space dimension equals rank, so the requested dimension is 2. The eight rows specify the ambient output space but do not replace the seven in the rank–nullity equation.

Exercise 4.6.11: From nullity to row-space dimension

Here $n=5$ and nullity is 3, so rank is $5-3=2$. The row-space dimension equals rank and is therefore 2.

Exercise 4.6.12: A second row-space dimension

There are four columns and nullity 2. Hence rank, and therefore row-space dimension, equals $4-2=2$.

Exercise 4.6.13: The largest possible rank

There can be no more pivots than rows or columns, so an $m \times n$ matrix has rank at most $\min(m,n)$. Both a 7×5 and a 5×7 matrix therefore have largest possible rank 5. This is attainable by placing a 5×5 identity matrix in the first five rows or columns and filling remaining entries with zeros.

Exercise 4.6.14: The largest row-space dimension

The row-space dimension is rank. For sizes 5×4 and 4×5, the maximum rank is 4 in each case. Identity blocks with an additional zero row or zero column show that this bound is attained.

Exercise 4.6.15: The smallest nullity for a wide matrix

A 3×7 matrix has rank at most 3, so its nullity is $7-\operatorname{rank} A \geq 7-3=4$. The matrix $[\,I_3\ 0\,]$ attains rank 3 and nullity 4. Thus the smallest possible nullity is 4.

Exercise 4.6.16: The smallest nullity for a tall matrix

A 7×5 matrix can have rank 5, so its nullity can be $5 - 5 = 0$. For example stack two zero rows below I_5. Nullity cannot be negative, so zero is the minimum.

Exercise 4.6.17: Five rank statements

(a) **True.** The columns of A^T are exactly the transposed rows of A; their spans are the same coordinate vectors.

(b) **False.** The first original rows need not be independent. For $A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ there are three pivots but the first three rows are dependent.

(c) **True.** Both dimensions equal rank, including for rectangular matrices.

(d) **False.** The sum equals the number of *columns*, n, not the number of rows m.

(e) **True.** Finite-precision rounding can turn an exact zero pivot into a small nonzero value or the reverse. Exact elementary operations preserve rank; its apparent computed value may change, as the Numerical Note explains.

Exercise 4.6.18: Five statements about row operations

(a) **False.** Use pivot columns of A for $\operatorname{Col} A$, not generally those of its echelon form. For example swapping the rows of $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ changes its column span.

(b) **False.** The row span is preserved, but the particular relations among the rows can change. Rows $(1, 0), (1, 0)$ satisfy $r_2 - r_1 = 0$; a row replacement gives rows $(1, 0), (0, 0)$, for which that relation no longer holds.

(c) **True.** Nullity equals the number of free variables, namely the number of nonpivot columns.

(d) **True.** Transposing the rows of A^T gives the columns of A.

(e) **True.** Every new row is a combination of old rows, and invertibility of elementary operations gives the reverse inclusion of row spans.

Exercise 4.6.19: A one-dimensional kernel of a five-row system

All homogeneous solutions are multiples of one nonzero solution, so nullity is 1. With six columns, rank is $6 - 1 = 5$, equal to the number of rows. Thus there is a pivot in every row, and $Ax = b$ is consistent for every right-hand side $b \in \mathbb{R}^5$. The answer is yes.

Exercise 4.6.20: Two free variables in an eight-variable system

A consistent system with eight variables and two free variables has six pivots. There are also six equations, so every row has a pivot. Therefore $Ax = b$ is consistent for every $b \in \mathbb{R}^6$; changing constants alone cannot make it inconsistent. The answer is no.

Exercise 4.6.21: Onto with just one free direction

Consistency for every right-hand side in a nine-row system means rank 9. With ten variables the nullity is $10 - 9 = 1$. Thus all homogeneous solutions are multiples of one vector. It is impossible to find two nonzero homogeneous solutions that are not multiples of each other.

Exercise 4.6.22: Too many variables for a one-dimensional kernel

For ten equations in twelve variables, rank is at most ten. Nullity is therefore at least $12 - 10 = 2$. It cannot be 1, so not all homogeneous solutions can be multiples of one fixed nonzero vector.

Exercise 4.6.23: Replace redundant equations

The two given nonproportional solutions are independent and span the kernel, so nullity is 2. There are eight variables, giving rank $8 - 2 = 6$. Row reduction therefore leaves six nonzero rows and six redundant zero rows. Keeping just the six nonzero homogeneous equations describes exactly the same solutions. Fewer than six cannot suffice: a matrix with fewer rows would have rank below six and nullity above two.

Exercise 4.6.24: A unique solution for one right-hand side versus all

A 7×6 matrix may have rank 6, giving zero kernel. Then every consistent system has a unique solution; for example $A = \begin{pmatrix} I_6 \\ 0 \end{pmatrix}$ and $b = 0$ do. But rank is at most $6 < 7$, so the column space cannot be all of $\mathbb{R}^7$. Thus a unique solution for some right-hand side is possible, while a unique solution for every right-hand side is impossible.

Exercise 4.6.25: One missing pivot row leaves some constants unattainable

Twelve variables and three free variables mean rank 9. The system has ten rows, so it lacks a pivot in one row. Its column space is a proper subspace of $\mathbb{R}^{10}$, and some right-hand sides lie outside it. Thus the scientist cannot guarantee a solution after changing the constants; indeed there necessarily exist choices of constants that make the system inconsistent.

Exercise 4.6.26: Full rank for a tall matrix

When $m > n$, the largest possible rank is n. The columns are independent if and only if all n columns are pivot columns, which is equivalent to rank n. This is precisely full rank for the given shape of matrix.

Exercise 4.6.27: Locate the four fundamental spaces

Rows of A and inputs to A have n entries; columns of A and inputs to A^T have m entries. Consequently

$$\begin{array}{c|cc} \mathbb{R}^n & \operatorname{Row} A = \operatorname{Col} A^T, & \operatorname{Nul} A \\ \mathbb{R}^m & \operatorname{Col} A = \operatorname{Row} A^T, & \operatorname{Nul} A^T \end{array}$$

The six names reduce to four fundamental spaces after the two displayed identifications. For special matrices some of these four may coincide as sets (for example several can be the zero space); "four" refers to the four roles, not a guarantee of four different sets in every example.

Exercise 4.6.28: Two rank–nullity identities

(a) Apply rank–nullity to A.

Since $\dim \operatorname{Row} A = \operatorname{rank} A$ and $\dim \operatorname{Nul} A = n - \operatorname{rank} A$, their sum is n.

(b) Apply rank–nullity to A^T.

This transpose has m columns and rank equal to that of A. Thus $\dim \operatorname{Nul} A^T = m - \operatorname{rank} A$; adding $\dim \operatorname{Col} A = \operatorname{rank} A$ gives m.

Exercise 4.6.29: Onto is equivalent to a trivial transpose kernel

The equation $Ax = b$ is solvable for every $b \in \mathbb{R}^m$ if and only if $\dim \operatorname{Col} A = m$. By Exercise 4.6.28(b), this is equivalent to $\dim \operatorname{Nul} A^T = 0$, which is equivalent to $\operatorname{Nul} A^T = \{0\}$. That last condition says exactly that $A^T x = 0$ has only the trivial solution.

Exercise 4.6.30: The augmented-rank consistency criterion

Adding b as a new column increases rank precisely when b is outside the span of the old columns. Therefore $Ax = b$ is consistent if and only if

$$\operatorname{rank}[\, A \;\; b \,] = \operatorname{rank} A.$$

Equivalently, an inconsistent row in the reduced augmented matrix gives an extra pivot in its last column. If ranks agree, that extra pivot is absent.

Exercise 4.6.31: An outer product has at most one independent column

For $u = (2, -3, 5)$ and $v = (a, b, c)$,

$$uv^T = \begin{pmatrix} 2a & 2b & 2c \\ -3a & -3b & -3c \\ 5a & 5b & 5c \end{pmatrix}.$$

Its columns are au, bu, cu, so its column space is contained in $\operatorname{Span}\{u\}$, a one-dimensional space. Hence rank is at most 1. It is zero when $v = 0$, and otherwise is exactly 1, since $u \neq 0$.

Exercise 4.6.32: Factor a rank-one matrix

With $u = (1, 2)$ the matrix uv^T has first row v^T and second row $2v^T$. Thus choose $v = (1, -3, 4)$, giving

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} [\, 1 \;\; -3 \;\; 4 \,] = \begin{pmatrix} 1 & -3 & 4 \\ 2 & -6 & 8 \end{pmatrix}.$$

Exercise 4.6.33: Every nonzero column generates a rank-one column space

If rank is 1 and the first column u is nonzero, it is a basis for the column space. Thus the three columns have form $u, \alpha u, \beta u$. Taking $v = (1, \alpha, \beta)$ gives $A = uv^T$. If the first column is zero, choose any other nonzero column as u (one exists because rank is 1). Express each column as $v_j u$, including coefficient zero for a zero column, and set $v = (v_1, v_2, v_3)$. The same outer-product formula then holds.

Exercise 4.6.34: A rank-r matrix is a sum of r rank-one matrices

Since U is obtained by elementary row operations, $U = FA$ for an invertible product F of elementary matrices. Set $E = F^{-1}$ to get $A = EU$. Write $E = [\, e_1 \ \cdots \ e_m \,]$ and let u_i^T be row i of U. Matrix multiplication by columns and rows gives

$$A = EU = \sum_{i=1}^{m} e_i u_i^T = \sum_{i=1}^{r} e_i u_i^T.$$

The last equality uses the zero rows below the first r rows of the echelon matrix. Each of those first r rows is nonzero, and each column e_i of an invertible matrix is nonzero. Thus every displayed outer product has rank exactly 1, proving the assertion.

Exercise 4.6.35: Four fundamental bases and two invertible assembled matrices

$$A = \begin{pmatrix} 7 & -9 & -4 & 5 & 3 & -3 & -7 \\ -4 & 6 & 7 & -2 & -6 & -5 & 5 \\ 5 & -7 & -6 & 5 & -6 & 2 & 8 \\ -3 & 5 & 8 & -1 & -7 & -4 & 8 \\ 6 & -8 & -5 & 4 & 4 & 9 & 3 \end{pmatrix}.$$

(a) Supply and verify the three requested bases.

The pivot columns are 1, 2, 4, 6, so rank is 4. The following matrices give respectively a column-space basis in their columns, a null-space basis in their columns, and a row-space basis in their rows:

$$C = \begin{pmatrix} 7 & -9 & 5 & -3 \\ -4 & 6 & -2 & -5 \\ 5 & -7 & 5 & 2 \\ -3 & 5 & -1 & -4 \\ 6 & -8 & 4 & 9 \end{pmatrix}.$$

$$N = \begin{pmatrix} -13 & -10 & 3 \\ -11 & -1 & -2 \\ 2 & 0 & 0 \\ 0 & 11 & -7 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$R = \begin{pmatrix} 1 & 0 & \frac{13}{2} & 0 & 5 & 0 & -3 \\ 0 & 1 & \frac{11}{2} & 0 & \frac{1}{2} & 0 & 2 \\ 0 & 0 & 0 & 1 & -\frac{11}{2} & 0 & 7 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix}.$$

The rows of R are the nonzero rows of the reduced echelon form. The 3 columns of N result from its free-variable equations; exact multiplication gives $AN = 0$, and their free coordinates show independence. Thus these are bases, not only spanning lists.

(b) Supply the transpose-kernel basis and check invertibility.

$$M = \begin{pmatrix} 2 \\ 41 \\ 0 \\ -28 \\ 11 \end{pmatrix}.$$

Here $A^T M = 0$, and its 1 columns are independent. Form $S = [\, R^T \; N \,]$ and $T = [\, C \; M \,]$ using the displayed blocks. There are $4 + 3 = 7$ columns in $S \in M_{7\times 7}$ and $4 + 1 = 5$ columns in $T \in M_{5\times 5}$. Exact determinants are

$$\det S = -\frac{88369}{2}, \qquad \det T = 10360.$$

Both are nonzero, verifying that both square matrices are invertible. Conceptually, a vector in both $\operatorname{Row} A$ and $\operatorname{Nul} A$ has form $x = A^T y$ and satisfies $Ax = 0$, so $x^T x = y^T Ax = 0$. Since $x^T x$ is a sum of squares, $x = 0$. The same argument with A^T treats $\operatorname{Col} A$ and $\operatorname{Nul} A^T$; their dimensions add to the full ambient dimensions. This explains why assembling their bases produces invertible matrices.

Exercise 4.6.36: Four fundamental bases and two invertible assembled matrices

The problem permits a generated example. Here is one reproducible sample: with Python `random.Random(46)`, draw the entries of a 6×2 matrix J and then a 2×7 matrix K successively from `randint(-1,1)`, and set $A = JK$. This guarantees rank at most 2, which in particular is at most 4. The resulting matrix is displayed below; its first two columns are independent, so its actual rank is 2.

$$A = \begin{pmatrix} 1 & 1 & 0 & -1 & 1 & 1 & 1 \\ 2 & 1 & 1 & -1 & 2 & 0 & 1 \\ -2 & -1 & -1 & 1 & -2 & 0 & -1 \\ 0 & -1 & 1 & 1 & 0 & -2 & -1 \\ 2 & 1 & 1 & -1 & 2 & 0 & 1 \\ -2 & -1 & -1 & 1 & -2 & 0 & -1 \end{pmatrix}.$$

(a) Supply and verify the three requested bases.

The pivot columns are 1, 2, so rank is 2. The following matrices give respectively a column-space

basis in their columns, a null-space basis in their columns, and a row-space basis in their rows:

$$C = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ -2 & -1 \\ 0 & -1 \\ 2 & 1 \\ -2 & -1 \end{pmatrix}.$$

$$N = \begin{pmatrix} -1 & 0 & -1 & 1 & 0 \\ 1 & 1 & 0 & -2 & -1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

$$R = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & -1 & 0 & 2 & 1 \end{pmatrix}.$$

The rows of R are the nonzero rows of the reduced echelon form. The 5 columns of N result from its free-variable equations; exact multiplication gives $AN = 0$, and their free coordinates show independence. Thus these are bases, not only spanning lists.

(b) Supply the transpose-kernel basis and check invertibility.

$$M = \begin{pmatrix} 0 & 2 & 0 & 0 \\ 1 & -1 & -1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Here $A^T M = 0$, and its 4 columns are independent. Form $S = [\, R^T \; N \,]$ and $T = [\, C \; M \,]$ using the displayed blocks. There are $2 + 5 = 7$ columns in $S \in M_{7\times 7}$ and $2 + 4 = 6$ columns in $T \in M_{6\times 6}$. Exact determinants are

$$\det S = 23, \qquad \det T = -21.$$

Both are nonzero, verifying that both square matrices are invertible. Conceptually, a vector in both $\operatorname{Row} A$ and $\operatorname{Nul} A$ has form $x = A^T y$ and satisfies $Ax = 0$, so $x^T x = y^T Ax = 0$. Since $x^T x$ is a sum of squares, $x = 0$. The same argument with A^T treats $\operatorname{Col} A$ and $\operatorname{Nul} A^T$; their dimensions add to the full ambient dimensions. This explains why assembling their bases produces invertible matrices.

Exercise 4.6.37: Reconstruct the original matrix from pivot columns

Use the C and R displayed in Exercise 4.6.35. Those are precisely the pivot columns of the original A and the nonzero rows of its reduced echelon form. Multiplication gives

$$CR = \begin{pmatrix} 7 & -9 & -4 & 5 & 3 & -3 & -7 \\ -4 & 6 & 7 & -2 & -6 & -5 & 5 \\ 5 & -7 & -6 & 5 & -6 & 2 & 8 \\ -3 & 5 & 8 & -1 & -7 & -4 & 8 \\ 6 & -8 & -5 & 4 & 4 & 9 & 3 \end{pmatrix} = A.$$

For example, the third column of R is $(13/2, 11/2, 0, 0)$, so the third column of CR is $(13/2)a_1 + (11/2)a_2 = a_3$. The reduced matrix records the coefficients of every original column in terms of the pivot columns. Thus this product restores all columns exactly.

Exercise 4.6.38: Three generated rank factorizations and a general proof

For a reproducible family, use `random.Random(4600+r)` for each $r = 5, 4, 3$. Form C by stacking I_r over a $(5 - r) \times r$ array of successive `randint(-2,2)` draws, and form R by placing an $r \times (7 - r)$ array of subsequent draws after I_r. Set $A = CR$. The first r columns of A are C; its reduced echelon form has nonzero rows R. Thus these are actual examples of the exercise's construction, with exact rank r, rather than uncontrolled random ranks.

The rank-5 sample.

$$C = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \qquad R = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 2 & -1 \\ 0 & 1 & 0 & 0 & 0 & -1 & -2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 0 \end{pmatrix}.$$

$$CR = A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 2 & -1 \\ 0 & 1 & 0 & 0 & 0 & -1 & -2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 0 \end{pmatrix}.$$

The rank-4 sample.

$$C = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -2 & 0 & 2 & 1 \end{pmatrix}, \qquad R = \begin{pmatrix} 1 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 1 & 0 & 0 & -2 & -1 & -1 \\ 0 & 0 & 1 & 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & 1 & -1 & 0 & 2 \end{pmatrix}.$$

$$CR = A = \begin{pmatrix} 1 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 1 & 0 & 0 & -2 & -1 & -1 \\ 0 & 0 & 1 & 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & 1 & -1 & 0 & 2 \\ -2 & 0 & 2 & 1 & -3 & 2 & 2 \end{pmatrix}.$$

The rank-3 sample.

$$C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & 2 & -2 \\ 1 & 2 & 2 \end{pmatrix}, \qquad R = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & -2 & 0 \\ 0 & 1 & 0 & 2 & -2 & 2 & -1 \\ 0 & 0 & 1 & 2 & -1 & -2 & 1 \end{pmatrix}.$$

$$CR = A = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & -2 & 0 \\ 0 & 1 & 0 & 2 & -2 & 2 & -1 \\ 0 & 0 & 1 & 2 & -1 & -2 & 1 \\ 2 & 2 & -2 & 2 & -2 & 4 & -4 \\ 1 & 2 & 2 & 9 & -6 & -2 & 0 \end{pmatrix}.$$

Prove the observed equality for every matrix.

Let C consist of the original pivot columns of a rank-r matrix A, and R of the nonzero rows of its reduced echelon form U. In U, the pivot columns are the first r coordinate vectors above the zero rows. Therefore the entries of column j of R express column j of U as a combination of its pivot columns. Elementary row operations preserve all column dependence relations: if $U = EA$ with E invertible, then $U\alpha = 0$ if and only if $A\alpha = 0$. The same coefficients thus express column j of A using columns of C. This means column j of CR is column j of A, for every j; hence $CR = A$.

4.7 Change of basis

A vector stays the same when a basis changes; its list of coordinates changes. We use $P_{C\leftarrow B}$ to mean *from B-coordinates to C-coordinates*:

$$[x]_C = P_{C\leftarrow B}[x]_B, \qquad P_{C\leftarrow B} = [\,[b_1]_C \ \cdots \ [b_n]_C\,].$$

In $\mathbb{R}^n$, if B_0, C_0 are the matrices of the basis vectors in standard coordinates, then $P_{C\leftarrow B} = C_0^{-1}B_0$. In practice reduce $[\,C_0 \ B_0\,]$ to $[\,I \ P_{C\leftarrow B}\,]$. Reversing the change uses the inverse matrix; swapping the labels without inverting is incorrect.

Practice problem 4.7.P1: Read the coordinate labels on the columns

The columns of P are $[f_1]_G, [f_2]_G$. If $v = af_1+bf_2$, linearity of coordinates gives $[v]_G = a[f_1]_G+b[f_2]_G = P(a,b)^T = P[v]_F$. Thus the correct choice is **(ii)**.

Practice problem 4.7.P2: Reverse the change from the example

Example 1 gives $P_{C\leftarrow B} = \left(\begin{smallmatrix} 4 & -6 \\ 1 & 1 \end{smallmatrix}\right)$. Its determinant is $4 + 6 = 10$, so the inverse is

$$P_{B\leftarrow C} = \frac{1}{10}\begin{pmatrix} 1 & 6 \\ -1 & 4 \end{pmatrix}.$$

Multiplying it by the original matrix gives I_2, confirming that changing there and back restores every coordinate vector.

Exercise 4.7.1: Construct columns directly from the given equations

(a) Record the C-coordinates of each b-vector.

From $b_1 = 6c_1 - 2c_2$ and $b_2 = 9c_1 - 4c_2$,

$$P_{C\leftarrow B} = \begin{pmatrix} 6 & 9 \\ -2 & -4 \end{pmatrix}.$$

(b) Apply that change to the specified vector.

For $x = -3b_1 + 2b_2$, its B-coordinate vector is $(-3, 2)$, so

$$[x]_C = \begin{pmatrix} 6 & 9 \\ -2 & -4 \end{pmatrix}\begin{pmatrix} -3 \\ 2 \end{pmatrix} = \begin{pmatrix} -18+18 \\ 6-8 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}.$$

Exercise 4.7.2: A singular matrix reveals a printed inconsistency

The supplied relations are $b_1 = -2c_1 + 4c_2$ and $b_2 = 3c_1 - 6c_2$. But then $b_2 = -\frac{3}{2}b_1$, so $\{b_1, b_2\}$ is dependent and cannot be a basis. These relations are incompatible with the printed assumption that B and C are both bases. This is visible in the formal column matrix

$$Q = \begin{pmatrix} -2 & 3 \\ 4 & -6 \end{pmatrix}, \qquad \det Q = 12 - 12 = 0.$$

(a) State what the formal calculation does and does not mean.

The columns are the given C-coordinates, but Q is not a change-of-basis matrix between two bases and has no inverse. There is no valid instance satisfying all the printed assumptions.

(b) Evaluate the specified linear combination nevertheless.

The vector $x = 2b_1 + 3b_2$ is meaningful even for dependent generators:

$$x = (-4 + 9)c_1 + (8 - 18)c_2 = 5c_1 - 10c_2.$$

Thus $[x]_C = (5, -10)$. The coefficient list $(2, 3)$ is one representation using b_1, b_2, but cannot be called a unique B-coordinate vector.

Exercise 4.7.3: The output coordinate system matches the columns

The columns are $[u_1]_W, [u_2]_W$, so multiplying by the coefficients of x in U forms the W-coordinates of x. Hence $[x]_W = P[x]_U$, choice **(ii)**. The reverse equation would need P^{-1}.

Exercise 4.7.4: A three-vector version of the same rule

Here P has columns $[d_1]_A, [d_2]_A, [d_3]_A$. Therefore the input coefficients are D-coordinates, and the output is an A-coordinate vector: $[x]_A = P[x]_D$. The correct choice is **(i)**.

Exercise 4.7.5: Translate three linear-combination equations

(a) Put the B-coordinates of a_1, a_2, a_3 into columns.

The equations give

$$P_{B \leftarrow A} = \begin{pmatrix} 4 & -1 & 0 \\ -1 & 1 & 1 \\ 0 & 1 & -2 \end{pmatrix}.$$

(b) Multiply by the A-coordinate vector.

For $x = 3a_1 + 4a_2 + a_3$,

$$[x]_B = \begin{pmatrix} 4 & -1 & 0 \\ -1 & 1 & 1 \\ 0 & 1 & -2 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 8 \\ 2 \\ 2 \end{pmatrix}.$$

For example, the first output entry is $4(3) - 1(4) + 0(1) = 8$.

Exercise 4.7.6: Change from F to D

The columns must be $[f_1]_D, [f_2]_D, [f_3]_D$, giving

$$P_{D \leftarrow F} = \begin{pmatrix} 2 & 0 & -3 \\ -1 & 3 & 0 \\ 1 & 1 & 2 \end{pmatrix}.$$

For $x = f_1 - 2f_2 + 2f_3$, multiply by $(1, -2, 2)$ to obtain

$$[x]_D = (2 - 6,\ -1 - 6,\ 1 - 2 + 4) = (-4, -7, 3).$$

This answers (a) and (b), keeping the direction of the coordinate change explicit.

Exercise 4.7.7: Compute both directions in standard coordinates

The standard-coordinate basis matrices are $B_0 = \begin{pmatrix} 7 & -3 \\ 5 & -1 \end{pmatrix}$ and $C_0 = \begin{pmatrix} 1 & -2 \\ -5 & 2 \end{pmatrix}$. Solve $C_0 P = B_0$ column by column, or reduce $[\,C_0\ B_0\,]$, to obtain

$$P_{C \leftarrow B} = \begin{pmatrix} -3 & 1 \\ -5 & 2 \end{pmatrix}, \qquad P_{B \leftarrow C} = \begin{pmatrix} -2 & 1 \\ -5 & 3 \end{pmatrix}.$$

The first identity is checked by $C_0 P_{C \leftarrow B} = B_0$; the second is the inverse, checked by $P_{B \leftarrow C} P_{C \leftarrow B} = I_2$. For example, column one of $C_0 P$ reconstructs b_1 as a combination of the c-vectors. These checks ensure that the two directions have not been interchanged.

Exercise 4.7.8: Compute both directions in standard coordinates

The standard-coordinate basis matrices are $B_0 = \begin{pmatrix} -1 & 1 \\ 8 & -7 \end{pmatrix}$ and $C_0 = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}$. Solve $C_0 P = B_0$ column by column, or reduce $[\,C_0\ B_0\,]$, to obtain

$$P_{C \leftarrow B} = \begin{pmatrix} 9 & -8 \\ -10 & 9 \end{pmatrix}, \qquad P_{B \leftarrow C} = \begin{pmatrix} 9 & 8 \\ 10 & 9 \end{pmatrix}.$$

The first identity is checked by $C_0 P_{C \leftarrow B} = B_0$; the second is the inverse, checked by $P_{B \leftarrow C} P_{C \leftarrow B} = I_2$. For example, column one of $C_0 P$ reconstructs b_1 as a combination of the c-vectors. These checks ensure that the two directions have not been interchanged.

Exercise 4.7.9: Compute both directions in standard coordinates

The standard-coordinate basis matrices are $B_0 = \begin{pmatrix} 4 & 8 \\ 4 & 4 \end{pmatrix}$ and $C_0 = \begin{pmatrix} 2 & -2 \\ 2 & 2 \end{pmatrix}$. Solve $C_0 P = B_0$ column by column, or reduce $[\,C_0\ B_0\,]$, to obtain

$$P_{C \leftarrow B} = \begin{pmatrix} 2 & 3 \\ 0 & -1 \end{pmatrix}, \qquad P_{B \leftarrow C} = \begin{pmatrix} \frac{1}{2} & \frac{3}{2} \\ 0 & -1 \end{pmatrix}.$$

The first identity is checked by $C_0 P_{C \leftarrow B} = B_0$; the second is the inverse, checked by $P_{B \leftarrow C} P_{C \leftarrow B} = I_2$. For example, column one of $C_0 P$ reconstructs b_1 as a combination of the c-vectors. These checks ensure that the two directions have not been interchanged.

Exercise 4.7.10: Compute both directions in standard coordinates

The standard-coordinate basis matrices are $B_0 = \begin{pmatrix} 6 & 4 \\ -12 & 2 \end{pmatrix}$ and $C_0 = \begin{pmatrix} 4 & 3 \\ 2 & 9 \end{pmatrix}$. Solve $C_0P = B_0$ column by column, or reduce $[\,C_0\ B_0\,]$, to obtain

$$P_{C\leftarrow B} = \begin{pmatrix} 3 & 1 \\ -2 & 0 \end{pmatrix}, \qquad P_{B\leftarrow C} = \begin{pmatrix} 0 & -\frac{1}{2} \\ 1 & \frac{3}{2} \end{pmatrix}.$$

The first identity is checked by $C_0P_{C\leftarrow B} = B_0$; the second is the inverse, checked by $P_{B\leftarrow C}P_{C\leftarrow B} = I_2$. For example, column one of C_0P reconstructs b_1 as a combination of the c-vectors. These checks ensure that the two directions have not been interchanged.

Exercise 4.7.11: Two statements about the direction of a change

(a) False.

The columns of $P_{C\leftarrow B}$ are the C-coordinates of the B vectors, $[b_j]_C$, not the B-coordinates of the C vectors. The latter columns belong to its inverse $P_{B\leftarrow C}$.

(b) True.

If C is the standard basis of $\mathbb{R}^n$, $[x]_C = x$ and $[b_j]_C = b_j$. Thus $P_{C\leftarrow B} = [\,b_1\ \cdots\ b_n\,] = P_B$.

Exercise 4.7.12: Independence and an algorithm's direction

(a) True.

A change-of-coordinates matrix between bases is invertible: composing it with the reverse change gives identity. By the Invertible Matrix Theorem, its columns are therefore independent.

(b) False.

Reducing $[\,c_1\ c_2\ b_1\ b_2\,]$ to $[\,I\ P\,]$ computes $P = C_0^{-1}B_0$. It therefore satisfies $[x]_C = P[x]_B$, the reverse of the printed assertion. The printed equation would use P^{-1} instead.

Exercise 4.7.13: Change from a polynomial basis to coefficients

The standard coordinate vector of $a + bt + ct^2$ is (a, b, c). Consequently, reading coefficients of the three given polynomials gives

$$P_{C\leftarrow B} = \begin{pmatrix} 1 & 3 & 0 \\ -2 & -5 & 2 \\ 1 & 4 & 3 \end{pmatrix}.$$

To find $[-1 + 2t]_B$, solve $Pz = (-1, 2, 0)$. Exact elimination gives $z = (5, -2, 1)$. Check directly:

$$5(1 - 2t + t^2) - 2(3 - 5t + 4t^2) + (2t + 3t^2) = -1 + 2t.$$

Thus the coordinate vector is $(5, -2, 1)$, and the displayed matrix is the requested change to the standard basis.

Exercise 4.7.14: Represent t^2 using three different polynomials

Reading constant, linear, and quadratic coefficients gives

$$P = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ -3 & -5 & 0 \end{pmatrix}.$$

Solve $Pz = (0, 0, 1)$. Its first two equations are $z_1 + 2z_2 + z_3 = 0$ and $z_2 + 2z_3 = 0$, so $z_2 = -2z_3$ and $z_1 = 3z_3$. The last equation gives $-9z_3 + 10z_3 = 1$, hence $z_3 = 1$. Therefore

$$t^2 = 3(1 - 3t^2) - 2(2 + t - 5t^2) + (1 + 2t).$$

The coefficients $(3, -2, 1)$ can be checked by canceling constant and linear terms.

Exercise 4.7.15: Justify the construction of the change matrix

(a) B spans V, because B is a basis. Therefore v has an expansion $v = \sum_j x_j b_j$.

(b) The coordinate mapping determined by a basis is linear. It takes that expansion to $[v]_C = \sum_j x_j [b_j]_C$.

(c) This is the definition of a matrix-vector product as a linear combination of the matrix's columns.

(d) The column $(x_1, \ldots, x_n)$ is $[v]_B$, by the definition and uniqueness of coordinates relative to B.

These observations prove $[v]_C = P_{C \leftarrow B}[v]_B$ for every vector v.

Exercise 4.7.16: Justify uniqueness of the change matrix

(a) $[b_1]_B = e_1$, so $Q[b_1]_B = Qe_1$ is the first column of Q.

(b) For a general k, column k must be $[b_k]_C$.

(c) Indeed $[b_k]_B = e_k$, and multiplying Q by e_k selects its kth column.

Thus every column of Q is forced. It must equal $[\,[b_1]_C \ \cdots \ [b_n]_C\,] = P_{C \leftarrow B}$.

Exercise 4.7.17: Change between powers of cosine and multiple angles

Write $x_j(t) = \cos^j t$, $y_j(t) = \cos(jt)$ for $j = 0, \ldots, 6$.

(a) Form P from the multiple-angle identities, then invert it.

The angle-addition formulas recursively give the polynomials $T_0(u) = 1$, $T_1(u) = u$, $T_{j+1}(u) =$

$2uT_j(u) - T_{j-1}(u)$ with $y_j(t) = T_j(\cos t)$. Taking their coefficients as columns gives

$$P = \begin{pmatrix} 1 & 0 & -1 & 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & -3 & 0 & 5 & 0 \\ 0 & 0 & 2 & 0 & -8 & 0 & 18 \\ 0 & 0 & 0 & 4 & 0 & -20 & 0 \\ 0 & 0 & 0 & 0 & 8 & 0 & -48 \\ 0 & 0 & 0 & 0 & 0 & 16 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 32 \end{pmatrix}.$$

$$P^{-1} = \begin{pmatrix} 1 & 0 & \frac{1}{2} & 0 & \frac{3}{8} & 0 & \frac{5}{16} \\ 0 & 1 & 0 & \frac{3}{4} & 0 & \frac{5}{8} & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & \frac{15}{32} \\ 0 & 0 & 0 & \frac{1}{4} & 0 & \frac{5}{16} & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & \frac{3}{16} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{32} \end{pmatrix}.$$

Exact multiplication gives $PP^{-1} = P^{-1}P = I_7$.

(b) Interpret every column, rather than just listing numbers.

Since $P = P_{B \leftarrow C}$, its inverse converts B-coordinates to C-coordinates. The jth x-vector has B-coordinates equal to the corresponding standard coordinate vector, so its C-coordinates are a column of P^{-1}. Reading those seven columns gives

$$\begin{aligned} 1 &= \cos 0t, \\ \cos t &= \cos t, \\ \cos^2 t &= \tfrac{1}{2} + \tfrac{1}{2}\cos 2t, \\ \cos^3 t &= \tfrac{3}{4}\cos t + \tfrac{1}{4}\cos 3t, \\ \cos^4 t &= \tfrac{3}{8} + \tfrac{1}{2}\cos 2t + \tfrac{1}{8}\cos 4t, \\ \cos^5 t &= \tfrac{5}{8}\cos t + \tfrac{5}{16}\cos 3t + \tfrac{1}{16}\cos 5t, \\ \cos^6 t &= \tfrac{5}{16} + \tfrac{15}{32}\cos 2t + \tfrac{3}{16}\cos 4t + \tfrac{1}{32}\cos 6t. \end{aligned}$$

Exercise 4.7.18: Integrate after changing the function basis

The integrand has B-coordinates $(0, 0, 0, 5, -6, 5, -12)$ in the notation of Exercise 4.7.17. Multiplication by P^{-1} gives its C-coordinates

$$(-6,\ 55/8,\ -69/8,\ 45/16,\ -3,\ 5/16,\ -3/8).$$

Thus it equals

$$-6 + \tfrac{55}{8}\cos t - \tfrac{69}{8}\cos 2t + \tfrac{45}{16}\cos 3t - 3\cos 4t + \tfrac{5}{16}\cos 5t - \tfrac{3}{8}\cos 6t.$$

Integrate these simple cosine terms using $\int \cos(jt)\,dt = \sin(jt)/j$ for $j > 0$:

$$\int (5\cos^3 t - 6\cos^4 t + 5\cos^5 t - 12\cos^6 t)\,dt$$

$$= -6t + \tfrac{55}{8}\sin t - \tfrac{69}{16}\sin 2t + \tfrac{15}{16}\sin 3t - \tfrac{3}{4}\sin 4t + \tfrac{1}{16}\sin 5t - \tfrac{1}{16}\sin 6t + C.$$

Differentiating the result restores the multiple-angle expression, which Exercise 4.7.17 identifies with the original polynomial in cosine.

Exercise 4.7.19: Find bases when a change matrix is prescribed

Let $V_0 = [\, v_1 \; v_2 \; v_3 \,]$ and let P be the supplied matrix.

(a) Its columns give the new u-vectors in v-coordinates.

For $P = P_{V \leftarrow U}$ we need $U_0 = V_0 P$. Multiplication gives

$$U_0 = \begin{pmatrix} -6 & -6 & -5 \\ -5 & -9 & 0 \\ 21 & 32 & 3 \end{pmatrix}.$$

Take its columns as u_1, u_2, u_3. Since V_0 and P are invertible (the latter has determinant -1), their product is invertible, so these columns are a basis.

(b) Solve the reversed requirement.

Now $P = P_{W \leftarrow V}$ means $V_0 = W_0 P$, so $W_0 = V_0 P^{-1}$. This gives

$$W_0 = \begin{pmatrix} 28 & 38 & 21 \\ -9 & -13 & -7 \\ -3 & 2 & 3 \end{pmatrix}.$$

Take these three columns as w_1, w_2, w_3. Their matrix is invertible for the same reason, and $W_0 P = V_0$ checks the required direction.

Exercise 4.7.20: Compose changes in the same order as their actions

(a) Apply the first change, then the second.

For every $x \in V$,

$$[x]_D = P_{D \leftarrow C}[x]_C = P_{D \leftarrow C} P_{C \leftarrow B}[x]_B.$$

By uniqueness of the change matrix, $P_{D \leftarrow B} = P_{D \leftarrow C} P_{C \leftarrow B}$. The rightmost factor acts first, exactly as in composition of functions.

(b) A concrete exact check.

Use the bases with standard-coordinate matrices

$$B_0 = \begin{pmatrix} 7 & -3 \\ 5 & -1 \end{pmatrix}, \quad C_0 = \begin{pmatrix} 1 & -2 \\ -5 & 2 \end{pmatrix}, \quad D_0 = \begin{pmatrix} -1 & 1 \\ 8 & -7 \end{pmatrix}.$$

The program calculation $C_0^{-1}B_0$, $D_0^{-1}C_0$, $D_0^{-1}B_0$ (using exact matrix inversion) gives

$$\begin{pmatrix} 2 & -12 \\ 3 & -14 \end{pmatrix} \begin{pmatrix} -3 & 1 \\ -5 & 2 \end{pmatrix} = \begin{pmatrix} 54 & -22 \\ 61 & -25 \end{pmatrix}.$$

The right side is $D_0^{-1}B_0$, independently verifying the composition rule.

4.8 Applications to difference equations

A signal in this section is a sequence $\{y_k\}$ defined for every integer k, including negative integers. For an order-n constant-coefficient homogeneous equation with nonzero first and last coefficients, specifying n consecutive values determines exactly one signal (Theorem 16). Its solution space therefore has dimension n (Theorem 17). To prove that n proposed solutions form a basis, verify the equations when necessary and prove independence. One convenient independence test is the Casorati matrix whose columns are the signals evaluated at n consecutive indices. If that matrix is invertible at one index, the signals are independent. Fewer than n solutions cannot span an n-dimensional space.

Practice problem 4.8.P1: A three-signal basis

The problem already supplies that 2^k, $3^k \sin(k\pi/2)$, and $3^k \cos(k\pi/2)$ solve the third-order equation. Their values at $0, 1, 2$ form the matrix

$$C(0) = \begin{pmatrix} 1 & 0 & 1 \\ 2 & 3 & 0 \\ 4 & 0 & -9 \end{pmatrix}, \qquad \det C(0) = -39 \neq 0.$$

A relation among the three signals would give $C(0)c = 0$, forcing all coefficients to vanish. The solution space has dimension three; these three independent solutions are therefore a basis.

Exercise 4.8.1: Substitute two exponential signals

For $y_k = r^k$, the left side of $y_{k+2} + 2y_{k+1} - 8y_k = 0$ is $r^k(r^2 + 2r - 8) = r^k(r-2)(r+4)$. It vanishes for $r = 2$ and $r = -4$. Both are nonzero, so these signals are defined at negative indices as well. Thus 2^k and $(-4)^k$ are solutions for every integer k.

Exercise 4.8.2: Two roots with opposite signs

For $y_k = r^k$, the equation $y_{k+2} - 25y_k = 0$ becomes $r^k(r^2 - 25) = 0$. The roots $r = 5$ and $r = -5$ make it hold for every integer k, verifying the two proposed solutions.

Exercise 4.8.3: Show the first two verified solutions span

By Exercise 4.8.1 the signals solve the equation. Their values at $k = 0, 1$ form $\left(\begin{smallmatrix} 1 & 1 \\ 2 & -4 \end{smallmatrix}\right)$, with determinant $-6 \neq 0$. They are independent. The equation is of order 2 and has nonzero last coefficient -8, so its solution space has dimension two. Thus the two signals form a basis, and all solutions are $y_k = c_1 2^k + c_2(-4)^k$.

Exercise 4.8.4: A basis for the second equation

The solutions $5^k, (-5)^k$ from Exercise 4.8.2 give initial-value matrix $\left(\begin{smallmatrix} 1 & 1 \\ 5 & -5 \end{smallmatrix}\right)$, whose determinant is -10. Thus they are independent and, since the solution space has dimension two, form a basis. Every solution is $y_k = c_1 5^k + c_2(-5)^k$.

Exercise 4.8.5: A repeated root produces a factor of k

For $u_k = (-2)^k$, the substitution factor is $r^2 + 4r + 4 = (r+2)^2$, so u solves the equation. For $v_k = k(-2)^k$, substitution gives

$$v_{k+2} + 4v_{k+1} + 4v_k = (-2)^k[4(k+2) - 8(k+1) + 4k] = 0.$$

The initial-value matrix $\left(\begin{smallmatrix} 1 & 0 \\ -2 & -2 \end{smallmatrix}\right)$ has determinant $-2 \neq 0$, proving independence. There are two independent solutions in a two-dimensional solution space; hence all solutions are $y_k = (c_1 + c_2 k)(-2)^k$.

Exercise 4.8.6: A cosine-sine pair

For either sine or cosine, adding π to the angle changes its sign. Thus the signals $u_k = 4^k \cos(k\pi/2)$ and $v_k = 4^k \sin(k\pi/2)$ satisfy $u_{k+2} = -16u_k$ and $v_{k+2} = -16v_k$. The initial-value matrix is $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 4 \end{smallmatrix}\right)$ with determinant 4. They are independent and, by the order-two dimension theorem, a basis. Every solution is $4^k[c_1 \cos(k\pi/2) + c_2 \sin(k\pi/2)]$.

Exercise 4.8.7: Test the three supplied solutions by initial values

The exercise permits us to assume the three listed signals solve its order-three equation. Evaluate them at $k = 0, 1, 2$ in the listed order:

$$C(0) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -2 \\ 1 & 4 & 4 \end{pmatrix}, \qquad \det C(0) = 12 \neq 0.$$

Therefore the three signals are independent: a relation holding at every index must in particular vanish at these three indices, where the invertible matrix forces all coefficients to be zero. Since the solution space has dimension three, the proposed signals form a basis.

Exercise 4.8.8: Test the three supplied solutions by initial values

The exercise permits us to assume the three listed signals solve its order-three equation. Evaluate them at $k = 0, 1, 2$ in the listed order:

$$C(0) = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 2 & 3 \\ 1 & 4 & 9 \end{pmatrix}, \qquad \det C(0) = 12 \neq 0.$$

Therefore the three signals are independent: a relation holding at every index must in particular vanish at these three indices, where the invertible matrix forces all coefficients to be zero. Since the solution space has dimension three, the proposed signals form a basis.

Exercise 4.8.9: Test the three supplied solutions by initial values

The exercise permits us to assume the three listed signals solve its order-three equation. Evaluate them at $k = 0, 1, 2$ in the listed order:

$$C(0) = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 5 \\ 4 & -25 & 0 \end{pmatrix}, \qquad \det C(0) = 145 \neq 0.$$

Therefore the three signals are independent: a relation holding at every index must in particular vanish at these three indices, where the invertible matrix forces all coefficients to be zero. Since the solution space has dimension three, the proposed signals form a basis.

Exercise 4.8.10: Test the three supplied solutions by initial values

The exercise permits us to assume the three listed signals solve its order-three equation. Evaluate them at $k = 0, 1, 2$ in the listed order:

$$C(0) = \begin{pmatrix} 1 & 0 & 1 \\ -2 & -2 & 3 \\ 4 & 8 & 9 \end{pmatrix}, \qquad \det C(0) = -50 \neq 0.$$

Therefore the three signals are independent: a relation holding at every index must in particular vanish at these three indices, where the invertible matrix forces all coefficients to be zero. Since the solution space has dimension three, the proposed signals form a basis.

Exercise 4.8.11: Too few solutions for a third-order equation

The equation has order 3 and nonzero last coefficient 4, so its solution space has dimension 3. Only two signals are listed. Even if independent, their span has dimension at most 2, so they cannot form a basis for the whole solution space. The answer is no.

Exercise 4.8.12: Too few solutions for a fourth-order equation

Here the highest and lowest shifts differ by four and the last coefficient is $36 \neq 0$. The solution space therefore has dimension four. Two listed signals cannot span it. Thus they are not a basis, despite being solutions.

Exercise 4.8.13: Two distinct fractional roots

The auxiliary polynomial factors as $r^2 - r + 2/9 = (r - 1/3)(r - 2/3)$. Hence $(1/3)^k$ and $(2/3)^k$ solve the equation. Their initial-value matrix has determinant $2/3 - 1/3 = 1/3 \neq 0$, so they are independent. The dimension-two theorem proves that they span:

$$y_k = c_1(1/3)^k + c_2(2/3)^k.$$

Exercise 4.8.14: Two positive integer roots

The auxiliary polynomial is $r^2 - 5r + 6 = (r - 2)(r - 3)$. The solutions $2^k, 3^k$ have initial-value determinant $3 - 2 = 1 \neq 0$. They are therefore independent; since the solution space has dimension two, all solutions are $y_k = c_1 2^k + c_2 3^k$.

Exercise 4.8.15: A leading coefficient different from one

The auxiliary polynomial is $6r^2 + r - 2 = (2r - 1)(3r + 2)$, with roots $1/2, -2/3$. Dividing the original equation by 6 gives an equivalent order-two equation with nonzero last coefficient, so its solution

space has dimension two. The initial-value determinant is $-2/3 - 1/2 = -7/6 \neq 0$. Thus a basis is $\{(1/2)^k, (-2/3)^k\}$ and every solution is their linear combination.

Exercise 4.8.16: A familiar difference equation

The polynomial $r^2 - 25$ has roots $5, -5$. Their exponential signals solve the equation and have initial-value determinant $-10 \neq 0$. Consequently they form a basis by the dimension-two theorem: $y_k = c_1 5^k + c_2(-5)^k$.

Exercise 4.8.17: Income approaches its equilibrium

For $a = 9/10, b = 4/9$ the equation becomes

$$Y_{k+2} - \tfrac{13}{10} Y_{k+1} + \tfrac{2}{5} Y_k = 1.$$

A constant solution $Y_k = T$ satisfies $(1 - 13/10 + 2/5)T = 1$, giving $T = 10$. The homogeneous polynomial factors as $(r - 4/5)(r - 1/2)$. Its two distinct roots give independent exponential solutions, hence span the homogeneous solution space. Subtracting the particular solution 10 from any nonhomogeneous solution gives a homogeneous solution. Therefore

$$Y_k = 10 + c_1(4/5)^k + c_2(1/2)^k.$$

Both geometric terms tend to zero as $k \to +\infty$, so $Y_k \to 10$, regardless of the fixed initial values.

Exercise 4.8.18: A second stable income model

For $a = 9/10, b = 1/2$, the equation is $Y_{k+2} - (27/20)Y_{k+1} + (9/20)Y_k = 1$. A constant particular solution is again 10, since $1 - 27/20 + 9/20 = 1/10$. The characteristic roots are $3/4$ and $3/5$, as $(r - 3/4)(r - 3/5) = r^2 - (27/20)r + 9/20$. Thus the full solution is

$$Y_k = 10 + c_1(3/4)^k + c_2(3/5)^k.$$

The distinct-root initial-value matrix is invertible, so these terms give every homogeneous solution, not just examples. In particular this model also tends to 10 as k increases.

Exercise 4.8.19: Two real roots for the bending-moment recurrence

For $y_{k+2} + 4y_{k+1} + y_k = 0$, the roots of $r^2 + 4r + 1$ are $r = -2 + \sqrt{3}$ and $s = -2 - \sqrt{3}$. They are distinct and nonzero. The initial-value matrix for r^k, s^k has determinant $s - r = -2\sqrt{3} \neq 0$. The order-two theorem therefore gives the general solution

$$y_k = c_1 r^k + c_2 s^k.$$

These two terms can equivalently be written with exponent $k - 1$ by absorbing one nonzero factor into each arbitrary coefficient.

Exercise 4.8.20: Impose the two boundary values

Assume $N \geq 2$; for $N = 1$ the two stated boundary values would conflict. Write $y_k = Ar^{k-1} + Bs^{k-1}$ using the roots in Exercise 4.8.19. Then $y_1 = 5000$ and $y_N = 0$ give $A + B = 5000$ and $Ar^{N-1} + Bs^{N-1} = 0$. Eliminating one coefficient gives

$$y_k = 5000\,\frac{r^{k-1}s^{N-1} - s^{k-1}r^{N-1}}{s^{N-1} - r^{N-1}}.$$

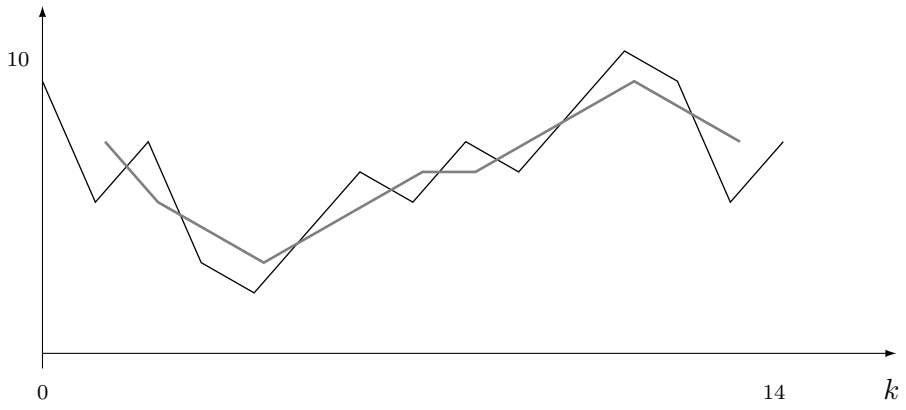
The denominator is nonzero because $|r| \neq |s|$ and $N - 1 > 0$. At $k = 1$ the fraction is 1; at $k = N$ its numerator is zero. Every term solves the recurrence, so their combination does too. Thus this is the required, uniquely determined bending-moment sequence on the support indices.

Exercise 4.8.21: Compute and plot a moving average

Each output is the average of the entry and its two neighbors: $z_k = (y_{k-1} + y_k + y_{k+1})/3$. For instance $z_1 = (9 + 5 + 7)/3 = 7$ and $z_2 = (5 + 7 + 3)/3 = 5$. Continuing through the supplied fifteen entries gives

$$(z_1, \ldots, z_{13}) = (7, 5, 4, 3, 4, 5, 6, 6, 7, 8, 9, 8, 7).$$

The original signal has indices 0 through 14, but the smoothed signal here only has indices 1 through 13; there is no invented endpoint average. In the plot below, the solid broken line is the original signal and the thicker gray broken line is the moving average, using the same axes.



Exercise 4.8.22: A filter separates two frequencies

The filter of Example 3 is $z_k = (\sqrt{2}/4)y_{k+2} + (1/2)y_{k+1} + (\sqrt{2}/4)y_k$. The decimals .35 and .7 in the example abbreviate exact radicals; treating them as rounded decimal inputs would introduce artificial error.

(a) Compute the exact output.

The input is $y_k = 2\cos(k\pi/4) + \cos(3k\pi/4)$. For $u_k = \cos(k\theta)$, angle addition gives

$$u_{k+2} + u_k = 2\cos\theta\cos((k+1)\theta).$$

Thus its filtered output is $[1/2+(\sqrt{2}/2)\cos\theta]\cos((k+1)\theta)$. For $\theta=\pi/4$ the bracket is 1; for $\theta=3\pi/4$ it is 0. Therefore

$$z_k = 2\cos((k+1)\pi/4),$$

with successive values, starting at $k=0$, $\sqrt{2}, 0, -\sqrt{2}, -2, -\sqrt{2}, 0, \sqrt{2}, 2, \ldots$.

(b) Explain the relation to the example.

The filter is linear, so it processes the two input components separately. Example 3 passes the lower-frequency component with a one-index advance and removes the higher-frequency component. The coefficient 2 simply doubles the former output; the latter still contributes zero.

Exercise 4.8.23: A loan paid off by a final smaller payment

(a) Model interest before each payment.

The balance obeys $y_{k+1} = 1.01y_k - 450$, with $y_0 = 10000$, until the last payment. Solving gives $y_k = 45000 - 35000(1.01)^k$ while full payments are made.

(b) A complete balance table and a reproducible program.

The table uses full-precision internal balances and displays dollars rounded to cents; the recurrence itself does not round the balance after each month.

Month	Balance	Month	Balance	Month	Balance
0	10,000.00	9	6,721.02	18	3,134.84
1	9,650.00	10	6,338.23	19	2,716.19
2	9,296.50	11	5,951.61	20	2,293.35
3	8,939.46	12	5,561.12	21	1,866.28
4	8,578.86	13	5,166.74	22	1,434.94
5	8,214.65	14	4,768.40	23	999.29
6	7,846.79	15	4,366.09	24	559.29
7	7,475.26	16	3,959.75	25	114.88
8	7,100.02	17	3,549.34	26	0.00

The following Python code generates the table and adjusts the last payment:

```
from decimal import Decimal, getcontext
getcontext().prec = 50
y = Decimal('10000'); payment = Decimal('450'); k = 0
print(k, f'{y:.2f}')
while y * Decimal('1.01') > payment:
    y = y * Decimal('1.01') - payment
    k += 1
    print(k, f'{y:.2f}')
last = y * Decimal('1.01')
print(k + 1, '0.00', 'last payment', f'{last:.2f}')
```

(c) Find the last payment and the total amount paid.

After payment 25, the unrounded balance is about 114.8801743968. One more month's interest makes the amount due 116.0289761408. Thus the final payment is at month 26, for approximately \$116.03; there have been 25 full \$450 payments. The total paid is $25(450) + 116.0289761408 \approx \$11{,}366.03$. Continuing to subtract a full \$450 after payoff would produce a negative balance and would not model the actual final payment.

Exercise 4.8.24: Monthly deposits and compound interest

(a) Apply interest, then make the new deposit.

The recurrence is $y_{k+1} = 1.005y_k + 200$ with $y_0 = 1000$. Its closed form is $y_k = 41000(1.005)^k - 40000$: substituting it checks both the initial value and the recurrence.

(b) The complete table for months zero through sixty.

Month	Balance	Month	Balance	Month	Balance
0	1,000.00	21	5,527.22	42	10,554.34
1	1,205.00	22	5,754.86	43	10,807.11
2	1,411.02	23	5,983.63	44	11,061.15
3	1,618.08	24	6,213.55	45	11,316.45
4	1,826.17	25	6,444.62	46	11,573.04
5	2,035.30	26	6,676.84	47	11,830.90
6	2,245.48	27	6,910.23	48	12,090.06
7	2,456.71	28	7,144.78	49	12,350.51
8	2,668.99	29	7,380.50	50	12,612.26
9	2,882.33	30	7,617.40	51	12,875.32
10	3,096.75	31	7,855.49	52	13,139.70
11	3,312.23	32	8,094.77	53	13,405.39
12	3,528.79	33	8,335.24	54	13,672.42
13	3,746.43	34	8,576.92	55	13,940.78
14	3,965.17	35	8,819.80	56	14,210.49
15	4,184.99	36	9,063.90	57	14,481.54
16	4,405.92	37	9,309.22	58	14,753.95
17	4,627.95	38	9,555.77	59	15,027.72
18	4,851.09	39	9,803.55	60	15,302.86
19	5,075.34	40	10,052.56		
20	5,300.72	41	10,302.83		

Here is the program used, retaining internal precision until display:

```
from decimal import Decimal, getcontext
getcontext().prec = 50
y = Decimal('1000')
for k in range(61):
    print(k, f'{y:.2f}')
    y = y * Decimal('1.005') + Decimal('200')
```

(c) Separate deposits from interest.

At months $24, 48, 60$, the balances are respectively \$6,213.55, \$12,090.06, and \$15,302.86, rounded to cents. By month 60 the deposits total $1000 + 60(200) = 13000$. Thus interest accounts for $15302.8562545 - 13000 \approx \$2{,}302.86$.

Exercise 4.8.25: A quadratic particular solution

For $y_k = k^2$, substitution gives

$$(k+2)^2 + 3(k+1)^2 - 4k^2 = 10k + 7,$$

which is the required right-hand side. The homogeneous polynomial is $r^2 + 3r - 4 = (r-1)(r+4)$, with independent exponential solutions $1, (-4)^k$. Any two particular solutions differ by a homogeneous solution, so the general solution is $y_k = k^2 + c_1 + c_2(-4)^k$.

Exercise 4.8.26: A linear particular solution

For $y_k = 1 + k$, the left side is $(k+3) - 6(k+2) + 5(k+1) = -4$. The homogeneous roots of $r^2 - 6r + 5$ are $1, 5$, so its solutions are $c_1 + c_2 5^k$. Hence all solutions are $y_k = 1 + k + c_1 + c_2 5^k$. The constant 1 could be absorbed into c_1, but retaining it makes the supplied particular solution visible.

Exercise 4.8.27: Another linear particular solution

For $y_k = k - 2$, compute $y_{k+2} - 4y_k = k - 4(k-2) = 8 - 3k$. The homogeneous roots are $2, -2$. Thus the full solution is $y_k = k - 2 + c_1 2^k + c_2(-2)^k$. The two exponential solutions are independent because their values at indices zero and one have determinant $-4 \neq 0$.

Exercise 4.8.28: An affine particular solution with roots of magnitude five

For $y_k = 1 + 2k$, the left side is $(1 + 2(k+2)) - 25(1 + 2k) = -48k - 20$, exactly as required. The associated homogeneous equation has roots $5, -5$. Therefore the general solution is $y_k = 1 + 2k + c_1 5^k + c_2(-5)^k$.

Exercise 4.8.29: A fourth-order equation as a first-order vector recurrence

Choose the state $x_k = (y_k, y_{k+1}, y_{k+2}, y_{k+3})^T$. The first three entries of x_{k+1} just shift those coordinates left. Solving the recurrence for y_{k+4} gives $y_{k+4} = 2y_k - 6y_{k+1} + 8y_{k+2} - 3y_{k+3}$. Thus

$$x_{k+1} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 2 & -6 & 8 & -3 \end{pmatrix} x_k.$$

The state definition is part of the answer: reversing the order of its coordinates would change the companion matrix.

Exercise 4.8.30: A third-order companion system

Set $x_k = (y_k, y_{k+1}, y_{k+2})^T$. Since $y_{k+3} = 5y_{k+2} - 8y_k$, the desired system is

$$x_{k+1} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -8 & 0 & 5 \end{pmatrix} x_k.$$

The middle coefficient is zero because there is no y_{k+1} term in the original equation.

Exercise 4.8.31: Remove an irrelevant common shift

The equation $y_{k+3} + 5y_{k+2} + 6y_{k+1} = 0$ has no y_k term. Put $j = k + 1$. Because the equation holds for *all* integers k, it is equivalent to $y_{j+2} + 5y_{j+1} + 6y_j = 0$ for all integers j. Its order is therefore 2, not 3: the largest and smallest shifts differ by two and both endpoint coefficients are nonzero.

Exercise 4.8.32: The order depends on the last nonzero coefficient

For $y_{k+3} + a_1y_{k+2} + a_2y_{k+1} + a_3y_k = 0$: if $a_3 \neq 0$ the order is 3; if $a_3 = 0, a_2 \neq 0$ a shift makes the order 2; if $a_3 = a_2 = 0, a_1 \neq 0$ it is 1. If all three are zero, it says $y_{k+3} = 0$ at every integer, which is just the zero-signal condition, of order 0 in this shift-width convention. Without a nonzero hypothesis on the final coefficient, one cannot simply call every equation in the displayed form third order.

Exercise 4.8.33: Independence does not require an invertible Casorati sample for arbitrary signals

Let $y_k = k^2$ and $z_k = 2k|k|$. If $ay_k + bz_k = 0$ for every integer k, using $k = 1$ gives $a + 2b = 0$ and using $k = -1$ gives $a - 2b = 0$. Subtracting gives $b = 0$, then $a = 0$. Thus the signals are independent on all integers (they would be dependent if restricted to nonnegative indices). The requested matrices are

$$C(0) = \begin{pmatrix} 0 & 0 \\ 1 & 2 \end{pmatrix}, \quad C(-1) = \begin{pmatrix} 1 & -2 \\ 0 & 0 \end{pmatrix}, \quad C(-2) = \begin{pmatrix} 4 & -8 \\ 1 & -2 \end{pmatrix}.$$

All are singular. Invertibility of one Casorati matrix is a sufficient independence test, but its failure alone does not prove dependence for arbitrary signals. No common second-order recurrence hypothesis was given here.

Exercise 4.8.34: Sampling can destroy independence

Take $f(t) = 1$, $g(t) = t$, and $h(t) = \sin(\pi t)$. They are independent functions: a relation $a + bt + c\sin(\pi t) = 0$ for every real t, evaluated at $0, 1, 1/2$, successively gives $a = b = c = 0$. But sampling at integers turns h into the zero signal. The three sampled signals are therefore dependent. Thus the answer is no: distinct continuous behaviors can become indistinguishable at the chosen sample points.

Exercise 4.8.35: A linear difference operator on signals

For two signals u, v and a scalar c, evaluate each index:

$$\begin{aligned}(T(u+v))_k &= (u_{k+2}+v_{k+2}) + a(u_{k+1}+v_{k+1}) + b(u_k+v_k)\\ &= (Tu)_k + (Tv)_k,\\ (T(cu))_k &= cu_{k+2} + acu_{k+1} + bcu_k = c(Tu)_k.\end{aligned}$$

The resulting sequences are defined at every integer, so T maps S into S. The two coordinate-wise identities prove linearity as an equality of signals.

Exercise 4.8.36: Add a homogeneous solution to a particular solution

Suppose $T(x_p) = z$ and $u \in \ker T$, so $T(u) = 0$. Linearity gives $T(u+x_p) = T(u)+T(x_p) = 0+z = z$. Thus $u + x_p$ solves the same nonhomogeneous equation. Conversely, if $T(x) = z$, then $T(x - x_p) = z - z = 0$, explaining why all solutions are precisely a particular solution plus kernel vectors.

Exercise 4.8.37: One-sided shifts have only a one-sided inverse

Apply the maps directly to an arbitrary sequence $y = (y_0, y_1, \ldots)$:

$$TD(y) = T(0, y_0, y_1, \ldots) = (y_0, y_1, \ldots) = y.$$

Hence $TD = I$. But $DT(y) = D(y_1, y_2, \ldots) = (0, y_1, y_2, \ldots)$, which need not equal y. For example $y = (1, 0, 0, \ldots)$ is sent to the zero sequence by DT. Thus $DT \neq I$, illustrating why the finite-dimensional square-matrix conclusion about one-sided inverses does not extend to this infinite-dimensional space.

4.9 Applications to Markov chains

We use column probability vectors. Entry p_{ij} is the probability of moving *from state j to state i*, so each column of the transition matrix sums to 1 and its entries are nonnegative. The distribution evolves by $x_{k+1} = Px_k$. A steady-state vector is a probability vector q with $Pq = q$; solve $(P-I)q = 0$ together with $\sum_i q_i = 1$. A stochastic matrix is regular if some positive integer power has every entry strictly positive. For a regular matrix, Theorem 18 guarantees a unique steady-state vector and convergence $P^k x_0 \to q$ for every initial probability vector.

Practice problem 4.9.P1: A city resident's next two locations

The migration matrix from Example 1 is $M = \left(\begin{smallmatrix} .95 & .03 \\ .05 & .97 \end{smallmatrix}\right)$, with city first and suburbs second. Starting from $x_0 = (1, 0)$, $x_1 = Mx_0 = (.95, .05)$. Thus (a) the probability of living in the suburbs next year is 5%. A second multiplication gives $x_2 = (.95^2 + .03(.05), .05(.95) + .97(.05)) = (.904, .096)$. Thus (b) the probability after two years is 9.6%.

Practice problem 4.9.P2: Test the steady-state equation rather than its appearance

For the given P, q, direct multiplication gives

$$\begin{pmatrix} .6 & .2 \\ .4 & .8 \end{pmatrix} \begin{pmatrix} .3 \\ .7 \end{pmatrix} = \begin{pmatrix} .18 + .14 \\ .12 + .56 \end{pmatrix} = \begin{pmatrix} .32 \\ .68 \end{pmatrix} \neq q.$$

Although q is a probability vector, it is not a steady-state vector.

Practice problem 4.9.P3: A long-run migration percentage

Solve $Mq = q$: the first equation reduces to $.05q_1 = .03q_2$. Together with $q_1 + q_2 = 1$, this gives $q = (3/8, 5/8)$. Every entry of M is positive, so it is regular; Theorem 18 makes this the limiting distribution. Thus 62.5% will live in the suburbs after many years.

Exercise 4.9.1: Radio listeners at two station breaks

(a) Build columns from current stations.

Order the states as news, music for both rows (next station) and columns (current station). The probabilities give

$$P = \begin{pmatrix} .7 & .6 \\ .3 & .4 \end{pmatrix}.$$

Column one describes a news listener; column two describes a music listener. Each column sums to 1.

(b) Use the stated initial audience.

Everyone initially listens to news, so $x_0 = (1, 0)$.

(c) Count the breaks, not the elapsed minutes.

At 9:25 there have been two breaks, at 8:30 and 9:00. Then $x_1 = (.7, .3)$ and $x_2 = (.7(.7) + .6(.3), .3(.7) + .4(.3)) = (.67, .33)$. Thus 33% will be listening to music.

Exercise 4.9.2: Three equally treated food choices

(a) Use foods 1, 2, 3 in both state lists.

The probability of retaining the current food is .6; the other two possibilities each have probability .2. Thus

$$P = \begin{pmatrix} .6 & .2 & .2 \\ .2 & .6 & .2 \\ .2 & .2 & .6 \end{pmatrix}.$$

(b) Apply two transitions after the initial trial.

Starting at food 1 gives $x_0 = (1, 0, 0)$ and $x_1 = (.6, .2, .2)$. The second component of x_2 is $.2(.6) + .6(.2) + .2(.2) = .12 + .12 + .04 = .28$. The requested probability is 28%; a single multiplication would answer the first trial after the initial trial, not the second.

Exercise 4.9.3: Healthy and ill students

(a) Define the state order and fill the columns.

Use healthy, ill. A healthy student becomes ill with probability .05; an ill student becomes healthy with probability $1 - .55 = .45$. Therefore

$$P = \begin{pmatrix} .95 & .45 \\ .05 & .55 \end{pmatrix}.$$

(b) Advance Monday's probability vector twice.

Monday's $x_0 = (.8, .2)$ gives Tuesday $x_1 = (.95(.8) + .45(.2), .05(.8) + .55(.2)) = (.85, .15)$. Wednesday gives $x_2 = (.875, .125)$. The ill percentages are 15% and 12.5%.

(c) Use a different initial condition for a currently healthy student.

From $(1, 0)$, after one day the vector is $(.95, .05)$, and after two days the healthy probability is $.95^2 + .45(.05) = .925$. Thus the requested probability is 92.5%.

Exercise 4.9.4: Good, indifferent, and bad weather

(a) Supply probabilities missing from the wording.

Use good, indifferent, bad. Each column sums to one, so the remaining indifferent-to-bad probability is .3 and bad-to-bad is .3. The matrix is

$$P = \begin{pmatrix} .4 & .5 & .3 \\ .3 & .2 & .4 \\ .3 & .3 & .3 \end{pmatrix}.$$

(b) Tomorrow's bad-weather probability.

Starting from $(.5, .5, 0)$, its bad component after one transition is $.3(.5) + .3(.5) + .3(0) = .3$, or 30%.

(c) Wednesday is two transitions from Monday.

From Monday $(0, .6, .4)$, Tuesday is $(.42, .28, .30)$. Wednesday's good component is $.4(.42) + .5(.28) + .3(.30) = .398$, so the answer is 39.8%.

Exercise 4.9.5: Normalize a two-state balance equation

For $P = \left(\begin{smallmatrix} .1 & .5 \\ .9 & .5 \end{smallmatrix}\right)$, $Pq = q$ reduces to $.9q_1 = .5q_2$, or $q_2 = (9/5)q_1$. Impose $q_1 + q_2 = 1$ to get $(14/5)q_1 = 1$.

Conclusion. $q = (5/14, 9/14)$. Both entries are nonnegative and sum to one.

Exercise 4.9.6: A second two-state equilibrium

Here the balance equation is $.6q_1 = .8q_2$, so $3q_1 = 4q_2$. Together with normalization this gives

$$q = (4/7, 3/7).$$

Substitution gives $Pq = q$, confirming the calculation.

Exercise 4.9.7: Three-state balance with two useful equations

From the first row, $.3q_1 = .1q_2 + .1q_3 = .1(1 - q_1)$, so $.4q_1 = .1$ and $q_1 = 1/4$. From the second, $.2q_2 = .2q_1 + .2q_3 = .2(1 - q_2)$, so $q_2 = 1/2$. Normalization then gives $q_3 = 1/4$.

Conclusion. $q = (1/4, 1/2, 1/4)$. The remaining row equation is satisfied as well.

Exercise 4.9.8: A steady state despite zero transition entries

The second and third row equations give $.5q_2 = .1q_3$ and $.9q_3 = .6q_1$. Thus $q_2 = q_3/5$ and $q_3 = 2q_1/3$. Normalizing gives

$$q_1(1 + 2/15 + 2/3) = 1, \qquad q = (5/9, 2/27, 10/27).$$

Substituting these fractions into the first row gives $5/9$; their sum is one, so they form the required probability vector.

Exercise 4.9.9: One zero can disappear under squaring

The given matrix is stochastic, since it has nonnegative entries and unit column sums. Its square is

$$P^2 = \begin{pmatrix} .2 & 1 \\ .8 & 0 \end{pmatrix}^2 = \begin{pmatrix} .84 & .2 \\ .16 & .8 \end{pmatrix}.$$

Every entry of this power is strictly positive. Therefore P is regular, even though P itself has a zero entry.

Exercise 4.9.10: An absorbing state prevents regularity

For $P = \left(\begin{smallmatrix} 1 & .3 \\ 0 & .7 \end{smallmatrix}\right)$, the first column is $(1, 0)$, and $P(1, 0) = (1, 0)$. Inductively the first column of every positive power remains $(1, 0)$. Thus each power has a zero entry, so P is not regular.

Exercise 4.9.11: Long-run radio preferences

For the matrix in Exercise 4.9.1, the balance equation is $.3q_N = .6q_M$. Thus $q_N = 2q_M$, and normalization gives (a) $q = (2/3, 1/3)$. All entries of P are positive, so it is regular. Theorem 18 implies (b) that the long-run fraction listening to the news is $2/3$, regardless of the initial audience. The finite two-break answer need not equal this limit.

Exercise 4.9.12: No long-run preference between symmetric foods

The matrix of Exercise 4.9.2 sends $(1/3, 1/3, 1/3)$ to itself; every row also sums to one. All entries are positive, so Theorem 18 makes this the unique limiting probability vector. Hence none of the three foods is preferred in the long run: each is chosen with probability $1/3$.

Exercise 4.9.13: The long-run illness rate

For the healthy/ill matrix, $.05q_H = .45q_I$ implies $q_H = 9q_I$. Normalization gives (a) $q = (.9, .1)$. Since this matrix has all positive entries, it is regular; thus (b) the long-run probability that a specific student is ill is .1, or 10%. Theorem 18 applies to every initial probability vector, including certainty of being healthy or ill today, so today's state does not change that limit.

Exercise 4.9.14: Long-run good weather

The third row of the weather matrix gives $q_B = .3$ immediately because $q_G + q_I + q_B = 1$. Then $q_G + q_I = .7$ and $q_I = .3q_G + .2q_I + .4(.3)$. Substitute $q_I = .7 - q_G$ to obtain $.8(.7 - q_G) = .3q_G + .12$, whence $q_G = .4$ and $q_I = .3$. The matrix is strictly positive, so the long-run good-weather probability is 40%.

Exercise 4.9.15: A migration percentage in the long run

With states California and rest of the United States, use

$$P = \begin{pmatrix} .9821 & .0029 \\ .0179 & .9971 \end{pmatrix}.$$

The steady-state equation gives $.0179q_{CA} = .0029q_{rest}$. Since $q_{rest} = 1 - q_{CA}$,

$$q_{CA} = \frac{.0029}{.0179 + .0029} = \frac{29}{208} \approx .13942308.$$

All entries are positive, hence the matrix is regular. The eventual share is approximately 13.9423%. The initial 11.7% affects the transient distributions, but not this limit, under the stipulated constant migration model.

Exercise 4.9.16: Allocate a car fleet across three locations

Order states as City Airport, Downtown, Metro Airport, using the printed *columns* as rental locations and rows as return locations:

$$P = \begin{pmatrix} .90 & .01 & .09 \\ .01 & .90 & .01 \\ .09 & .09 & .90 \end{pmatrix}.$$

The downtown balance gives $.10q_D = .01q_C + .01q_M = .01(1 - q_D)$, so $.11q_D = .01$ and $q_D = 1/11$. The full steady state is $(91/209, 1/11, 9/19)$; its entries sum to one and satisfy $Pq = q$. Because the matrix is positive, it is regular and this steady distribution describes a typical long-run day under the model. For about 2000 cars, downtown gets $2000/11 \approx 181.82$ cars, or approximately **182 cars** rented or ready to rent there.

Exercise 4.9.17: A stochastic matrix has eigenvalue one

(a) Every column of P sums to one, while every column of I sums to one. Thus every column of $P - I$ sums to zero. Adding all other rows to the bottom row produces that column-sum row, which is zero.

(b) The sum of all original rows of $P - I$ is zero. The coefficients of this relation are all 1, not all zero, so the rows are dependent.

(c) A dependent list of n rows spans a space of dimension less than n. Equivalently $\operatorname{rank}(P - I) < n$.

(d) By rank–nullity, $\dim \operatorname{Nul}(P - I) = n - \operatorname{rank}(P - I) > 0$. Thus there is a nonzero x satisfying $(P - I)x = 0$, or $Px = x$.

This proves existence of a nonzero fixed vector; by itself it does not prove its entries can be chosen nonnegative and normalized, which is the additional point mentioned in the exercise.

Exercise 4.9.18: An explicit steady state for every two-state matrix

For $P = \left(\begin{smallmatrix} 1-\alpha & \beta \\ \alpha & 1-\beta \end{smallmatrix}\right)$, the equation $Pq = q$ is $\alpha q_1 = \beta q_2$. If $\alpha + \beta > 0$, choose

$$q = \left(\frac{\beta}{\alpha + \beta}, \frac{\alpha}{\alpha + \beta}\right).$$

Both entries are nonnegative and sum to one, and the balance equation holds, so this is a steady-state vector. It is the unique normalized solution because the balance and sum equations determine both entries. If $\alpha = \beta = 0$, then $P = I_2$. Every vector $(t, 1 - t)$ with $0 \le t \le 1$ is a steady state; in particular $(1, 0), (0, 1)$ are two independent ones. Thus the parenthetical claim means two independent steady-state directions, not only two probability vectors in this case.

Exercise 4.9.19: A row of ones records total probability

(a) Interpret the scalar Sx.

For $S = [\,1 \ \cdots \ 1\,]$, $Sx = \sum_i x_i$. Therefore x is a probability vector exactly when its entries are nonnegative and $Sx = 1$.

(b) Interpret the entries of SP.

Entry j is the sum of column j of P, which equals 1. Hence $SP = S$.

(c) Preserve nonnegativity and the total separately.

Every entry of Px is a sum of products of nonnegative entries, so it is nonnegative. Also $S(Px) = (SP)x = Sx = 1$. By part (a), Px is a probability vector.

Exercise 4.9.20: The square is stochastic

Every column p_j of P is a probability vector. Column j of P^2 is Pp_j, which is a probability vector by Exercise 4.9.19(c). Thus each column of P^2 has nonnegative entries and sums to one, which is exactly the definition of a stochastic matrix.

Exercise 4.9.21: Observe convergence of all columns of powers

(a) Four powers of the supplied positive 4×4 matrix.

Keep full internal precision in the products and round only the displayed entries to four decimal places. The results are

$$P^2 = \begin{pmatrix} 0.2779 & 0.2780 & 0.2803 & 0.2941 \\ 0.3368 & 0.3355 & 0.3357 & 0.3335 \\ 0.1847 & 0.1861 & 0.1833 & 0.1697 \\ 0.2005 & 0.2004 & 0.2007 & 0.2027 \end{pmatrix}.$$

$$P^3 = \begin{pmatrix} 0.2817 & 0.2817 & 0.2817 & 0.2814 \\ 0.3356 & 0.3356 & 0.3355 & 0.3352 \\ 0.1817 & 0.1817 & 0.1819 & 0.1825 \\ 0.2010 & 0.2010 & 0.2010 & 0.2009 \end{pmatrix}.$$

$$P^4 = \begin{pmatrix} 0.2816 & 0.2816 & 0.2816 & 0.2816 \\ 0.3355 & 0.3355 & 0.3355 & 0.3355 \\ 0.1819 & 0.1819 & 0.1819 & 0.1819 \\ 0.2009 & 0.2009 & 0.2009 & 0.2009 \end{pmatrix}.$$

$$P^5 = \begin{pmatrix} 0.2816 & 0.2816 & 0.2816 & 0.2816 \\ 0.3355 & 0.3355 & 0.3355 & 0.3355 \\ 0.1819 & 0.1819 & 0.1819 & 0.1819 \\ 0.2009 & 0.2009 & 0.2009 & 0.2009 \end{pmatrix}.$$

Solving $(P - I)q = 0$ together with $q_1 + q_2 + q_3 + q_4 = 1$ gives

$$q \approx \begin{pmatrix} 0.2816326912 \\ 0.3355052569 \\ 0.181921411 \\ 0.2009406409 \end{pmatrix}.$$

The columns are rapidly becoming nearly equal to this fixed probability vector.

(b) Powers of the more slowly mixing 3×3 matrix.

$$Q^{10} = \begin{pmatrix} 0.8222 & 0.4044 & 0.5385 \\ 0.0324 & 0.3966 & 0.1666 \\ 0.1453 & 0.1990 & 0.2949 \end{pmatrix}.$$

$$Q^{20} = \begin{pmatrix} 0.7674 & 0.6000 & 0.6690 \\ 0.0637 & 0.2036 & 0.1326 \\ 0.1688 & 0.1964 & 0.1984 \end{pmatrix}.$$

$$Q^{30} = \begin{pmatrix} 0.7477 & 0.6815 & 0.7105 \\ 0.0783 & 0.1329 & 0.1074 \\ 0.1740 & 0.1856 & 0.1821 \end{pmatrix}.$$

$$Q^{40} = \begin{pmatrix} 0.7401 & 0.7140 & 0.7257 \\ 0.0843 & 0.1057 & 0.0960 \\ 0.1756 & 0.1802 & 0.1783 \end{pmatrix}.$$

$$Q^{50} = \begin{pmatrix} 0.7372 & 0.7269 & 0.7315 \\ 0.0867 & 0.0951 & 0.0913 \\ 0.1761 & 0.1780 & 0.1772 \end{pmatrix}.$$

$$Q^{60} = \begin{pmatrix} 0.7360 & 0.7320 & 0.7338 \\ 0.0876 & 0.0909 & 0.0894 \\ 0.1763 & 0.1771 & 0.1767 \end{pmatrix}.$$

$$Q^{70} = \begin{pmatrix} 0.7356 & 0.7340 & 0.7347 \\ 0.0880 & 0.0893 & 0.0887 \\ 0.1764 & 0.1767 & 0.1766 \end{pmatrix}.$$

$$Q^{80} = \begin{pmatrix} 0.7354 & 0.7348 & 0.7351 \\ 0.0881 & 0.0887 & 0.0884 \\ 0.1764 & 0.1766 & 0.1765 \end{pmatrix}.$$

Its second equilibrium equation gives $q_2 = q_3/2$, and its first gives $.03q_1 = .05q_2 + .10q_3$. Normalization then yields

$$q = (25/34, 3/34, 3/17) \approx (.73529412, .08823529, .17647059).$$

The columns approach this vector more slowly, so an arbitrary fixed power such as 100 is not a universal certificate of four-place accuracy.

(c) Explain the observation with Theorem 18.

P is positive, hence regular. Q^2 has all entries positive (the only zero in Q acquires a positive two-step route), so Q is regular too. Column j of P^k is $P^k e_j$, the evolving probability vector from the initial state e_j. Theorem 18 says each such vector tends to the same unique steady state. The same reasoning applies to Q^k and explains the observed column limits for any regular stochastic matrix.

Exercise 4.9.22: A reproducible comparison of two numerical methods

What we must show. Run both methods, report measured times, and distinguish numerical approximation from a mathematical convergence guarantee. A matrix program such as NumPy has no fixed largest permitted order; the practical limit depends on memory and desired runtime. The

following experiment uses positive random matrices through order 1024, the largest order tested here, rather than claiming an intrinsic software size limit. Normalize each random positive column to obtain a regular stochastic matrix. For method 1 solve $(P - I)q = 0$ with its final equation replaced by $\sum_i q_i = 1$. For method 2 use column one of P^{100}.

The following table records actual execution times in seconds on this workspace, with one BLAS thread; input generation is outside the timing. Each method is invoked by one matrix-program call after the shared setup. Human typing time is user dependent and is not fabricated as an execution measurement.

Order	Direct solve	Power 100	Maximum difference
64	0.000149	0.000120	$6.09e-16$
128	0.000173	0.000829	$3.93e-16$
256	0.000788	0.005619	$5.89e-16$
512	0.004078	0.043638	$7.01e-16$
1024	0.028829	0.329893	$7.84e-16$

Here is the complete experimental core; start Python with `OPENBLAS_NUM_THREADS=1` for the stated thread setting.

```
import numpy as np
from time import perf_counter
rng = np.random.default_rng(4922)
for n in [64, 128, 256, 512, 1024]:
    P = rng.uniform(.1, 1, (n, n)); P /= P.sum(axis=0)
    A = P - np.eye(n); A[-1, :] = 1
    b = np.zeros(n); b[-1] = 1
    start = perf_counter()
    q = np.linalg.solve(A, b)
    direct = perf_counter() - start
    start = perf_counter()
    v = np.linalg.matrix_power(P, 100)[:, 0]
    power = perf_counter() - start
    print(n, direct, power, np.max(abs(q-v)))
```

For these well-mixed positive samples, the two vectors agree to less than 8×10^{-16} in maximum coordinate error. This empirical accuracy is not guaranteed for every regular stochastic matrix: a slowly mixing matrix can require far more than 100 steps, as Exercise 4.9.21 illustrates. Solving the normalized linear system avoids choosing a mixing time and was faster for the larger tested orders. Computing a power is easy to request and can reveal transients, but a large power performs more matrix products and still needs a convergence check. For finding only the steady state, the direct normalized solve is preferable in this experiment.

4.S Chapter 4 supplementary exercises

These exercises connect spanning sets, bases, coordinate changes, and rank. When proving two sets equal, establish both inclusions. When proving a universal claim false, give one concrete counterexample. For controllability, form $[\,B\ AB\ \cdots\ A^{n-1}B\,]$ and check whether its columns span all of the state space $\mathbb{R}^n$.

Exercise 4.S.1: Twenty statements tying the chapter together

(a) **True.** The set is $\text{Span}\{v_1, \ldots, v_p\}$, a subspace of V by the span theorem.

(b) **True.** Adding another generator cannot remove combinations already obtainable. A spanning subset remains spanning after a vector is added.

(c) **False.** Start with the independent list $\{e_1\}$ in $\mathbb{R}^2$ and add e_1 again; alternatively add $2e_1$. The enlarged list is dependent.

(d) **False.** In $\mathbb{R}^2$, $\{e_1\}$ is independent but does not span $\mathbb{R}^2$, so it is not a basis.

(e) **True.** The Spanning Set Theorem removes any redundant vectors from a finite spanning list, leaving an independent spanning subset.

(f) **True.** A list of exactly $\dim V = p$ vectors that spans V is a basis by the Basis Theorem, hence is independent.

(g) **False.** A plane not through zero is not a subspace. For example the plane $z = 1$ misses the zero vector.

(h) **False.** In $A = [\,I_2\ I_2\,]$ the nonpivot columns are e_1, e_2, which are independent. Each is a combination of pivot columns, but that does not force a dependence among the nonpivot columns themselves.

(i) **True.** Rows $(1, 0), (1, 0)$ have the relation $r_2 - r_1 = 0$; replacing row two by row two minus row one changes the list to $(1, 0), (0, 0)$, which no longer has that same relation.

(j) **False.** Elementary row operations replace $Ax = 0$ by an equivalent system, so they preserve the null space.

(k) **False.** The matrix $\left(\begin{smallmatrix} 1 & 0 \\ 2 & 0 \end{smallmatrix}\right)$ has two nonzero rows but rank one. Count nonzero rows only in an echelon form.

(l) **False.** Nullity is the number of columns minus rank, $n - k$, not generally $m - k$.

(m) **True.** Row-equivalent matrices have the same row space, so their row-space dimensions, and hence ranks, agree.

(n) **False.** Nonzero rows need not be independent, as the matrix in part (k) shows. Nonzero rows of an echelon form do give a row-space basis.

(o) **True.** Row operations preserve row spaces, and both row spaces equal that of the common reduced echelon form.

(p) **True.** Choose a basis $h_1, \ldots, h_d$ for H, where $d \leq 3$, and use these as columns of a 3×3 matrix, filling any remaining columns with zeros. Its column space is H. If $H = \{0\}$, use the zero matrix.

(q) **False.** Rank m means onto, not necessarily one-to-one. For $A = [\,1\ 0\,]$, rank is $m = 1$ but $(0, 1)$ is a nonzero kernel vector.

(r) **True.** Onto $\mathbb{R}^m$ means the column space has dimension m, which is exactly rank m.

(s) **True.** The reverse change-of-coordinates matrix is the inverse, since changing from one basis to the other and back restores every vector.

(t) **False.** Column j of $P_{C \leftarrow B}$ is $[b_j]_C$, not $[c_j]_B$. The latter columns occur in the inverse matrix.

Exercise 4.S.2: A parametrization has a redundant generator

Collect the coefficients of a, b, c to get generators

$$u = (1, 2, -1, 3), \quad v = (-2, 5, -4, 1), \quad w = (5, -8, 7, 1).$$

But $w = u - 2v$, so the third generator is redundant. The first two are independent: if $\alpha u + \beta v = 0$, the first two coordinates give $\alpha - 2\beta = 0$, $2\alpha + 5\beta = 0$, hence $9\beta = 0$ and both vanish.

Conclusion. A basis is $\{(1, 2, -1, 3), (-2, 5, -4, 1)\}$. There are three parameters in the original expression, but only two independent directions.

Exercise 4.S.3: Describe a plane by a homogeneous equation

Both $u_1 = (-2, 4, -6)$ and $u_2 = (1, 2, -5)$ satisfy $x_1 + 2x_2 + x_3 = 0$. They are not multiples, so their span is a two-dimensional subspace contained in the plane defined by that equation. To verify equality without relying only on dimensions, let b satisfy $b_1 + 2b_2 + b_3 = 0$. Solve the first two equations of $au_1 + cu_2 = b$: they give

$$a = (b_2 - 2b_1)/8, \qquad c = b_1 + 2a.$$

With these choices the third coordinate is $-6a - 5c = -b_1 - 2b_2 = b_3$.

Conclusion. $W = \{b \in \mathbb{R}^3 : b_1 + 2b_2 + b_3 = 0\}$.

Exercise 4.S.4: A scalar multiplier cannot depend on the input variable

The equality $g(t) = tf(t)$ uses the function t, not one fixed real scalar. Linear dependence would require constants a, b, not both zero, with $a(3 + t) + b(3t + t^2) = 0$ for every t. Comparing the t^2 coefficient gives $b = 0$; comparing the constant gives $3a = 0$, so $a = 0$ as well. Thus the pair is independent, and the proposed discussion incorrectly treats a variable multiplier as a scalar.

Exercise 4.S.5: Delete redundant polynomials from the given list

The polynomials are $p_1 = 1 + t$, $p_2 = 1 - t$, $p_3 = 4$, $p_4 = t + t^2$, $p_5 = 1 + 2t + t^2$. First $p_3 = 2p_1 + 2p_2$, so deleting p_3 does not alter the span. Next $p_5 = p_1 + p_4$, so delete p_5. The remaining p_1, p_2, p_4 are independent: in $ap_1 + bp_2 + cp_4 = 0$, the quadratic coefficient gives $c = 0$; then the constant and linear coefficients give $a + b = 0$ and $a - b = 0$, forcing $a = b = 0$.

Conclusion. $\{1+t, 1-t, t+t^2\}$ is a basis for H.

Exercise 4.S.6: A basis of a known two-dimensional span

At least one of the four spanning polynomials is nonzero; choose it as p. Since their span has dimension two, at least one other polynomial q is not a scalar multiple of p. Find such a q by comparing coefficients. Two nonproportional nonzero vectors are independent, so $\{p, q\}$ is an independent pair in the two-dimensional space H. By the Basis Theorem it is already a basis. There is no need to row reduce all four coefficient vectors when the dimension is known in advance.

Exercise 4.S.7: The homogeneous information that certifies every right-hand side

For eighteen equations in twenty variables, every nonhomogeneous right-hand side is attainable precisely when rank is eighteen. By rank–nullity this is equivalent to nullity $20 - 18 = 2$. Thus it suffices, and is necessary, to know that the homogeneous solution set is two-dimensional: exactly two independent homogeneous solutions span it.

Exercise 4.S.8: Equal finite dimensions force equality of nested spaces

Let a basis of H have n vectors. Because $H \subseteq V$, these are n independent vectors in the n-dimensional space V. The Basis Theorem says they span V. But their span is also H, so $H = V$. The containment hypothesis matters: distinct subspaces can have equal dimensions without either containing the other.

Exercise 4.S.9: Rank and nullity of injective and surjective maps

(a) If T is one-to-one.

Its kernel is zero, so rank–nullity gives $\dim \operatorname{range} T = n$. This also requires $n \leq m$, since the range is contained in $\mathbb{R}^m$.

(b) If T maps onto $\mathbb{R}^m$.

Its range has dimension m. Thus its kernel has dimension $n - m$, which in particular requires $n \geq m$.

Exercise 4.S.10: A maximal independent set must span

Suppose the independent set S failed to span V. Choose $v \in V \setminus \operatorname{Span} S$. Then adjoining v still leaves an independent set: if $av + \sum_i c_i s_i = 0$ is a finite relation and $a \neq 0$, it would express v as a combination of elements of S, a contradiction. Thus $a = 0$, and independence of S forces all $c_i = 0$. This contradicts maximality. Hence S spans, and being independent as well, it is a basis. The argument covers possibly infinite S because linear combinations always involve only finitely many terms.

Exercise 4.S.11: A minimal finite spanning set is independent

Suppose a finite spanning set were dependent. Then a nontrivial relation would allow one vector with a nonzero coefficient to be solved for as a combination of the others. Deleting that vector would preserve every original linear combination, and thus preserve the span V. That contradicts minimality. Therefore the set is independent; since it already spans V, it is a basis.

Exercise 4.S.12: Rank cannot increase beyond either factor's rank

(a) Compare column spaces.

Every output $ABx = A(Bx)$ belongs to $\operatorname{Col} A$, so $\operatorname{Col}(AB) \subseteq \operatorname{Col} A$. Taking dimensions gives $\operatorname{rank}(AB) \leq \operatorname{rank} A$.

(b) Apply that result to the transposes in reversed order.

Using $(AB)^T = B^T A^T$ and transpose invariance of rank,

$$\operatorname{rank}(AB) = \operatorname{rank}(B^T A^T) \leq \operatorname{rank} B^T = \operatorname{rank} B.$$

Both inequalities follow. Repeated application proves the corresponding bound for a product with more than two factors.

Exercise 4.S.13: An invertible left factor preserves rank

The product inequality gives $\operatorname{rank}(PA) \leq \operatorname{rank} A$. Since $A = P^{-1}(PA)$, applying it again gives $\operatorname{rank} A \leq \operatorname{rank}(PA)$. The two opposite inequalities imply equality. Invertibility is what supplies the reverse factorization.

Exercise 4.S.14: An invertible right factor preserves rank

The product inequality first gives $\operatorname{rank}(AQ) \leq \operatorname{rank} A$. The identity $A = (AQ)Q^{-1}$ gives the reverse inequality $\operatorname{rank} A \leq \operatorname{rank}(AQ)$, hence equality. Equivalently one can apply Exercise 4.S.13 to $(AQ)^T = Q^T A^T$.

Exercise 4.S.15: A zero product limits two ranks

If $AB = 0$, every column of B is in $\operatorname{Nul} A$. Thus $\operatorname{Col} B \subseteq \operatorname{Nul} A$, and dimensions give

$$\operatorname{rank} B \leq \dim \operatorname{Nul} A = n - \operatorname{rank} A.$$

Rearrangement yields $\operatorname{rank} A + \operatorname{rank} B \leq n$. The shared n is the number of inputs of A and outputs of B, which makes the containment meaningful.

Exercise 4.S.16: Rank is subadditive

Take rank factorizations $A = C_A R_A$, $B = C_B R_B$, with ranks r, s. Then block multiplication gives

$$A + B = \begin{bmatrix} C_A & C_B \end{bmatrix} \begin{pmatrix} R_A \\ R_B \end{pmatrix}.$$

The left block matrix has $r+s$ columns and therefore rank at most $r+s$. By the product inequality, $\operatorname{rank}(A+B) \le r+s = \operatorname{rank} A + \operatorname{rank} B$. If either matrix has rank zero, it is zero and the inequality is immediate, so the proof also covers those degenerate cases.

Exercise 4.S.17: Find an invertible square submatrix of full rank

(a) Select independent columns.

Choose the r pivot columns of A to obtain an $m \times r$ matrix A_1. They are independent, so $\operatorname{rank} A_1 = r$.

(b) Select independent rows of that submatrix.

Its row rank also equals r, so choose r independent rows of A_1. The resulting $r \times r$ matrix A_2 has r independent rows, hence rank r and is invertible by the Invertible Matrix Theorem. This proves the requested existence; the case $r = 0$ has the usual empty square-matrix convention or may be excluded as trivial.

Exercise 4.S.18: Expand four controlled time steps

(a) Compute each state before collecting the blocks.

From $x_0 = 0$ and $x_{k+1} = Ax_k + Bu_k$,

$$\begin{aligned} x_1 &= Bu_0, \\ x_2 &= ABu_0 + Bu_1, \\ x_3 &= A^2Bu_0 + ABu_1 + Bu_2, \\ x_4 &= A^3Bu_0 + A^2Bu_1 + ABu_2 + Bu_3 \\ &= \underbrace{[\,B \;\; AB \;\; A^2B \;\; A^3B\,]}_{M} \begin{pmatrix} u_3 \\ u_2 \\ u_1 \\ u_0 \end{pmatrix}. \end{aligned}$$

The order of the input blocks is reversed to match the stated order in M. Each u_j has two entries, so the stacked input has eight entries, matching the 4×8 matrix M.

(b) Use full row rank to realize an arbitrary target.

Controllability means rank $M = 4$, so M maps $\mathbb{R}^8$ onto $\mathbb{R}^4$. Given $v \in \mathbb{R}^4$, solve $Mz = v$, then split its eight entries into four two-entry blocks (u_3, u_2, u_1, u_0). The computation in part (a) shows that this control sequence gives $x_4 = v$.

Exercise 4.S.19: Check the rank of the controllability matrix

Here the state dimension is 3 and the input has one coordinate, so form $M = [\,B \;\; AB \;\; A^2B\,]$. Exact arithmetic gives

$$M = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -\frac{9}{10} & \frac{81}{100} \\ 1 & \frac{1}{2} & \frac{1}{4} \end{pmatrix}, \qquad \det M = \frac{14}{25}.$$

Since this determinant is nonzero, M has rank 3 and its columns span the state space. Therefore the pair (A, B) is controllable.

Exercise 4.S.20: Check the rank of the controllability matrix

Here the state dimension is 3 and the input has one coordinate, so form $M = [\,B\ AB\ A^2B\,]$. Exact arithmetic gives

$$M = \begin{pmatrix} 1 & \frac{1}{2} & \frac{19}{100} \\ 1 & \frac{7}{10} & \frac{9}{20} \\ 0 & 0 & 0 \end{pmatrix}, \qquad \det M = 0.$$

The determinant is zero, so M does not have full rank and (A, B) is not controllable. Its actual rank is 2. Its last row is zero, while the leading 2×2 minor has determinant $1/5 \neq 0$, verifying rank two.

Exercise 4.S.21: Check the rank of the controllability matrix

Here the state dimension is 4 and the input has one coordinate, so form $M = [\,B\ AB\ A^2B\ A^3B\,]$. Exact arithmetic gives

$$M = \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 0 & -1 & \frac{8}{5} \\ 0 & -1 & \frac{8}{5} & -\frac{24}{25} \\ -1 & \frac{8}{5} & -\frac{24}{25} & -\frac{3}{125} \end{pmatrix}, \qquad \det M = 0.$$

The determinant is zero, so M does not have full rank and (A, B) is not controllable. Its actual rank is 3. For example its leading 3×3 minor has determinant -1, showing rank at least three; the zero full determinant then gives exactly three.

Exercise 4.S.22: Check the rank of the controllability matrix

Here the state dimension is 4 and the input has one coordinate, so form $M = [\,B\ AB\ A^2B\ A^3B\,]$. Exact arithmetic gives

$$M = \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 0 & -1 & \frac{1}{2} \\ 0 & -1 & \frac{1}{2} & \frac{229}{20} \\ -1 & \frac{1}{2} & \frac{229}{20} & -\frac{411}{40} \end{pmatrix}, \qquad \det M = -\frac{3}{10}.$$

Since this determinant is nonzero, M has rank 4 and its columns span the state space. Therefore the pair (A, B) is controllable.